



SECJ2203: Software Engineering

System Documentation (SD)

Koperasi Kakitangan KADA Kelantan BHD

Version 1.0

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Faculty of Computing

Prepared by: <TechMedico>

Revision Page

a. Overview

The current version of system documentation introduces the basis of KADA system. The introduction and specific requirements are included. The objective, scope, definitions, acronyms and abbreviations, references, and overview are all included in the introduction. To give a thorough grasp of the project, this portion of the system requirement specification also explains the persona, system characteristics such as the use case diagram, class diagram, state diagram, user stories, non-functional requirements and design constraints.

b. Target Audience

As the ones who would utilize the system, system developers who oversee creating and implementing the KADA system, as well as project stakeholders like KADA cooperative members, are the target audience for this SD because they are more knowledgeable about the system and may have more precise requirements.

c. Project Team Members

The System Documentation is designed for system developers, administrators and technical support teams that are responsible for implementing, maintaining and troubleshooting the KADA system. It is also created for board members and other stakeholders who require a high-level understanding of the system's operations and procedures for management and decision-making.

Member Name	Role	Task	Status
Elijah She Yu Sheng	Project manager	Specify design constraints. Define performance and other requirements. Draw state diagram	Complete
Choh Jing Yi	System analyst	Create user story for KKK admin Draw sequence diagram Draw activity diagram	Complete
Lee Yin Shen	System designer	Create user story for apply membership Draw sequence diagram Draw activity diagram Draw domain model	Complete
Joanne Ching Yin Xuan	Process analyst	Document launch phase	Complete

		Draw sequence diagram Draw activity diagram Draw use case diagram	
Lim Yu Han	Quality analyst	Create user story for apply loan Draw sequence diagram Draw activity diagram	Complete

d. **Version Control History**

Version	Primary Author(s)	Description of Version	Date Completed
1.0	ELIJAH SHE YU SHENG LEE YIN SHEN CHOH JING YI JOANNE CHING YU XUAN LIM YU HAN	Completed all sections in the system documentation	15/12/2024
1.1	ELIJAH SHE YU SHENG LEE YIN SHEN CHOH JING YI JOANNE CHING YU XUAN LIM YU HAN	Edited user diagram and class diagram	23/12/2024

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1. Introduction

1.1 Purpose

This SD serves as a structured document to guide the analysis, design, development, and deployment of the KADA system, which is intended to digitize the cooperative's operations. This SD comprises the project's requirements, architecture, and functionality; by referring to this document, project stakeholders will gain a clear knowledge of what the project is doing. For example, SD can serve as an SRS since it clearly lists the functional and non-functional, user and system requirements of the KADA system. It can also serve as a reminder to developers to always deliver output that meets stakeholder expectations. Additionally, the project management can make sure that every step is carried out in the proper and logical order, and the development team may utilize this SD as a blueprint to build the system effectively. For the quality analyst team to evaluate and enhance the system in accordance with the requirements, this SD also contains predicted performance. As specified in this SD, both developers and stakeholders make sure that things like the deadline and deliverables move smoothly.

1.2 Scope

We want to replace the current manual Kada website with our software solution, the KADA online system. By using a digital solution, our KADA online system will increase operational effectiveness, accessibility, and usability. With certain essential features, the KADA online system offers a unified platform for administrators, which is cooperative executives, members, and regular users.

1. New Member Registration

By completing the member application form found on the website, users can register as members online. Both members and administrators will greatly benefit from this, as members can monitor their membership status and administrators may approve applications by just clicking on the website rather than having to review numerous documents.

2. Loan management

Members can browse the list of available loan types, use the loan calculator in our system to determine the interest rate, and then only apply for the loan type they want. If they receive a loan offer, members can also monitor their loan status and repayment plans.

3. Saving account

Members can obtain and repay loans by managing their savings accounts, which are integrated into the system.

4. Financial reporting

Both members and administrators can retrieve monthly and annual reports from the system, which will enable members to monitor their financial performance and administrators to learn about the cooperative's performance. The purpose of this report is to increase financial transparency.

5. Admin reporting

1.3 Definitions, Acronyms and Abbreviation

This subsection should provide the definitions of all terms, acronyms, and abbreviations used in the SD.

1. KADA	Lembaga Kemajuan Pertanian Kemubu
2. SD	System Documentation
3. SRS	System Requirement Specification
4. SDD	System Design Document
5. STD	System Test Document
6. UX	User Experience
7. MTBF	Mean Time Between Failures
8. API	Application Programming Interface
9. MCC	Malaysian Cooperative Commission
10. RAM	Random Access Memory
11. SSD	Solid State Drive
12. TLS	Transport Layer Security
13. RBAC	Role-Based Access Control

1.4 References

- [1] A. Athuraliya, “The Easy Guide to UML Activity Diagrams | Creately,” creately.com, Jun. 20, 2022. <https://creately.com/guides/activity-diagram-tutorial/>
- [2] “How to Draw a Sequence Diagram in UML,” Lucidchart, 2019. <https://www.lucidchart.com/pages/how-to-draw-a-sequence-diagram-in-UML>
- [3] “Domain Modeling: What you need to know before coding,” Thoughtworks. <https://www.thoughtworks.com/insights/blog/agile-project-management/domain-modeling-what-you-need-to-know-before-coding>

1.5 Overview

This System Documentation includes special requirements for program development. We added five personas, each with their own user needs and tales. We will use them throughout this section to ensure that our development team has a clear understanding of the system requirements. The system features will then be explained in more depth using a use case diagram, as well as a brief description of each use case using a table, class diagram, and state machine diagram. Next, we'll use a sequence diagram and an activity diagram to represent each user story, allowing designers and developers to fully exploit this SD to help their work. Finally, we will conclude this document on the basis of system performance, needs, and design constraints.

2. Specific Requirements

This section outlines the software requirements in detail to guide the process of design, development, and testing. By describing all required inputs, outputs, and functionality specifically, it ensures that the system satisfies the needs of users and stakeholders.

2.1 Persona

The Persona section defines the target users of the system, detailing their characteristics and user stories when using the system, to ensure that the system is designed to meet their specific needs. Understanding the personas helps ensure that the system is appropriate for its target users.

2.1.1 Persona 1: KADA employee (New User)

User Need

A KADA employee needs to register as a new KADA member to enjoy benefits.

User Stories

1. As a KADA employee, I want to register myself as a new member of the cooperative, so that I can access the benefits provided by the corporation.
2. As a KADA member, I want to log in to the system using my unique username and password, so that I can access my respective dashboard

2.1.2 Persona 2: KADA member (Existing member)

User Need

A KADA member needs a way to apply loans easily to purchase equipment, machinery, or infrastructure required.

User Stories

1. As a KADA member, I want the system to check the approval status of my loan application, so that I can know if my loan request has been approved and understand any next steps or additional requirements to secure funding.
2. As a KADA member, I want to apply for a loan online through the system, so that I can complete the application process conveniently and quickly without needing to visit a physical office.

3. As a KADA member, I want to apply for a training program online, so that I can complete my application quickly and conveniently without needing to visit a physical office.
4. As a KADA member, I want to manage my saving account through the system, so that I can deposit money, pay for loans, apply for programs, and view my account information conveniently.
5. As a KADA member, I want to generate reports through the system, so that I can make informed decisions based on my financial situation.
6. As a KADA member, I want to view and verify financial information through the system, so that I can stay updated on my current financial status and ensure accuracy.

2.1.3 Persona 3: KADA admin

User Need

A KADA admin needs a way to efficiently manage member applications, loan requests, program participation, and system policies to ensure smooth operations.

User Stories

1. As a KADA admin, I want to view applicant lists for new members and, loans, applications, so that I can review their details and proceed with the approval process in an orderly manner.
2. As a KADA admin, I want to approve new members, so that I can ensure applicants meet the necessary requirements and maintain the integrity of the KADA membership.
3. As a KADA admin, I want to approve new loans, so that I can verify members' eligibility and ensure compliance with the loan policies.
4. As a KADA admin, I want to view

2.2 System Features

KADA System is a web-based software. The system provided a means for the members and admin to better manage the member application and loan application. It also has other features such as managing saving account, view financial status, generate finance report, and manage website.

2.2.1 Use Case Diagram

The system features are illustrated in Figure 1.1 below. The detail description of each module is tabulated in Table 1.1.

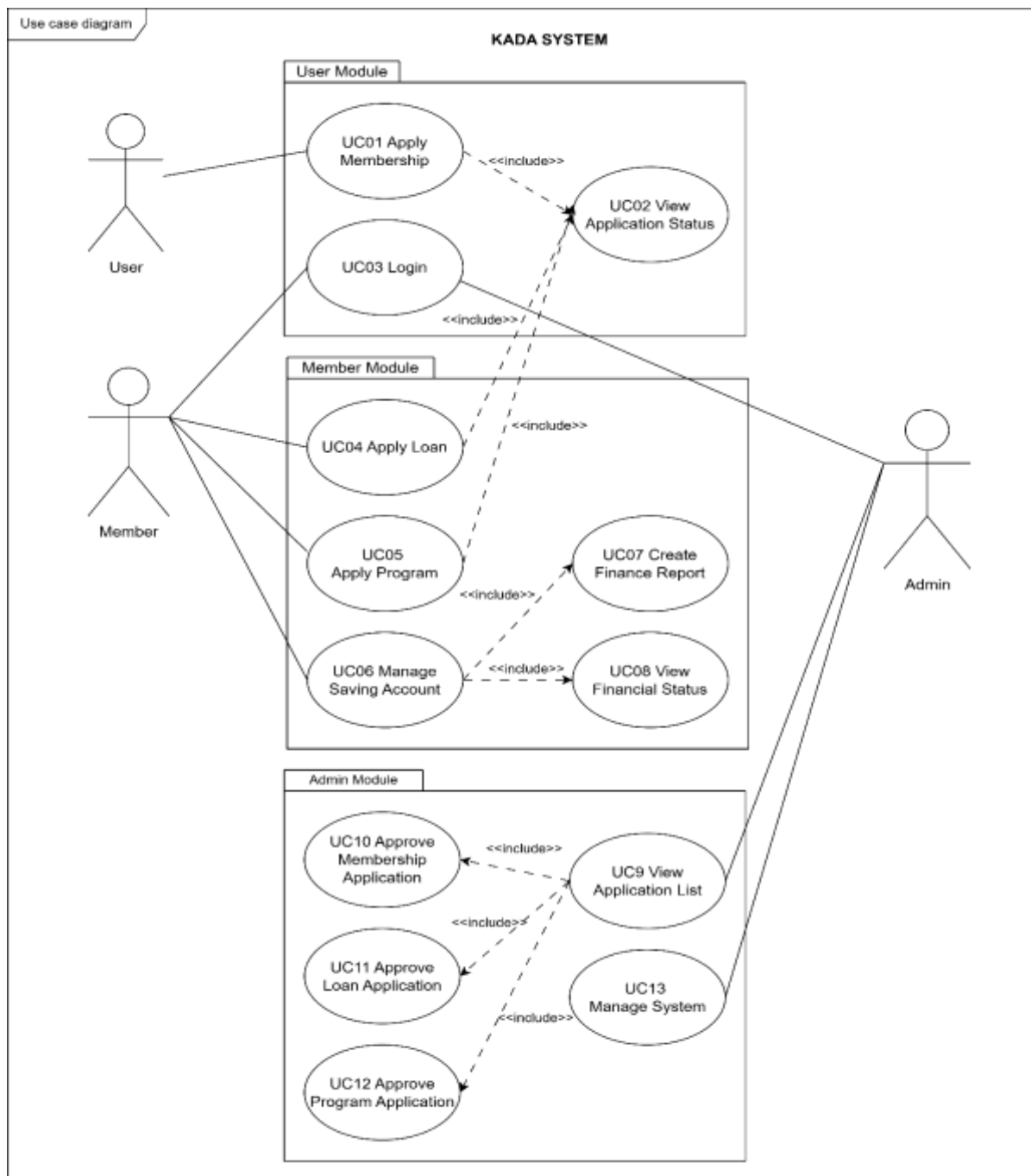


Figure 2.2.1.1: Use Case Diagram for <KADA Koperasi System>

Table 1.1: Description of Module and Functions for <Koperasi KADA System>

Use case	Function	Description
UC01	Apply Membership	A user can apply to be a member in KADA cooperative by filling in the online form and uploading the required documents.
UC02	View Application Status	If a user applies for membership, or a member apply for loan or program, they would be able to view their application status whether approved or rejected and further instructions.
UC03	Log In	The member and admin can login to the system by using the unique login username and password. Members will turn to the member page while admin will turn to admin page.
UC04	Apply Loan	The members can apply for loans by filling in the online form and uploading any required documents.
UC05	Manage Saving Account	The members can manage their saving accounts like depositing, paying for loans and program application and view account info.
UC06	Create Finance Report	Upon the management saving account operation, the members can create their monthly and annual finance report.
UC07	View Financial Status	Upon the saving account operation, the member can view their financial status, such as loan period, loan balance and loan amount.
UC08	Admin view Application List	The amin can view the list of membership and applications as well as the membership, and existing applications.
UC09	Approve Membership Application	The admin would be able to update the status of the membership application.
UC10	Approve Laon Application	The admin would be able to update the status of the loan application.
UC11	Admin view Report	The admin can view the report of the statistics of total membership applications and loan applications in a year.

Functional Requirement of Each Use Case

UC01 Apply Membership

- Req-1: The system shall validate the user input, include verified email address, ID card number and phone number.
- Req-2: The system shall display error message for invalid input.
- Req-3: The system shall send an email notification to the user upon the submission of the membership application.
- Req-4: The system shall ensure that all application data is securely stored and complies with relevant data protection regulations.
- Req-5: The system shall display a confirmation message to the user after successfully updating the database.

UC02 View Application Status

- Req-1: The system shall allow users to access their application status securely.
- The system shall display detailed application status, like pending, approved, and rejected.
- The system shall notify the users of any updates of their application status.

UC03 Log In

- Req-1: The system shall validate the user credentials, include verified username and password.
- Req-2: The system shall display error message for invalid input and login attempts.
- Req-3: The system shall provide an option for password recovery.
- Req-4: The system shall log unsuccessful login attempts for security purpose.

UC04 Apply Loan

- Req-1: The system shall validate the user input, include loan type, amount, and period.
- Req-2: The system shall display error message for invalid input such as exceeding loan amount limit or missing document.
- Req-3: The system shall send a confirmation email to users if successful submit the form.
- Req-4: The system shall allow secure upload of required documents during the application process.

UC05 Manage Saving Account

- Req-1: The system shall allow users to view their saving account details, including:
 - Current balance
 - Recent transactions
 - Loan balance
- Req-2: The system shall enable users to make deposits or payments securely via supported methods like credit card and online banking.
- Req-3: The system shall show confirmation message for successful transactions and error message if transaction fail.
- Req-4: The system shall provide transaction history for each user.

UC06 Create Finance Report

- Req-1: The system shall allow users to generate detailed financial reports, including:
 - Loan balances
 - Payment histories
 - Total contributions to savings
- Req-2: The system shall enable users to download reports in standard formats like PDF and Excel.
- Req-3: The system shall ensure all reports are accurate and updated in real-time with latest data.

UC07 View Financial Status

- Req-1: The system shall display a detailed financial summary for users, including:
 - Contributions to savings
 - Outstanding loan balances
 - Payment due dates
- Req-2: The system shall ensure all financial data are accurate and updated in real-time with the latest data.

UC08 View Application List

- Req-1: The system shall provide admins with a list of all membership, and loan, applications.
- Req-2: The system shall allow admins to sort and filter application based on:
 - Application type
 - Application status
 - Submission date
- Req-3: The system shall enable admins to view detailed information for each application.

UC09 Approve Membership Application

- Req-1: The system shall enable admins to review the submitted membership application forms.
- Req-2: The system shall allow admins to approve or reject the application with reasons for rejection.
- Req-3: The system shall notify applications of the decision via email and update member details in the list.

UC10 Approve Loan Application

- Req-1: The system shall enable admins to review the submitted loan application forms.
- Req-2: The system shall allow admins to approve or reject the application with reasons for rejection.
- Req-3: The system shall notify applications of the decision via email and update member details in the list.

UC11 Manage Website

- Req-1: The system shall allow admins to generate detailed applications report, including:
 - Loan applications
 - Membership applications
- Req-2: The system shall enable admin to download reports in standard formats like PDF .
- Req-3: The system shall ensure all reports are accurate and updated in real-time with latest data.

2.2.2 Domain Model: Class Diagram

The domain model of the KADA System is illustrated in Figure 1.2 below.

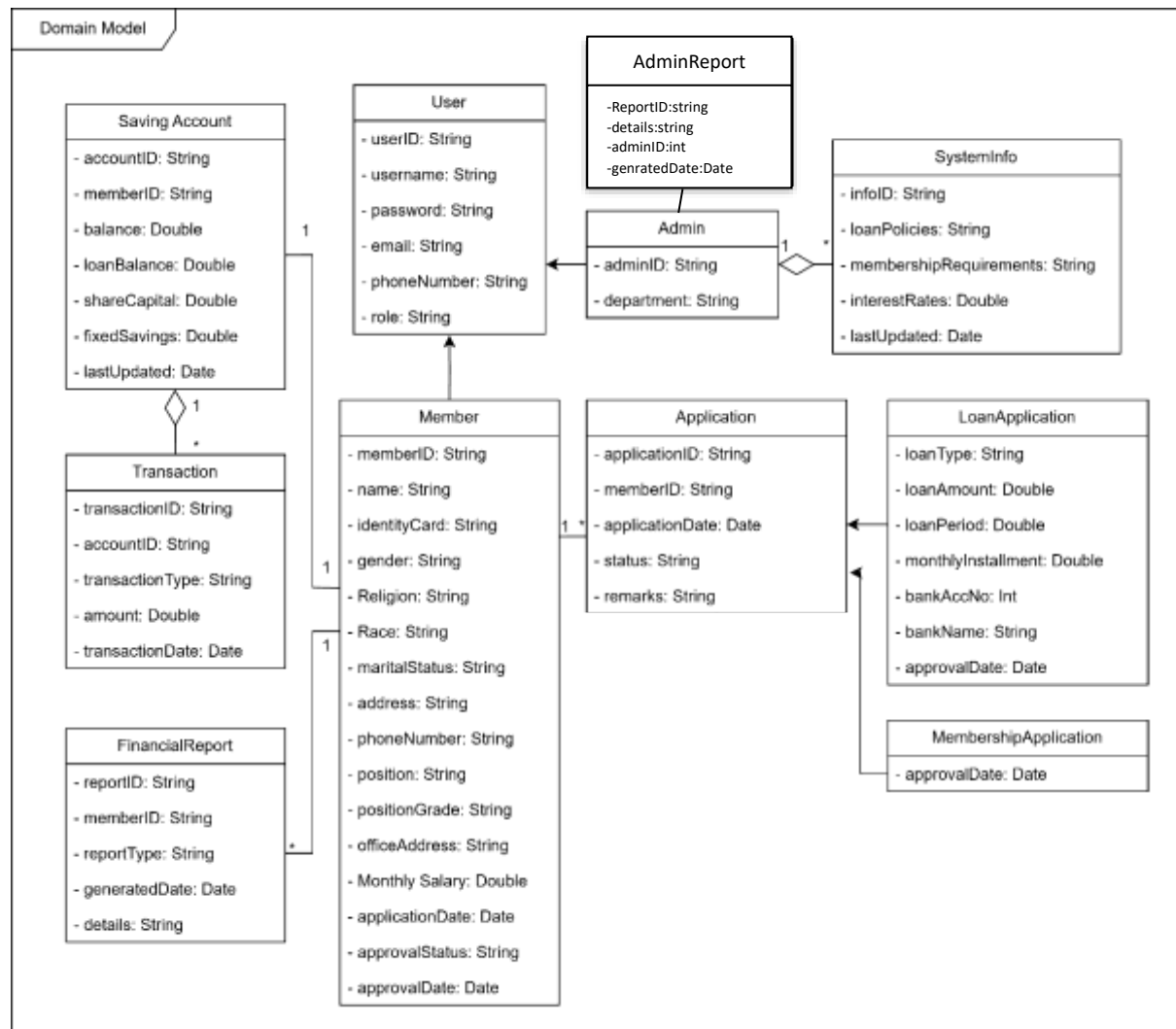


Figure 2.2.2.1: Domain Model for <KADA System>

Description of Classes

User Class

Represent as superclass for all users using the system, including members and admins.

- **userID:** A unique identifier for each user in the system.
- **username:** The username used for logging into the system.
- **password:** The password used for authentication.
- **email:** The email address associated with the user.
- **phoneNumber:** The contact phone number of the user.
- **role:** The role of the user, such as Member or Admin.

** The Member class is inherited from the User class. Each member is a specific type of user.

Admin Class

Represents an admin user responsible for managing the system.

- **adminID:** A unique identifier for each admin, referencing the userID.
- **department:** The department the admin belongs to (e.g., finance or operations).

** The Admin class is inherited from the User class. Each admin is a specific type of user.

Member Class

Represents a person registered as a member of the system.

- **memberID:** A unique identifier for each member, referencing the userID.
- **name:** The full name of the member.
- **identityCard:** The identity card number of the member (e.g., MyKad number).
- **gender:** The gender of the member (e.g., Male, Female).
- **race:** The race or ethnicity of the member.
- **maritalStatus:** The marital status of the member (e.g., Single, Married).
- **address:** The home address of the member.
- **phoneNumber:** The contact phone number of the member.
- **position:** The member's job position within the organization.
- **positionGrade:** The grade or level of the member's job position.
- **officeAddress:** The office address where the member works.
- **monthlySalary:** The member's monthly income.
- **applicationDate:** The date the member applied for membership.
- **approvalStatus:** The status of the membership application (e.g., Pending, Approved, Rejected).
- **approvalDate:** The date the membership application was approved.

SavingAccount Class

Represents a member's savings account within the system.

- **accountID:** A unique identifier for the saving account.
- **memberID:** A reference to the memberID that owns this account.
- **balance:** The current balance in the saving account.
- **loanBalance:** The total outstanding loan amount associated with this account.
- **shareCapital:** The share capital contributed by the member.
- **fixedSavings:** The fixed savings amount stored in the account.
- **lastUpdated:** The date when the account details were last updated.

** The SavingAccount class is associated with the Member class. The relation is one-to-one, where each member has exactly one saving account.

Transaction Class

Represents a financial transaction made on a saving account.

- **transactionID:** A unique identifier for the transaction.
- **accountID:** A reference to the accountID of the saving account associated with the transaction.
- **transactionType:** The type of transaction (e.g., Deposit, Withdrawal, Payment).
- **amount:** The monetary value of the transaction.
- **transactionDate:** The date the transaction was made.

Application Class

Represents a general application for membership, loans, or programs.

- **applicationID:** A unique identifier for the application.
- **memberID:** A reference to the memberID of the applicant.
- **applicationDate:** The date the application was submitted.
- **status:** The current status of the application (e.g., Pending, Approved, Rejected).
- **remarks:** Additional comments or feedback related to the application.

** The MembershipApplication and LoanApplication classes are inherited from the Application class. Each specific application type extends the general application class.

MembershipApplication Class

A specific type of application for membership.

- **approvalDate:** The date the membership application was approved.

LoanApplication Class

A specific type of application for loans.

- **loanType:** The type of loan applied for (e.g., Personal, Business).
- **loanAmount:** The amount of money requested in the loan.

- **loanPeriod:** The duration of the loan in months.
- **monthlyInstallment:** The monthly repayment amount.
- **bankAccNo:** The bank account number associated with the loan disbursement.
- **bankName:** The name of the bank where the loan will be deposited.
- **approvalDate:** The date the loan application was approved.

FinancialReport Class

Represents a financial report generated for a member.

- **reportID:** A unique identifier for the financial report.
- **memberID:** A reference to the memberID of the member who owns this report.
- **reportType:** The type of financial report (e.g., Monthly, Annual).
- **generatedDate:** The date the report was generated.
- **details:** A summary of the report's contents, such as savings and loan balances.

** The FinancialReport class is associated with the Member class. The relation is one-to-many, where a member can generate multiple financial reports.

AdminReport Class

Represents a financial report generated for a member.

- **reportID:** A unique identifier for the financial report.
- **adminID:** A reference to the memberID of the member who owns this report.
- **reportType:** The type of financial report (e.g., Monthly, Annual).
- **generatedDate:** The date the report was generated.
- **details:** A summary of the report's contents, such as savings and loan balances.

** The AdminReport class is associated with the Admin class. The relation is one-to-many, where a member can generate multiple financial reports.

2.2.3 State Diagram

The state diagram shows the states and transitions of an object throughout the system. The *MembershipApplication* class, *LoanApplication* class, *Program Application* class, *SavingAccount* class, *Transaction* class and the *Financial Report* class would require a state diagram that shows the transition of application status.

Figure 2.2.3.1 shows the state diagram for the *Membership, loan, and program application process*. Once the user submits the application, the application will be in Processing state. If there are incomplete applications that need addition documents from applicants, it will move to waiting for documents state. Then the result of the application will be updated to either Approve state or Rejected state.

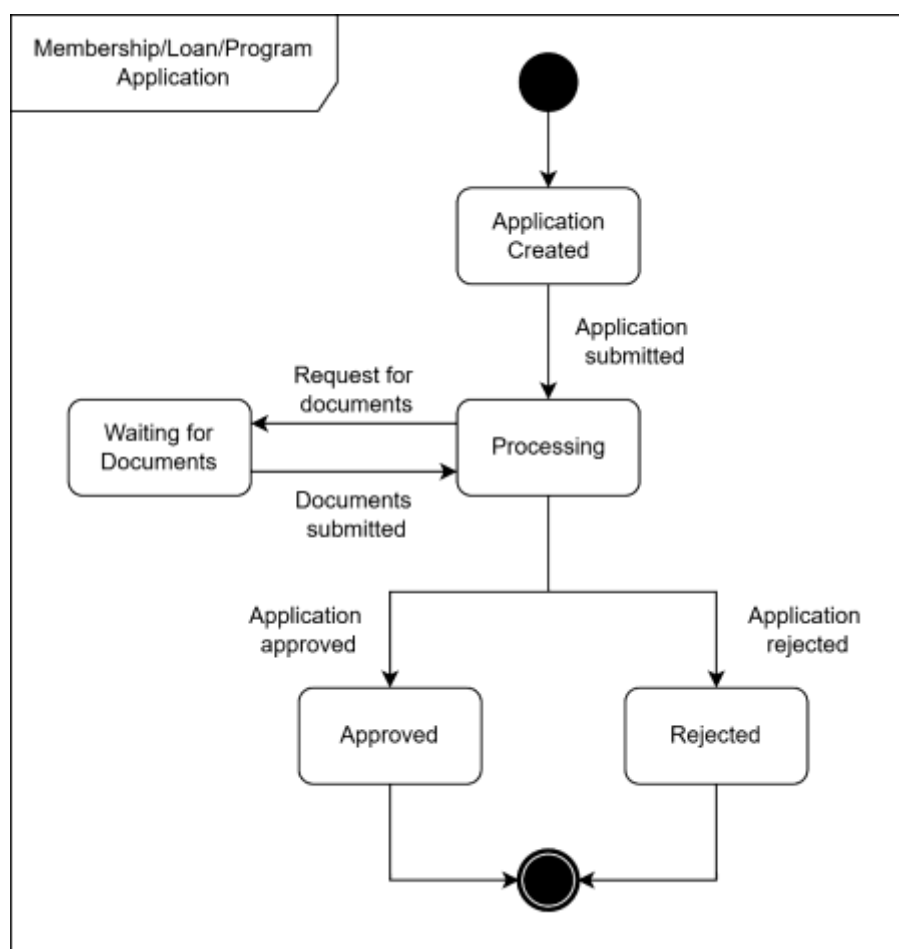


Figure 2.2.3.1: State Machine Diagram for Membership, Loan, and Program Application process

Figure 2.2.3.2 shows the state diagram for the *Saving Account Lifecycle*. Once the user becomes a member, their saving account will be created. The account will move to the Active State where allows members to manage the account. If there are issues like policy violations or inactive, the account will move to Suspended state. A member needs to reactive the account so that the account can move back to Active state. Lastly, the account can transition to Closed state when it is permanently closed.

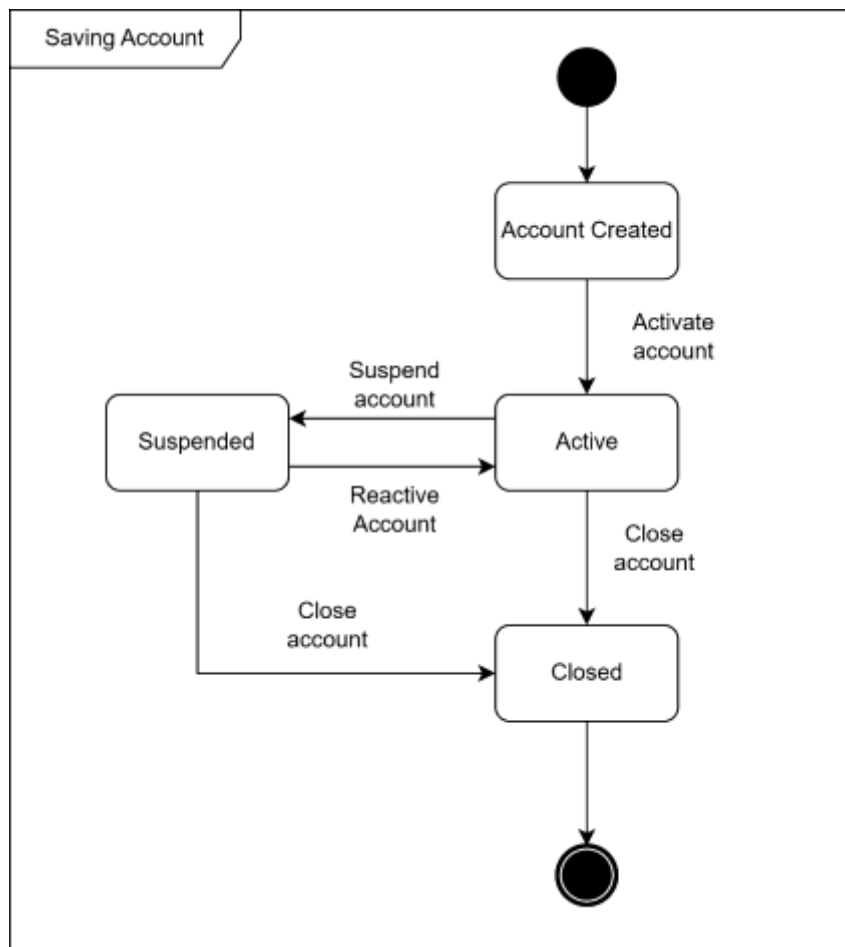


Figure 2.2.3.2: State Machine Diagram for Saving Account Lifecycle

Figure 2.2.3.3 shows the state diagram for the *Transaction Processing*. When a user initiates a transaction, it will enter the Transaction Created state. The system then begins processing the transaction and moves to Pending Processing State. If a user has insufficient funds or other reasons, it will move to Failed state. While if transaction successful, it will move to Completed state. A user can cancel the transaction any time, so that it will move to Cancelled state.

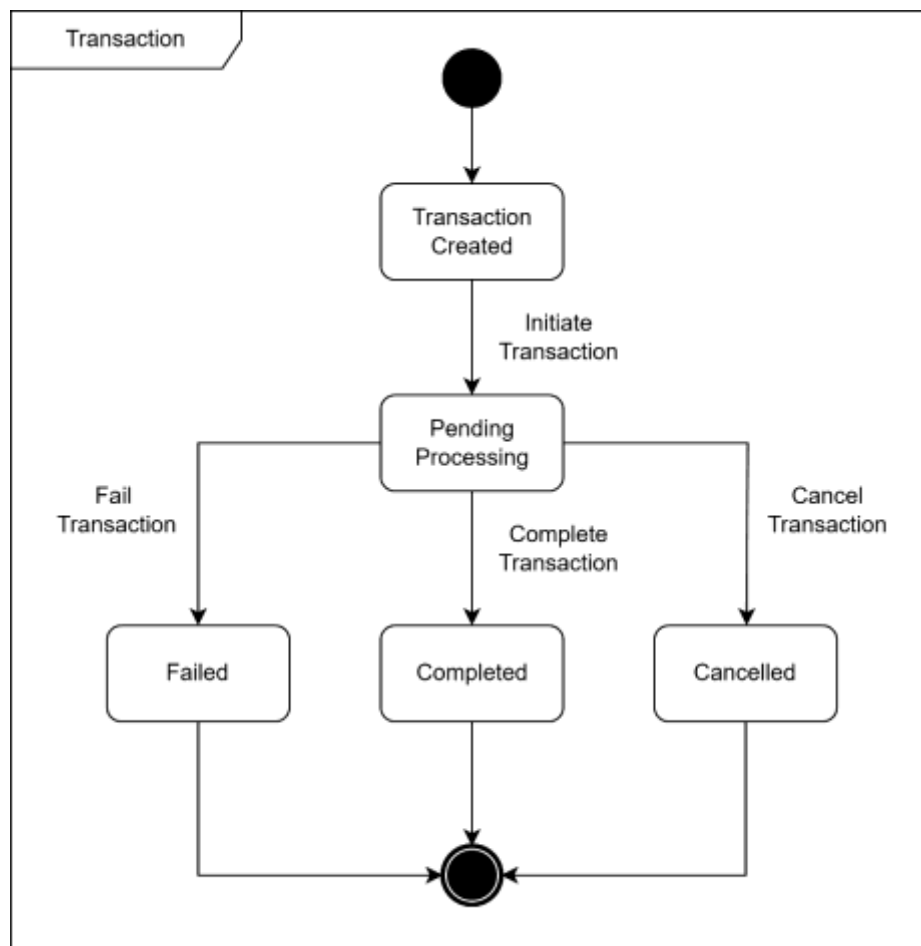


Figure 2.2.3.3: State Machine Diagram for Transaction Processing

Figure 2.2.3.4 shows the state diagram for the *Financial Report Generation* process. When a member requests a financial report, it will enter the Report Requested state. Then the system begins compiling the requested data and moves to Generating state. If the process completed successful, the report moves to Completed state. However, if errors occur like missing or corrupted data, the report will transition to Failed state. A member can cancel the generation any time so that it will move to Cancelled State.

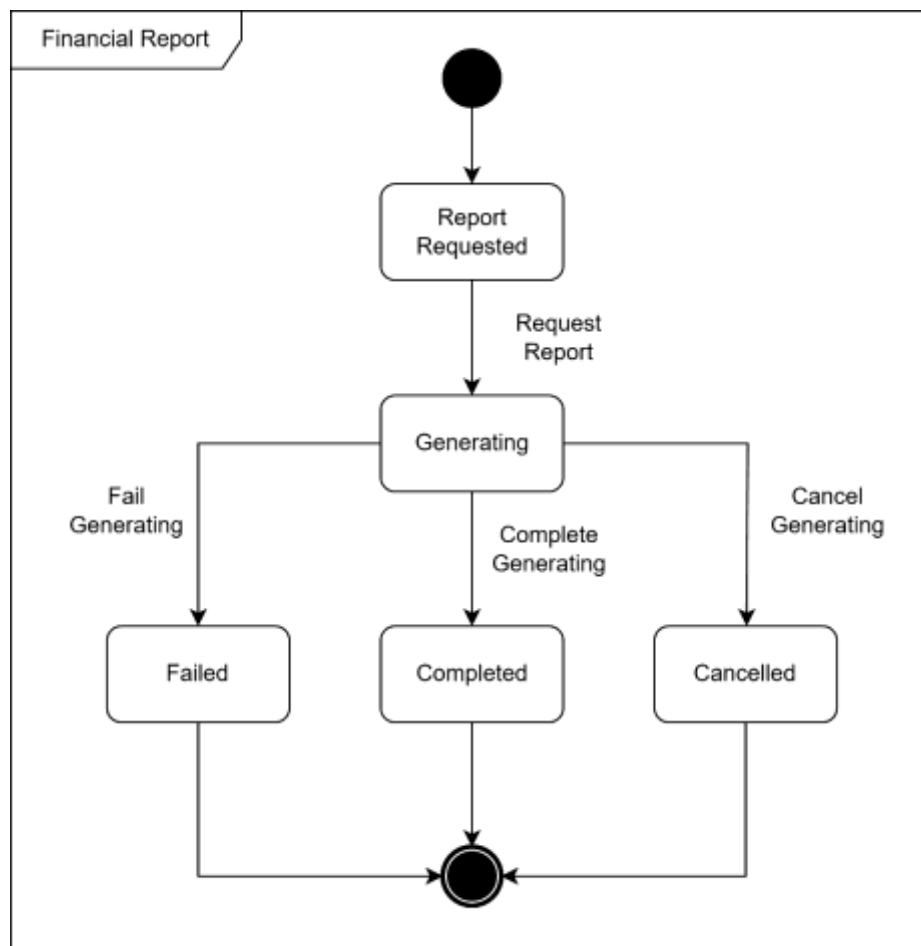


Figure 2.2.3.4: State Machine Diagram for Financial Report Generation

2.3 Launch Phase

The product backlog lists the parts and modules for the project according to the priorities. Table 1.2 shows the product backlog for the KADA Online System.

Table 1.2 Product Backlog for the KADA Online System

Sprint	Use Case Involved	Team Members Assigned
Sprint 1 User Module	UC001 Apply Membership UC002 View Application Status UC003 Log In	Choh Jing Yi Elijah She Yu Sheng Lee Yin Shen
Sprint 2 Member Module	UC004 Apply Loan Application UC005 Manage Saving Account	Choh Jing Yi Elijah She Yu Sheng Lee Yin Shen
Sprint 3 Admin Module	UC006 Create Finance Report UC007 View Financial Status UC008 Admin view Application List	Joanne Ching Yin Xuan Lim Yu Han
Sprint 4 Admin Module	UC009 Approve Membership Application UC010 Approve LOAN Application UC011 View Administrative	Joanne Ching Yin Xuan Lim Yu Han

2.4 User Story Details

This section highlights the goals and expectations of users for the KADA system, emphasizing the necessary actions to be taken. Each user story outlines the scope, event flow, and includes supporting diagrams, such as sequence and activity diagrams, to ensure the system fulfills user's needs.

2.4.1 US001: User Story Apply for Membership

Table 2.1: User Story Description for Applying for Membership

User story: Apply for Membership
ID: US001
User Story Description As a KADA employee I want to register myself as a new member of the cooperative So that I can access the benefits provided by the corporate
Flow of events: 1. The employee accesses the system through the website. 2. The employee chooses to register as a member. 3. The employee inputs all the required details. 4. The system presents the “Important Info before applying” statement and the employee agrees by ticking the confirmation box. 5. The employee applies for review. 6. The employee will get a proof of submission
Alternative flow 1: 1. If incomplete details are provided, the system prompts the employee to fill in the missing fields before proceeding. 2. If the application is rejected, the member cannot log into the system.
Acceptance Criteria Precondition: 1. The applicant must be a current employee of KADA. Postcondition: 1. If the application is approved, employee becomes a registered member and can access the benefits provided by the cooperative. 2. If the application is rejected, the employee cannot log into the system. 3. The system stores the accepted employee's data in the database
Exception flow: If an error occurs while updating the information, the system prompts the employee to try again later.

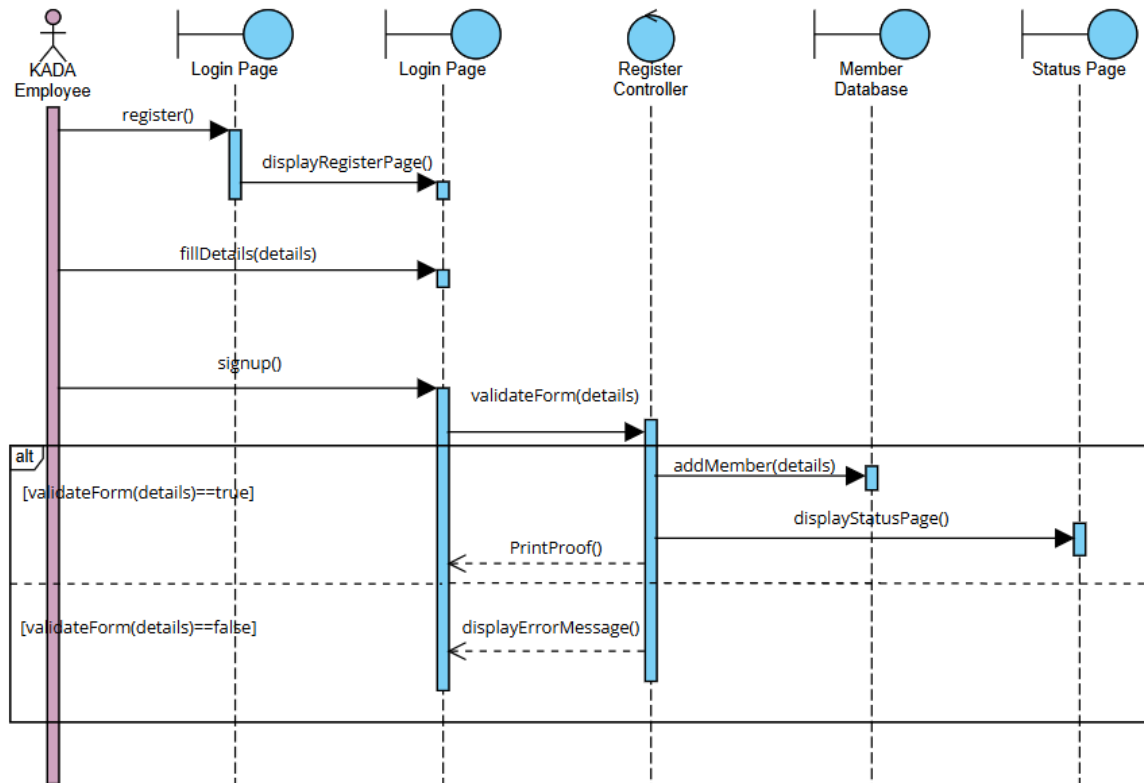


Figure 2.1: Sequence Diagram for Apply for Membership

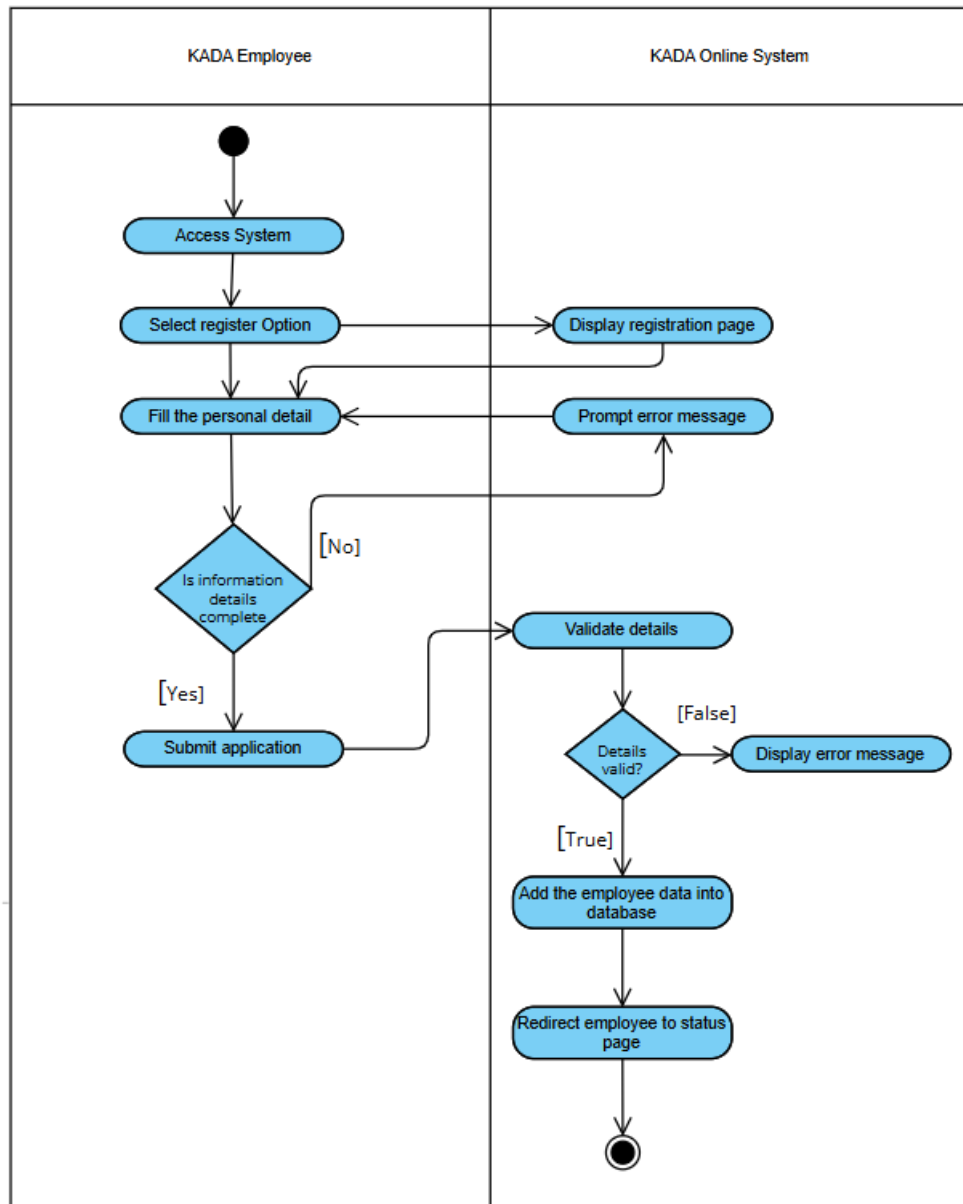


Figure 2.2: Activity Diagram for Apply Membership

2.4.2 US002: User Story View Application Status

Table 2.2: User Story Description for View Application Status

User story: View Application Status
ID: US002
User Story Description As a cooperative member I want to view my application status So that I get to know my current application status
Flow of events: 1. The employee logs into the system. 2. Click "Status" on the navigation bar 3. Employee reviews all statuses in this system including membership and loan application.
Alternative flow n: If the employee has no active applications, the system displays a message stating, "No active applications."
Acceptance Criteria Precondition: 1. The employee is logged into the system. 2. The employees' information is available in the system, including their application statuses. Postcondition: The employees successfully view their current application status, including membership, and loan applications.
Exception flow: If an error occurs while retrieving or displaying the status information, the system prompts the employee to try again later, with an error message: "Unable to retrieve application status. Please try again later."

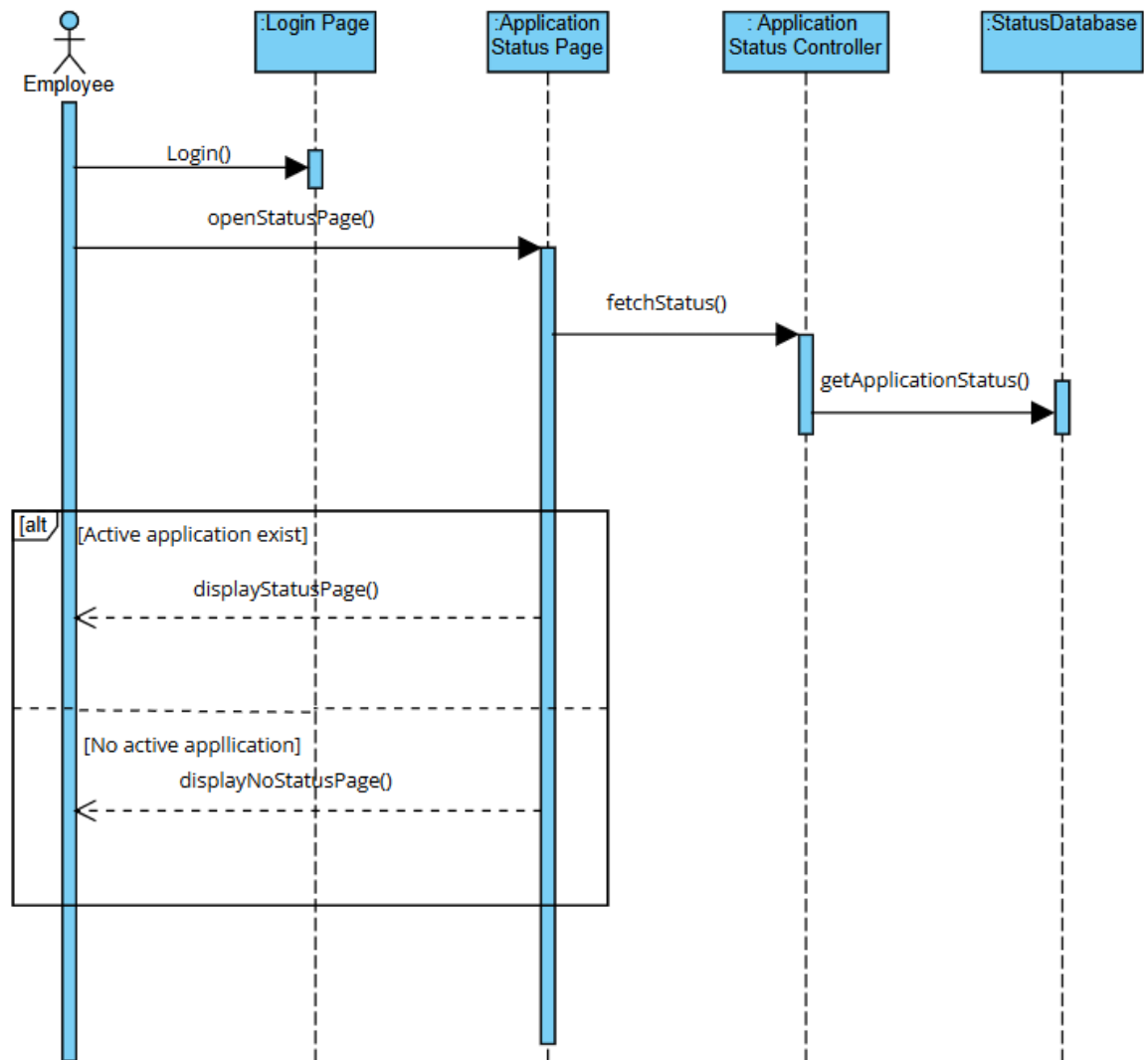


Figure 2.3: Sequence Diagram for Application Status

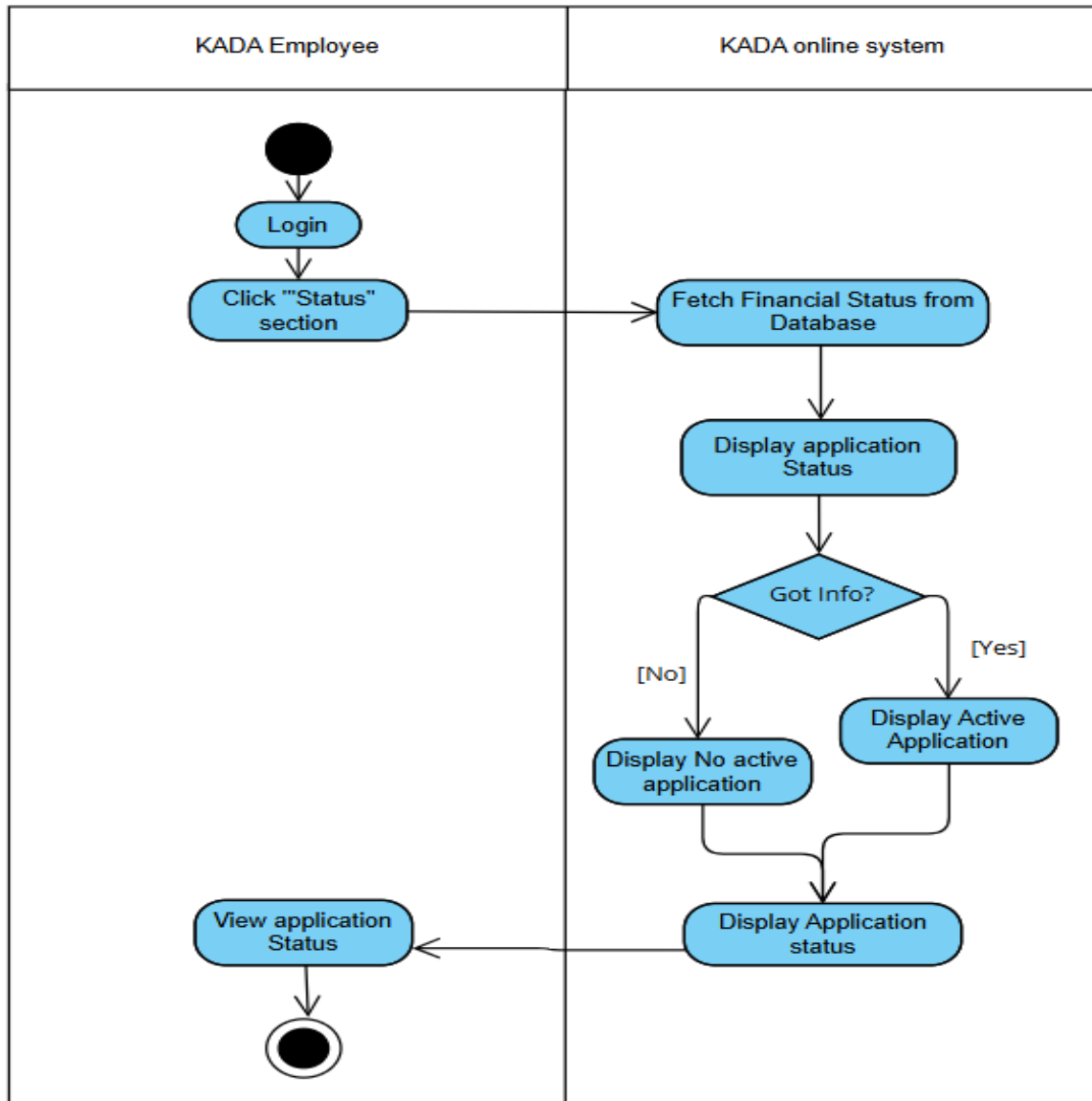


Figure 2.4: Activity Diagram for Application Status

2.4.3 US003: User Story Log In

Table 2.3: User Story Description for Log In

User story: Log In
ID: US003
User Story Description As a cooperative member or admin I want to log in to the system using my unique username and password So that I can access my respective dashboard
Flow of events: 1. Members or Admin opens the KADA cooperation website. 2. Members or Admin clicks “Log Masuk”. 3. Members or Admin key in user ID and password in their respective column. 4. System validates user ID and password.
Alternative flow 1: If invalid user ID and password entered, the system prompts the user to re-enter user ID and password.
Acceptance Criteria Precondition: The system contains unique usernames and passwords for all members and admins. Postcondition: 1.The member successfully logs in and is redirected to the member page. 2.The admin successfully logs in and is redirected to the admin page.
Exception flow: If the system encounters an error during the login process, the system displays the error message: "Unable to process login at the moment. Please try again later."

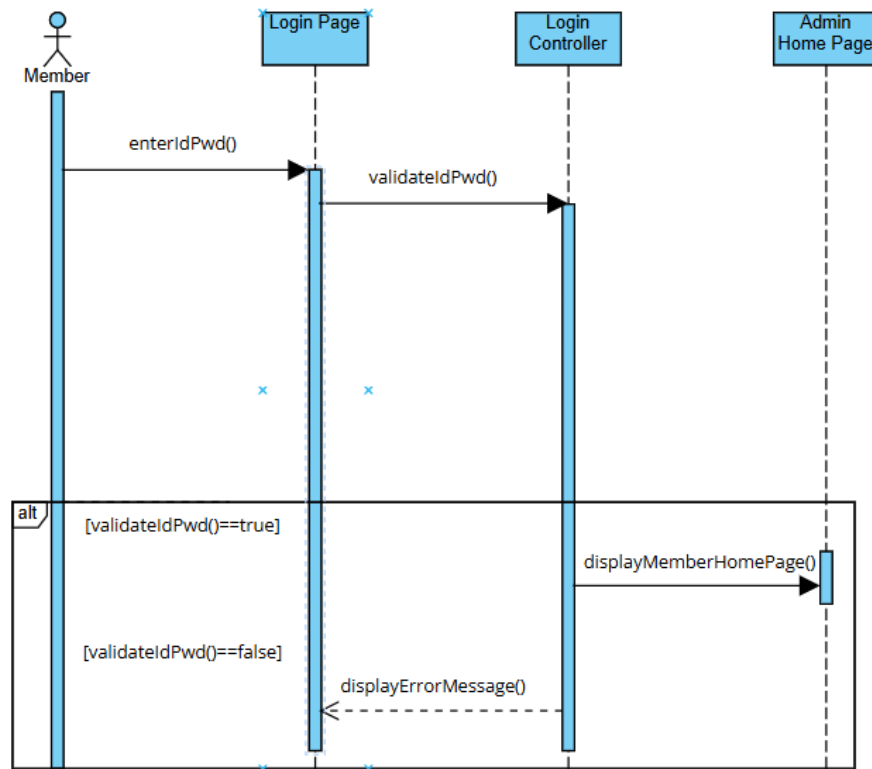


Figure 2.5: Sequence Diagram for Member Login to System

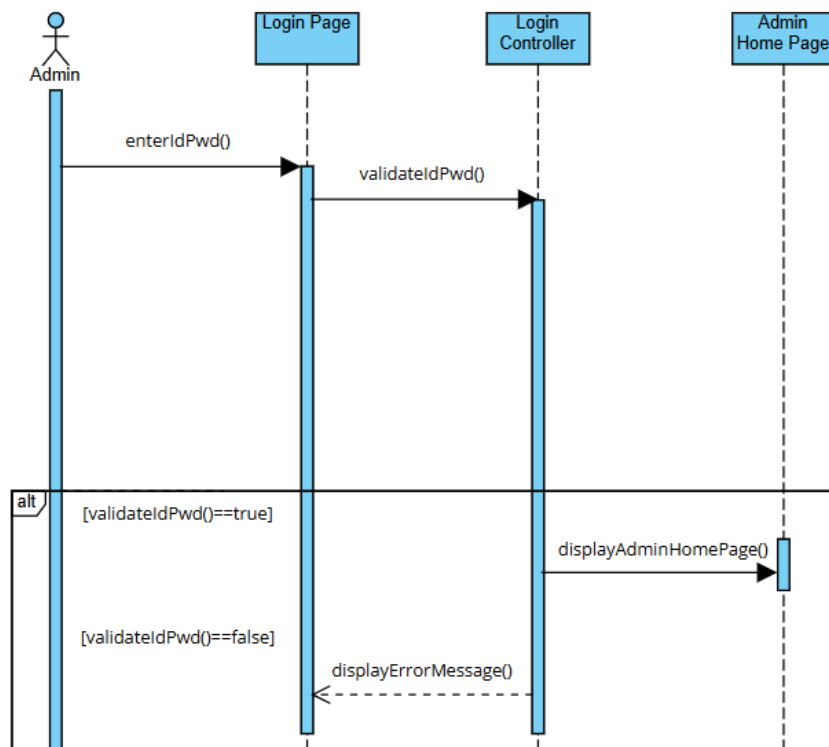


Figure 2.6: Sequence Diagram for Admin Login to System

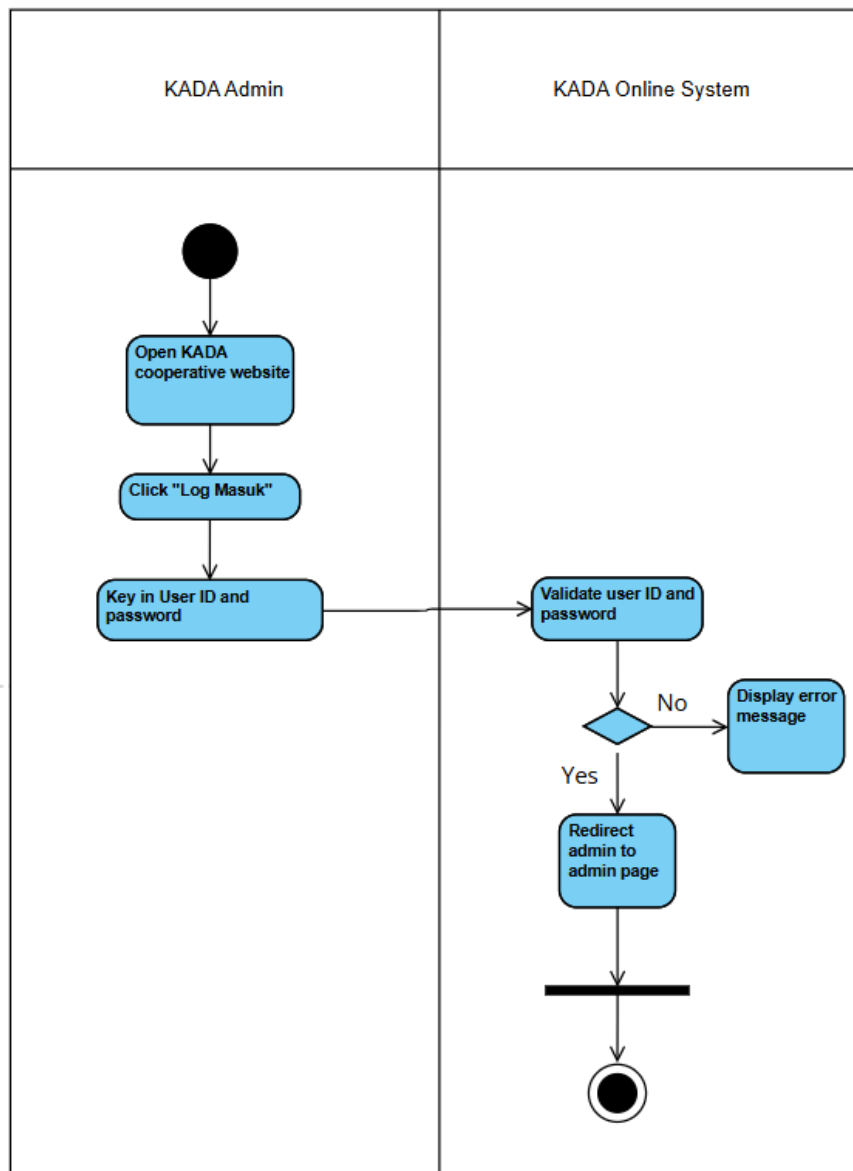


Figure 2.7: Activity Diagram for Admin Login to System

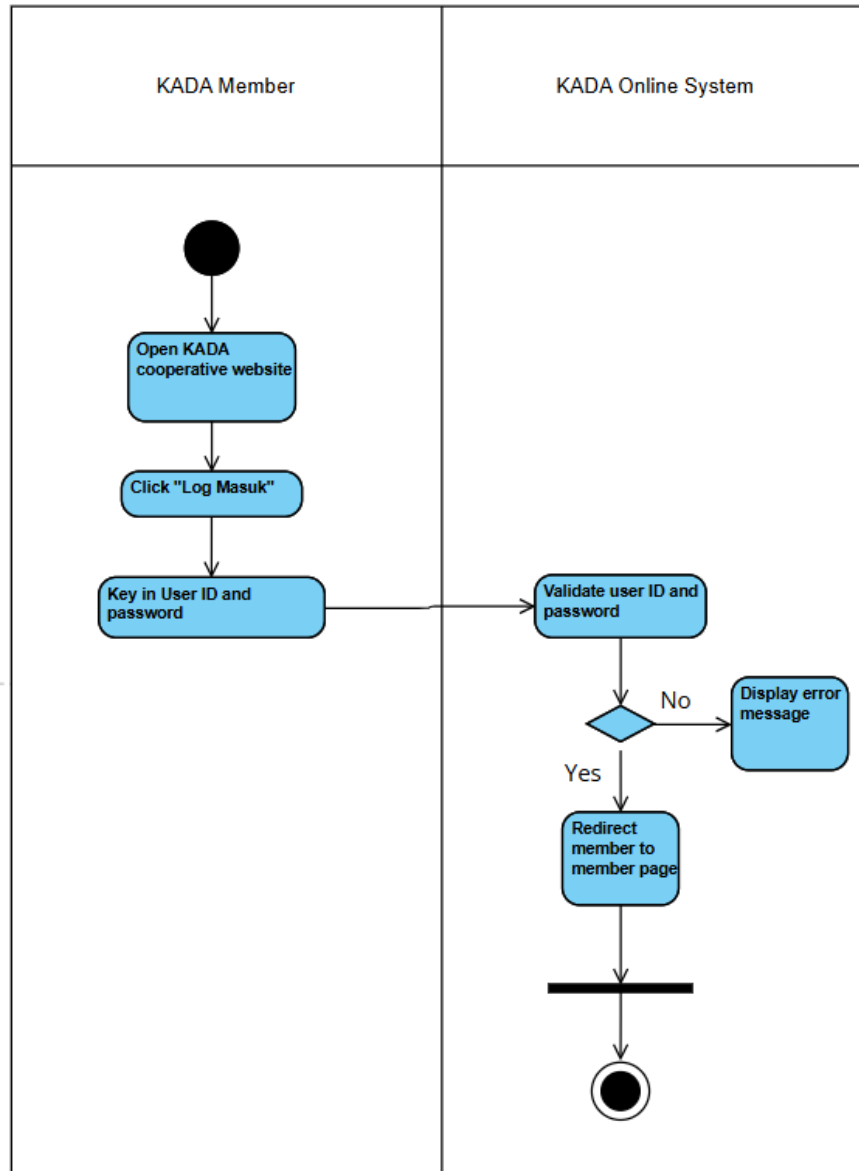


Figure 2.8: Activity Diagram for Member Login to System

2.3.4 US004: User Story Applying for a Loan

Table 2.4: User Story Description for Applying for a Loan

User story: Applying for a Loan
ID: US004
User Story Description As a cooperative member I want to apply for a loan through the system So that I get to know my current financial status
Flow of events: 1. The member logs into the system. 2. Click "Loan" on the navigation bar. 3. Click the "Apply" button under the Loan Apply section on the navigation bar. 4. The system will show the available Loan Type. 5. The members can navigate the loan repayment schedule to see the total financing and monthly installment. 6. The member can choose the desired loan type to apply for. 7. The member fills in all the personal details and enters the loan amount in the form. 8. The member presses the submit button to submit the application form. 9. The system will prompt a message to confirm the submission. 10. After submission, the system displays three options: 10.1 Redirects the member to the homepage. 10.2View Status: Allows the member to check the status of their loan application. 10.3 Print Proof: Enables members to print confirmation or proof of their loan application.
Alternative flow 1: If the member navigates away during the process, the system saves the entered data as a draft for later completion.
Acceptance Criteria: Precondition 1. The member must have an active account in the cooperative system. 2. The member must log in to access the "Loan" section. Postcondition 1. The loan application is successfully recorded in the system. 2. The member can view their application status in the "View Status" section.
Exception flow:

1. If an error occurs while submitting the application form, the system prompts the member with an error message: "An error occurred while submitting your loan application. Please try again later."

2. If mandatory fields are not filled, the system warns: "Please complete all required fields before submitting."

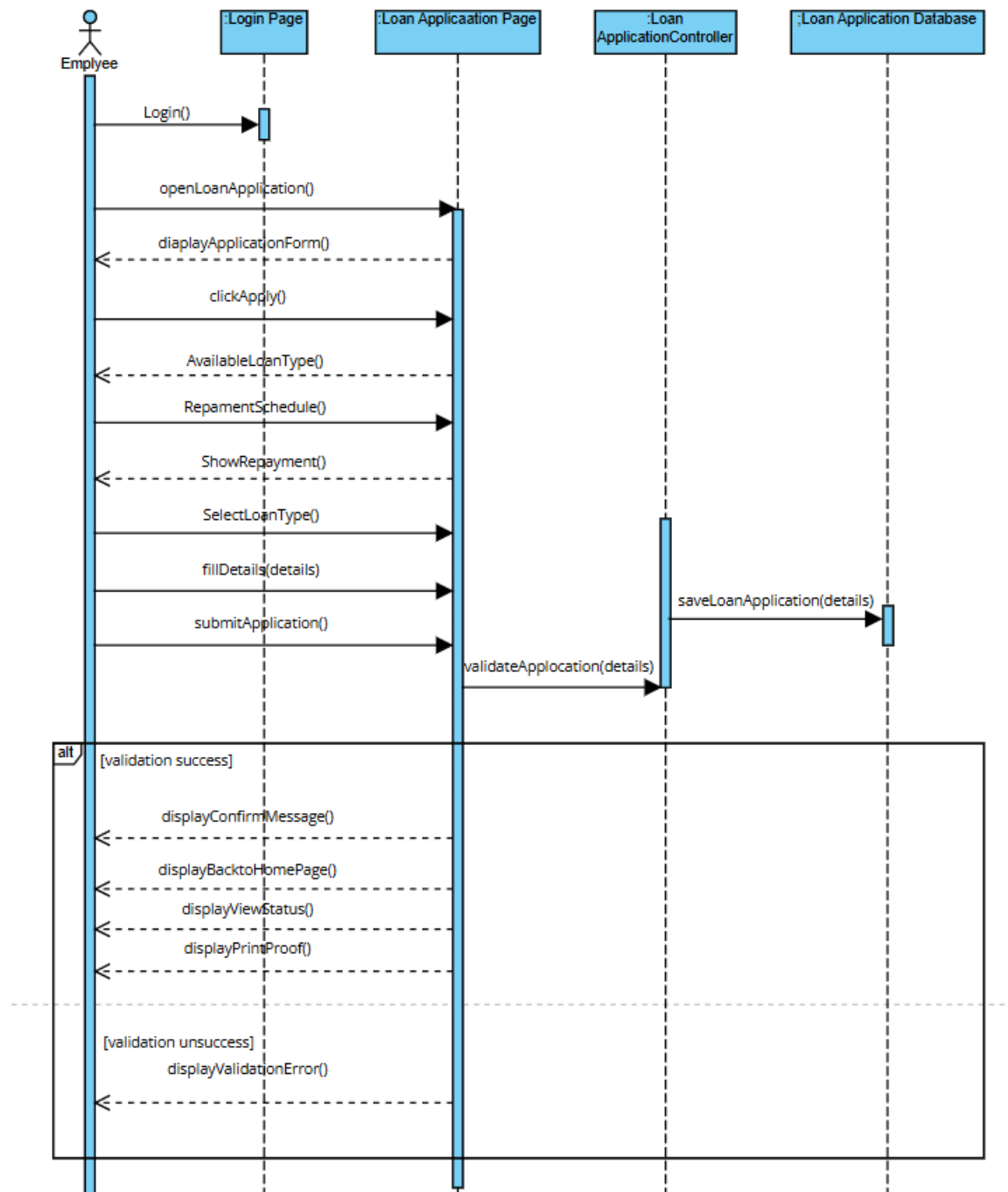


Figure 2.9: Sequence Diagram for Apply Loan

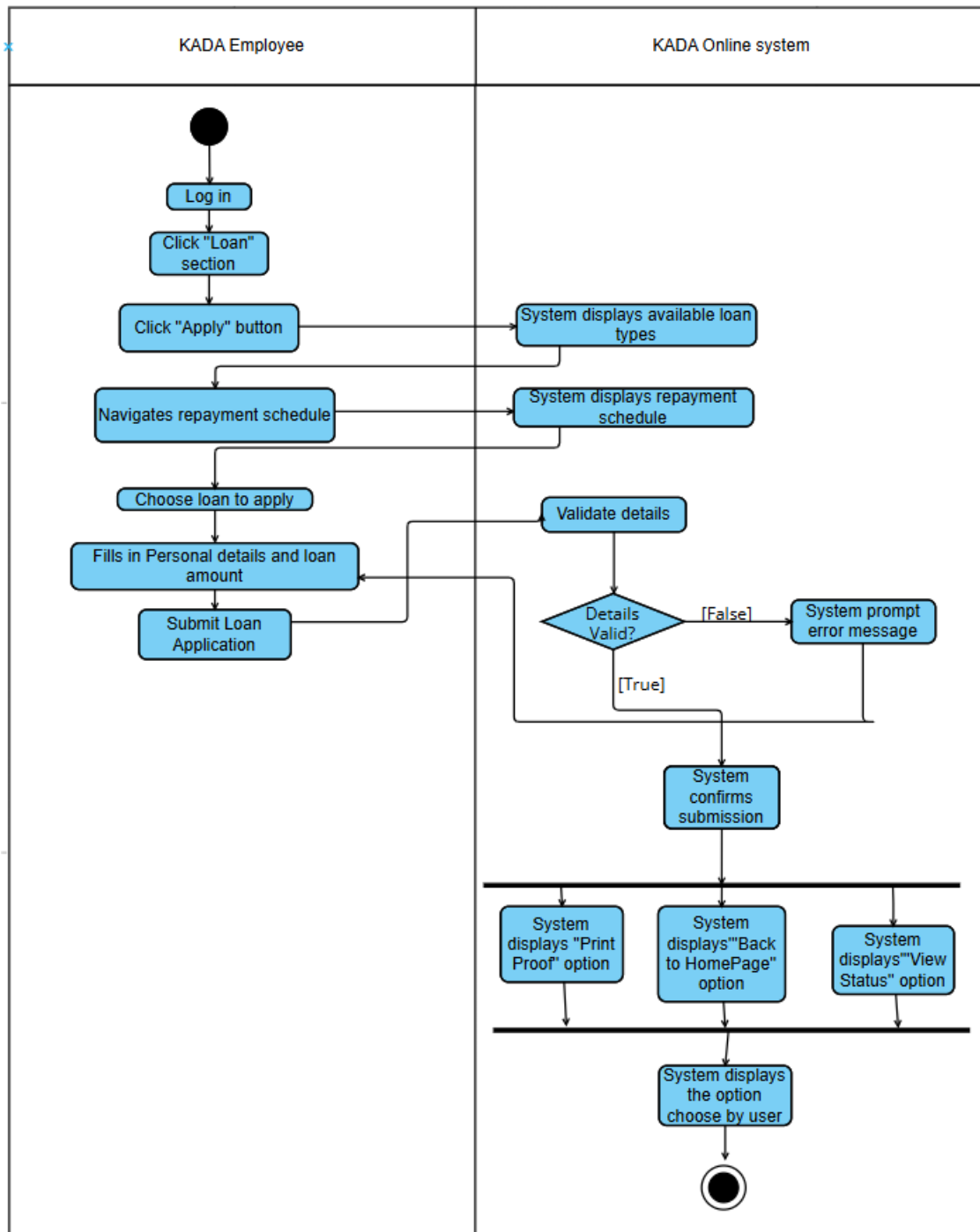


Figure 2.10: Activity Diagram for Apply Loan

2.3.5 US005: User Story Manage Saving Account

Table 2.5: User Story Description for Manage Saving Account

User story: Manage Saving Account
ID: US005
User Story Description As a cooperative member I want to manage my savings account through the system So that I can deposit money, pay for loans, apply for programs, and view my account information conveniently.
Flow of events: 1. The member logs into the system. 2. The system authenticates the member's credentials. 3. The members navigates to "Saving Account" section. 4. The members choose an option to: - view account details (current balance, recent transaction history and loan balances). - deposit money or making payments. - view transactions history. - apply for cooperative programs. 5. System process the member's selected actions and provides appropriate feedback. (success or error)
Alternative flow 1: Scenario: Failed Login 1. The member enters incorrect login credentials. 2. The system displays an error message, "Invalid username or password." 3. The member can retry logging in or reset their password.
Acceptance Criteria: 1. The members must be able to log in securely. 2. The system must allow the member to view account details, including current balance, recent transactions, and loan balance. 3. Deposits or payments must be processed securely, and feedback must be provided for both successful and failed transactions. 4. The transaction history must be accessible for each member. 5. The system must enable members to apply for cooperative programs easily. Precondition 1. The members must be registered with the cooperative and have an active savings account. 2. The system must be operational and accessible. Postcondition

1. The member's selected action (e.g., deposit, payment, program application) is processed and recorded.
2. Updated account information is available for future interactions.

Exception flow:

1. If the member attempts a payment exceeding their available balance, the system displays an error message: "Insufficient funds."

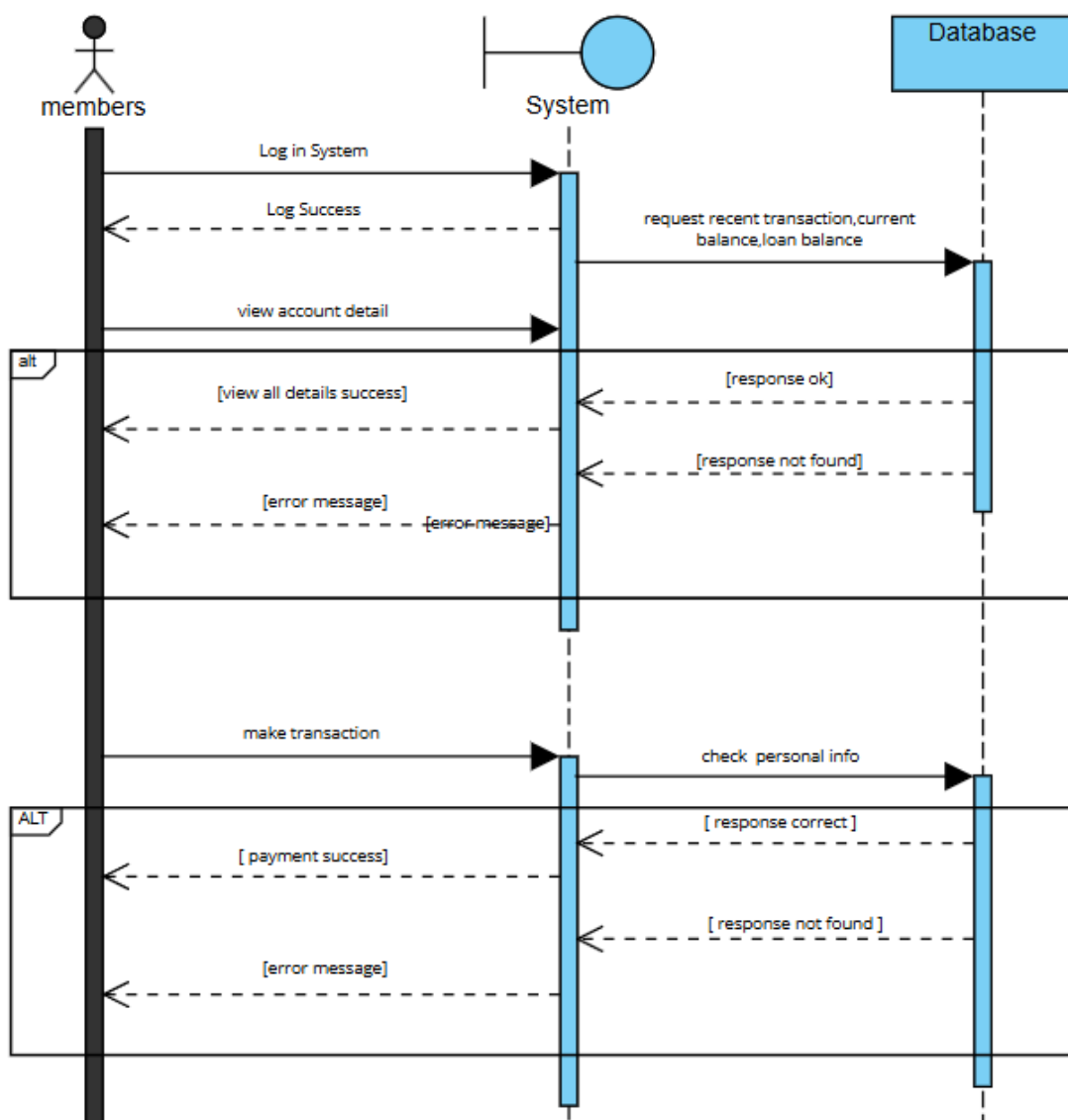


Figure 2.11: Sequence Diagram for Manage Saving Account

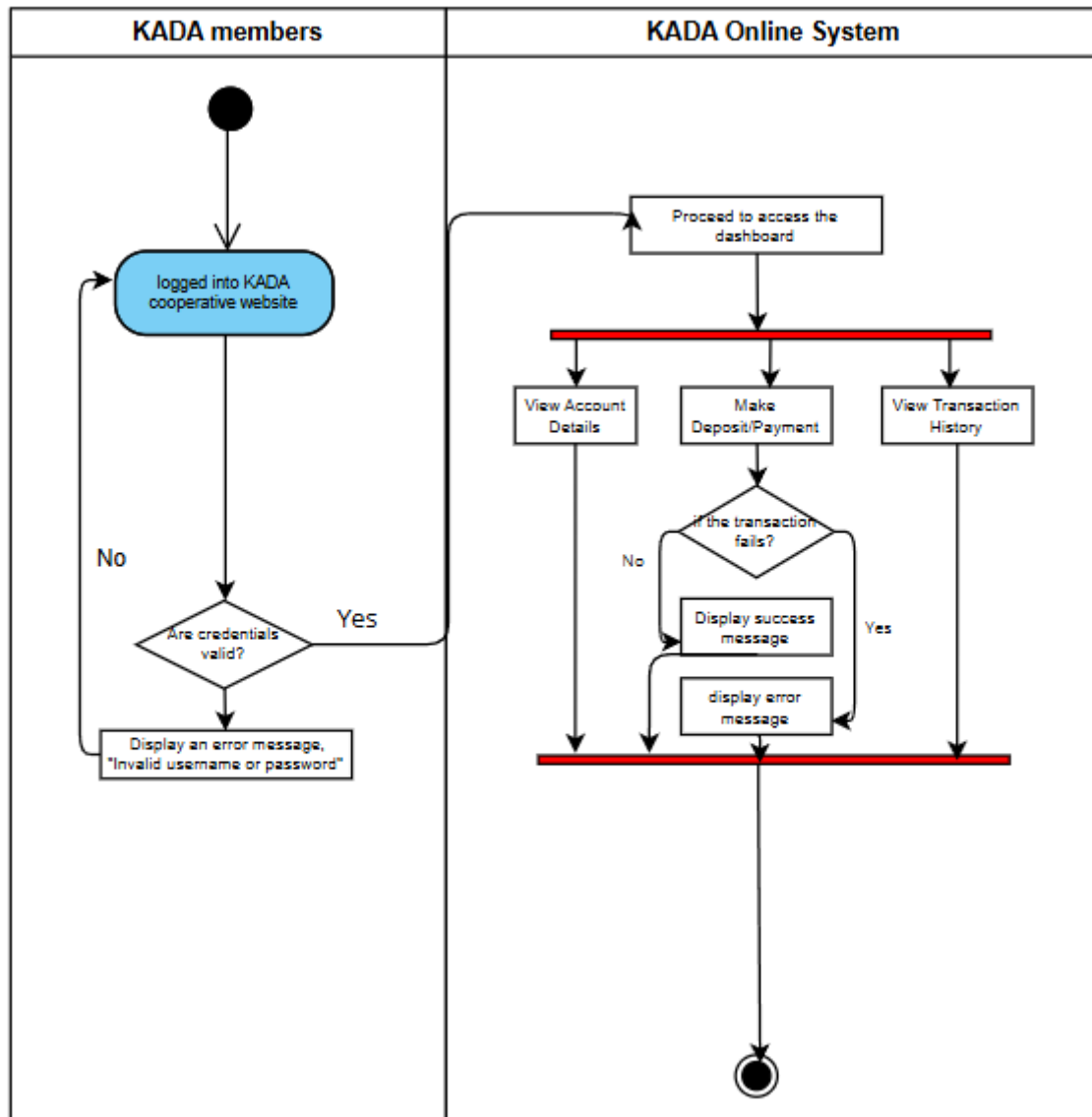


Figure 2.12: Activity Diagram for Manage Saving Account

2.3.6 US006: User Story Create Finance Report

Table 2.6: User Story Description for Create Finance Report

User story: Processing the report
ID: US007
User Story Description As a KADA members I want to generate the reports So that future decisions can be made for the members.
Flow of events: 1. The member navigates to the “Financial”. 2. The member selects the report type, either monthly or annually. 3. The system retrieves relevant data from the database, including new membership applications, Loan applications, fee claimed, and contributions claimed. 4. The system generates a financial summary and detailed breakdown of each category for the specified period. 5. The system displays the report, providing options to: view the report in detail, then download the financial summary report.
Alternative flow <i>n</i>: 1. If a report type is invalid, the system prompts the member to select the correct type again.
Acceptance Criteria Precondition: 1. The member is logged into the system. Postcondition: Financial reports are generated.
Exception flow: -

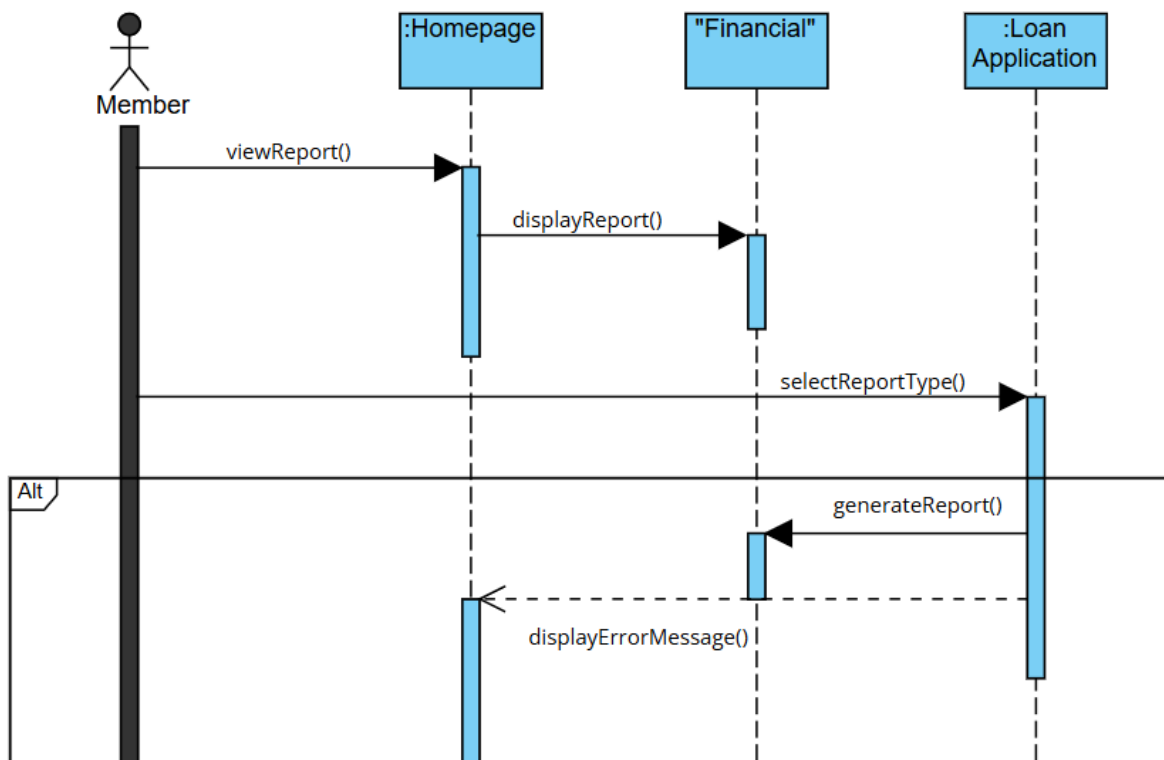


Figure 2.13: Sequence Diagram for Create Financial Report

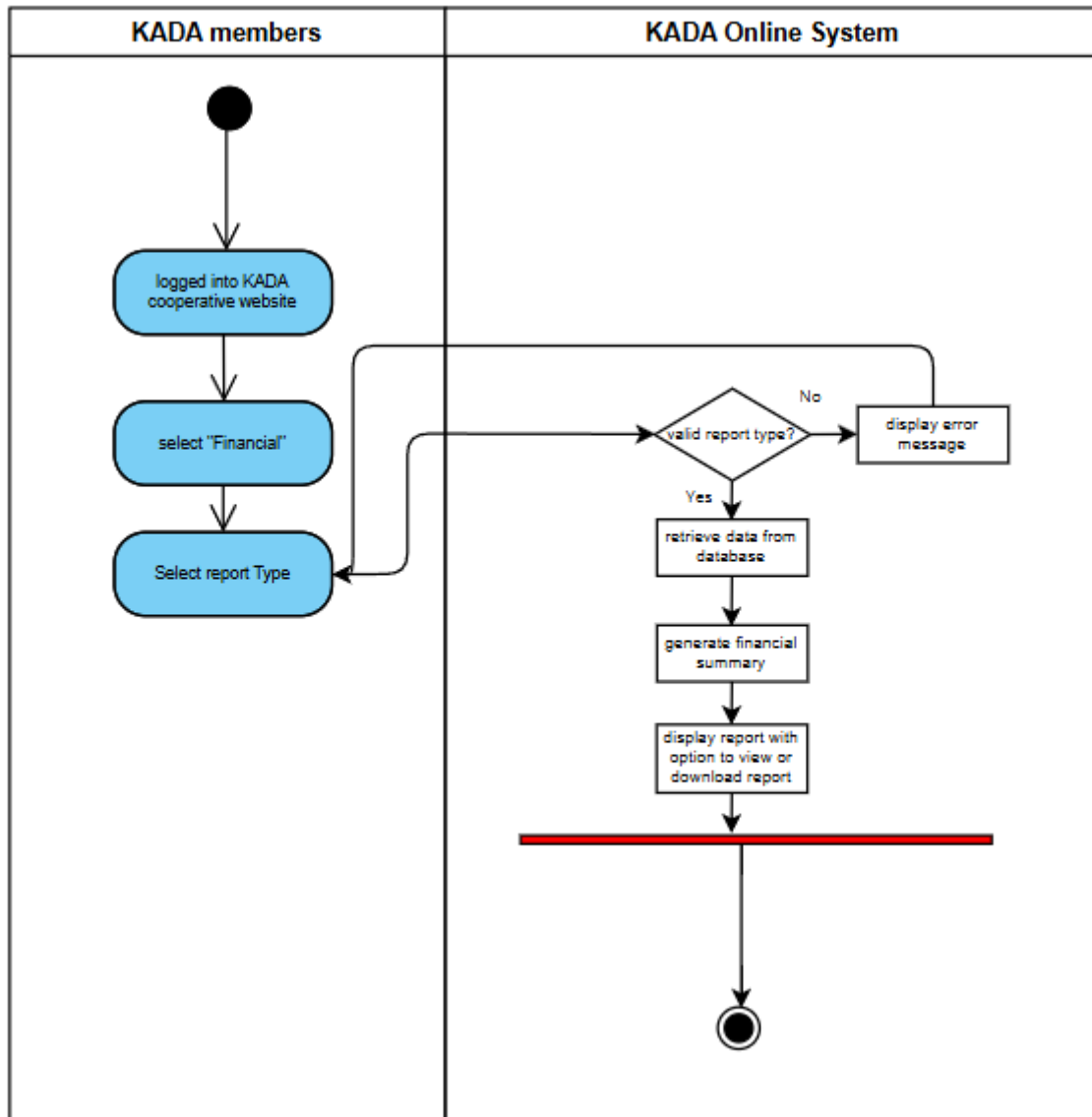


Figure 2.14: Activity Diagram for Create Financial Report

2.3.7 US007: User Story View Financial Status

Table 2.7: User Story Description for View Financial Status

User story: View Financial Status
ID: US007
User Story Description As a KADA member I want to view and verify financial information. So that I can know about my current financial status.
Flow of events: 1. The member logs into the member website. 2. The member navigates “Financial”. 3. The member successfully view the financial information displayed.
Alternative flow:
Acceptance Criteria Precondition: 1. The member is logged into the system. Postcondition: The members successfully view the current financial information.
Exception flow: -

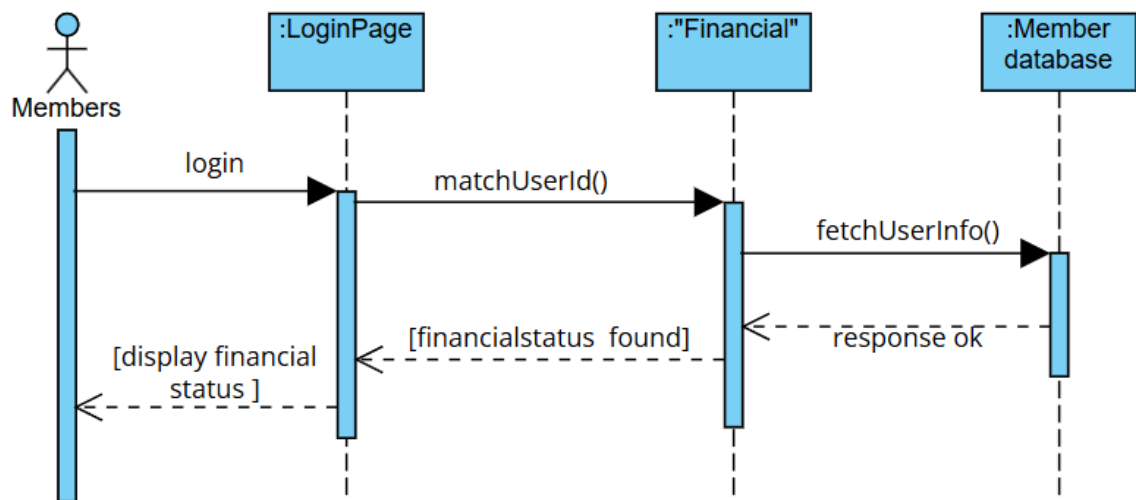


Figure 2.15: Sequence Diagram for View Financial Status

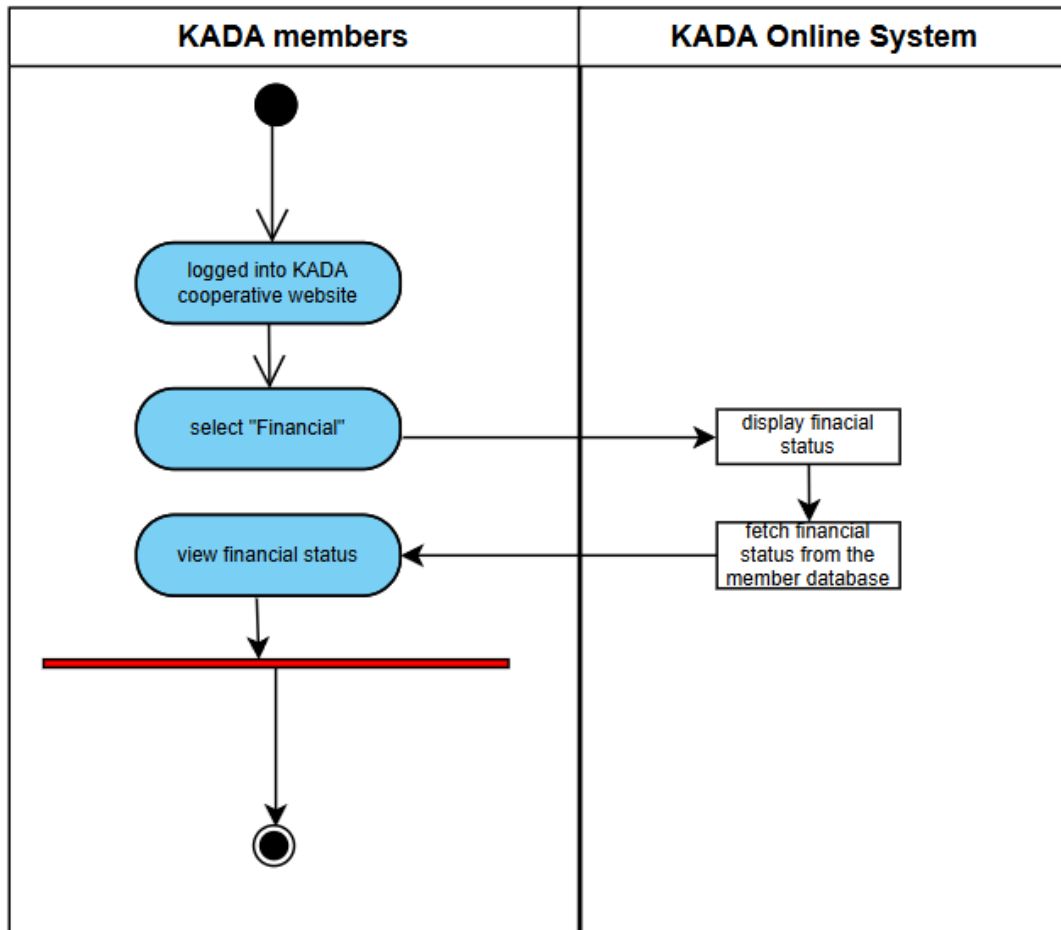


Figure 2.16: Activity Diagram for View Financial Status

2.3.8 US008: User Story View Application List

Table 2.8: User Story Description for View Application List

User story: Membership Management
ID: US008
User Story Description As a KADA admin I want to view applicant lists for new members, loans and program applications. So that I proceed with the approval process.
Flow of events: 1. The admin navigates “View Member List”. 2. The admin can select - “Manage Loan Application” to view loan application - “Manage Membership Application” to view membership application 3. The system retrieves information from databases.
Alternative flow: -
Acceptance Criteria Precondition: 1. The admin is logged into the system.
Exception flow: -

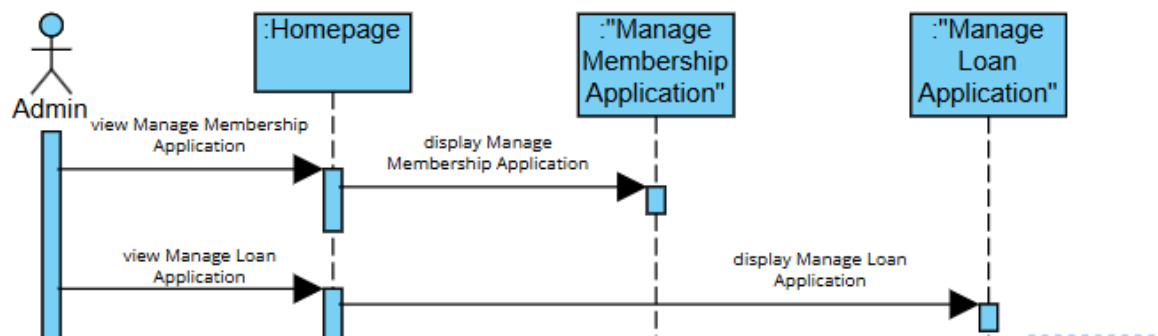


Figure 2.17: Sequence Diagram for View Application List

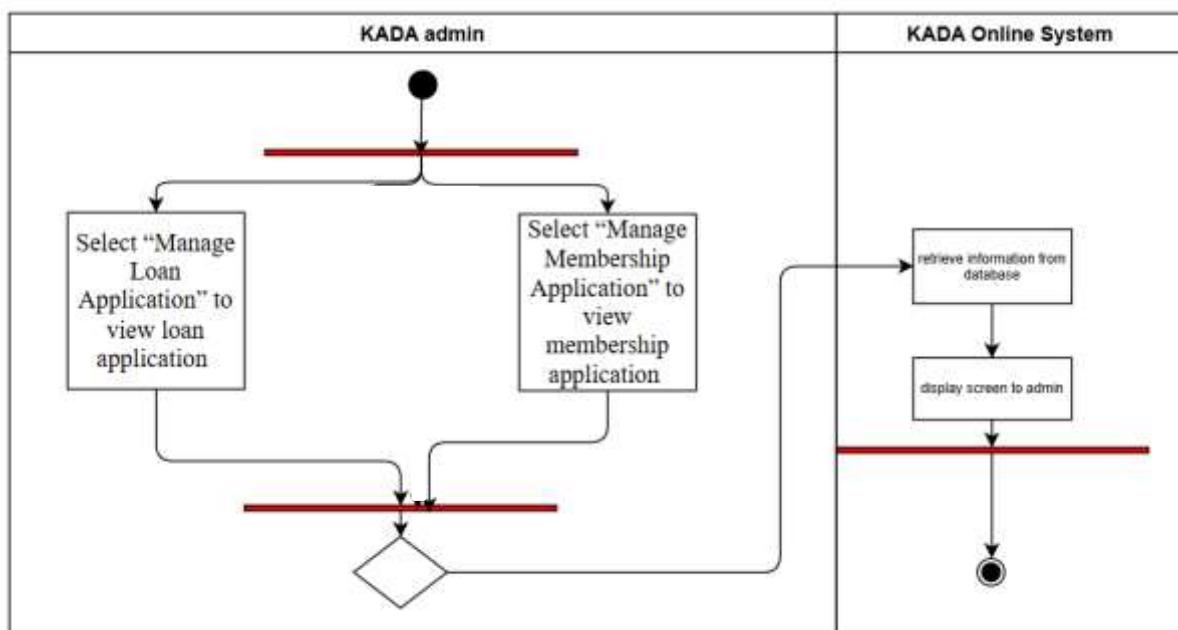


Figure 2.18: Activity Diagram for View Application List

2.3.9 US009: User Story Approve Membership Application

Table 2.9: User Story Description for Approve Membership Application

User story: Approve Membership Application
ID: US009
User Story Description As a KADA admin I want to approve a new member So that i need to check whether members meet the requirement to become a member.
Flow of events: 1. The admin navigates “Manage Membership Application”. 2. The admin selects “Batch Processing”. 3. The admin reviews the member information.
Alternative flow: 1. If the member meets the requirement, the status will change to “approve”. 2. If not, the status will change to “rejected”.
Acceptance Criteria Precondition: 1. The admin is logged into the system. Postcondition: After the status in the “Manage Membership Application” change, the status in “View Member List” will also be change.
Exception flow: -

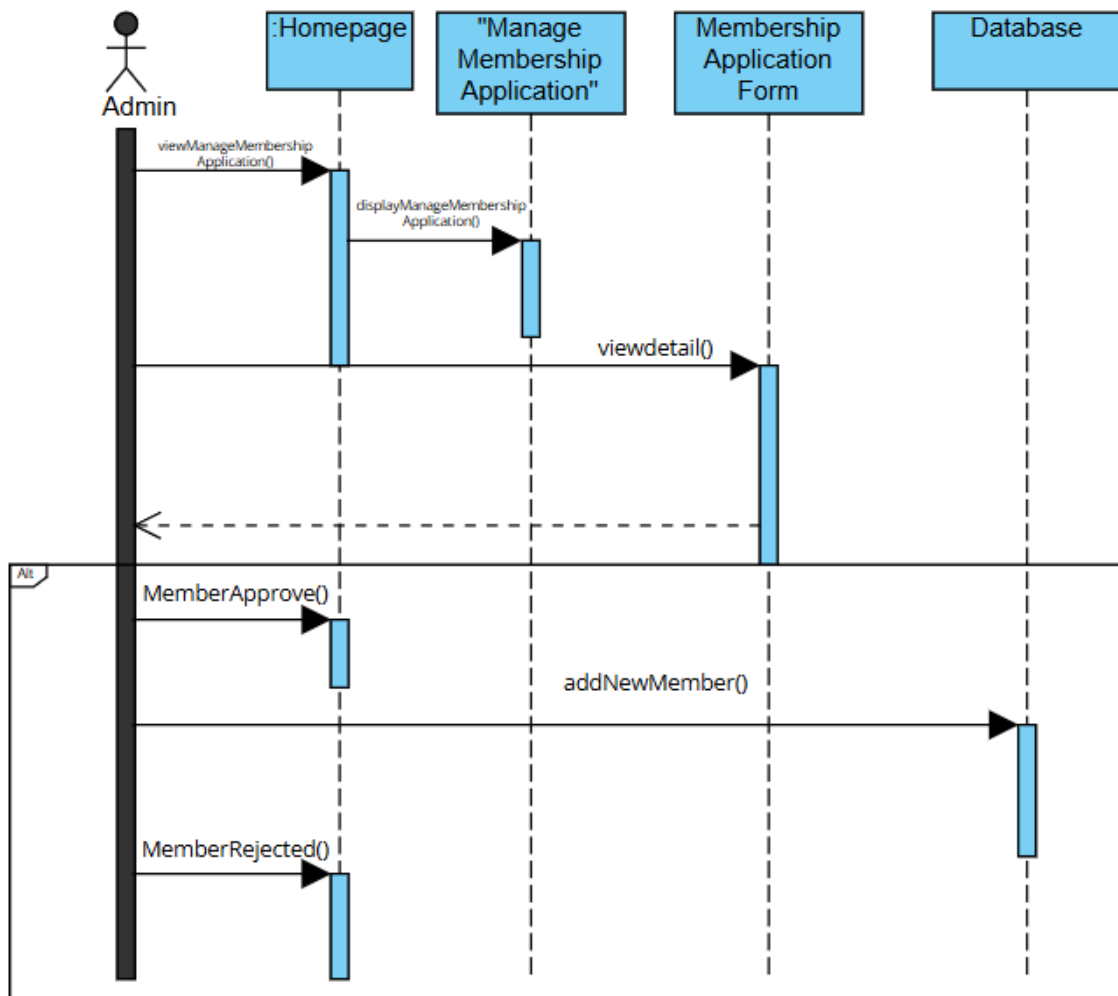


Figure 2.19: Sequence Diagram for Approve Membership Application

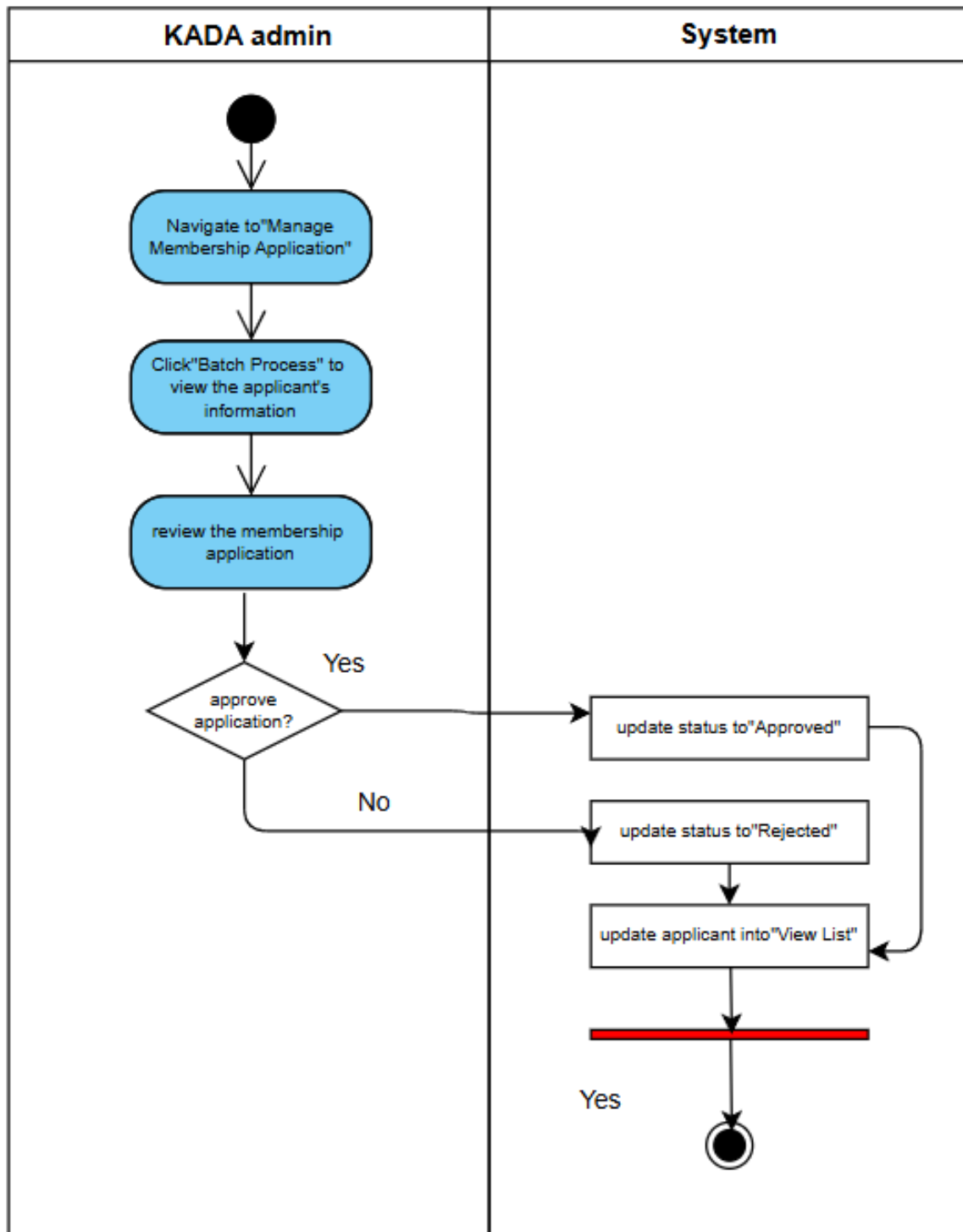


Figure 2.20: Activity Diagram for Approve Membership Application

2.3.10 US010: User Story Approve Loan Application

Table 2.10: User Story Description for Approve Loan Application

User story: Approve Loan Application
ID: US010
User Story Description As a KADA admin I want to approve a new loan So that I need to check whether members meet the requirement to apply for the loan.
Flow of events: 1. The admin navigates “Manage Loan Application”. 2. The admin selects “Batch Processing”. 3. The admin reviews the member information.
Alternative flow: 1. If the member meets the requirement, the status will change to “approve”. 2. If not, the status will change to “rejected”.
Acceptance Criteria Precondition: 1. The admin is logged into the system. Postcondition: After the status in the “Manage Loan Application” change, the status in “View Member List” will also be change.
Exception flow: -

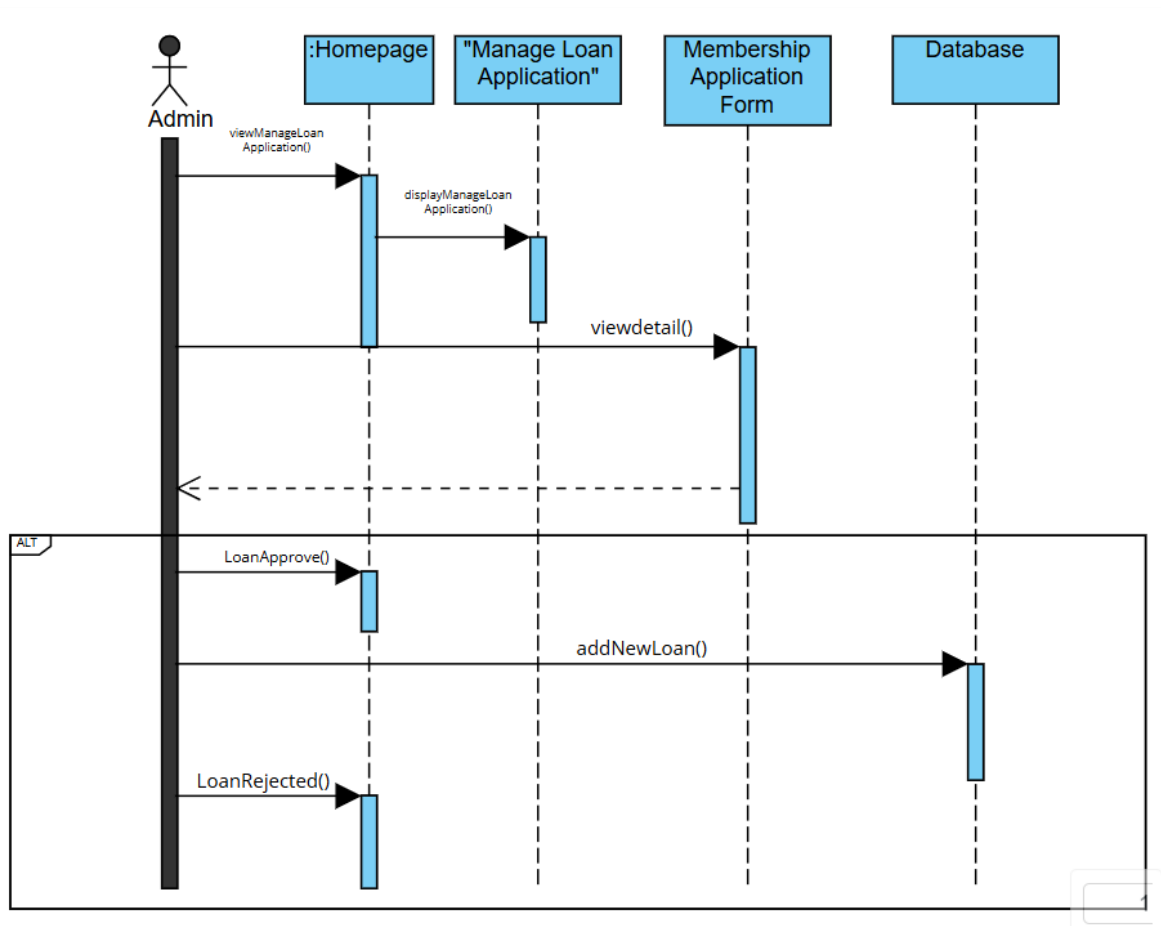


Figure 2.21: Sequence Diagram for Approve Loan Application

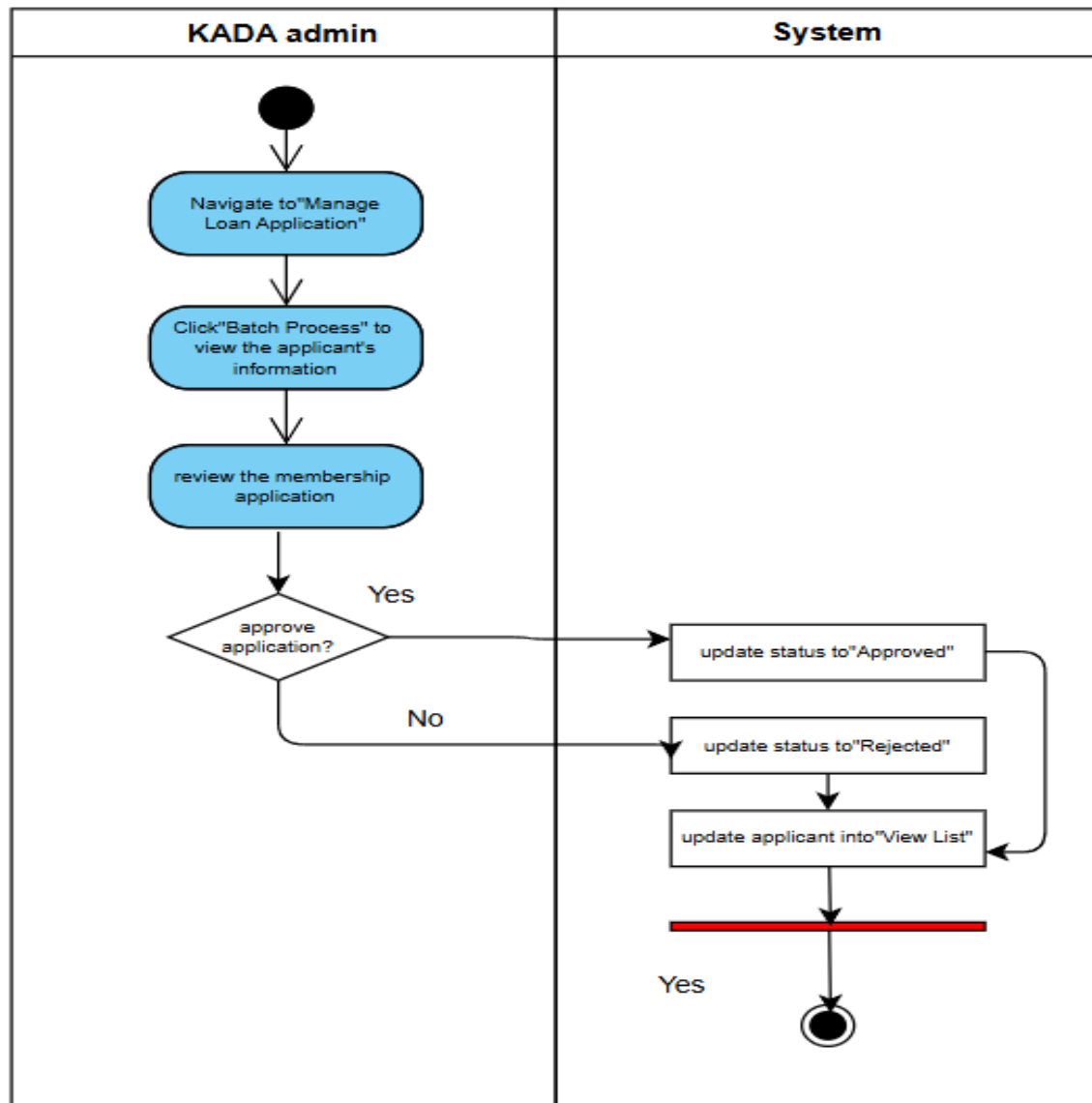


Figure 2.22: Activity Diagram for Approve Loan Application

2.4.11 US011 User Story Admin View Report

Table 2.11: User Story Description for Admin View Report

User story: Admin View Report
ID: US011
User Story Description: As an admin I want to view report So that I can view the loan applications and membership applications statistics
Flow of events: 1. The admin logs into the system. 2. Navigate to the "Jana Laporan" section. 3. The system retrieves relevant data from the database, including new membership applications, Loan applications 4. The system generates a application summary and detailed breakdown of each category for the specified period. 5. The system displays the report, providing options to: view the report in detail, then download the summary report.
Alternative flow: -
Acceptance Criteria: Precondition: 1. The member is logged into the system. Postcondition: Applications reports are generated.
Exception flow: -

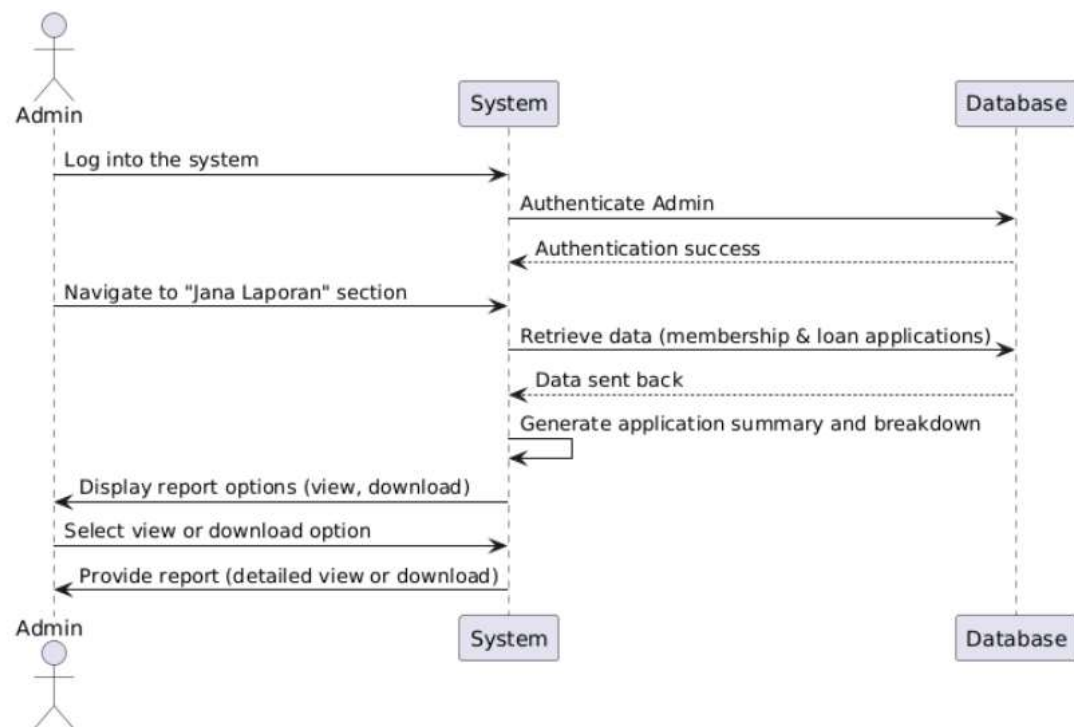


Figure 2.23: Sequence Diagram for Admin view report

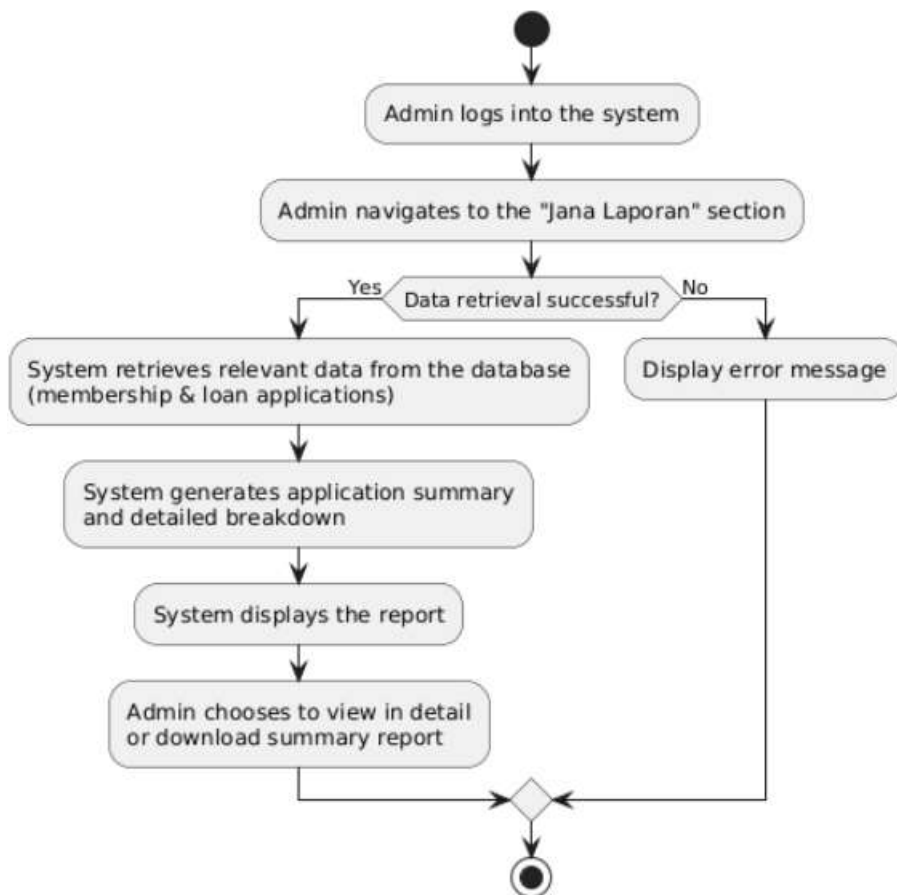


Figure 2.24: Activity Diagram for Admin view report

2.5 Performance and Other Requirements

2.5.1 Software System Attribute:

- Usability: The system should be user-friendly and adjustable font sizes.
- Reliability: The system has a meantime between failures (MTBF) of at least 500 hours and can recover from server outages in 15 minutes.
- Maintainability: Regular upgrades to system modules are possible without affecting the user's ongoing activity.
- Portability: The solution functions flawlessly on mobile platforms like iOS and Android as well as contemporary web browsers like Chrome and Edge.

- **Compatibility:** To streamline and expedite software development, the system will incorporate external APIs for reporting and payment processing with current KADA tools.
- **Security:** When users are completing private actions, such as applying for a loan or opening a savings account, the system will employ two-factor authentication (2FA) for admin and member login and AES-256 encryption to protect data.
- **Legal and Regulatory:** In order to oversee and control cooperative movements, the system must conform to the criteria set forth by the Malaysian Cooperative Commission (MCC).
- **Environmental Impact:** To lessen the system's carbon footprint, it will use green data centers and lessen its dependency on paper-based documentation.

2.5.2 Performance Requirements:

- **Response Time:** To guarantee effective communication, the KADA system must react to form submissions, such as member registration and loan applications, admin and member login requests, within one second, regardless of busy hours.
- **Throughput:** There will be no delay when the system manages at least 300 concurrent user sessions.
- **Capacity:** Up to 30,000 records of registered members' data can be stored in the database, which is sufficient for 10 years.
- **Availability:** During business hours, the system will have a minimum uptime of 99.5%, and during maintenance, it will have a 98% uptime.

2.5.3 Other Requirements

Localisation Requirements

- The system is specifically for the KADA cooperative in Malaysia.
- Supported languages: Malay only
- Date format: DD/MM/YY
- Currency: Ringgit Malaysia (RM) for all financial transactions.

Data Migration Requirements

- The KADA system will replace the existing manual and partially digitalized process, so an admin interface must be developed to ensure existing member profiles, loan records and financial reports are accurately transferred into the new system.

Accessibility Requirements

- Mobile responsiveness is required to provide usability on devices of various screen sizes.

Backup and Documentation Requirements

- Automatic backups must be scheduled daily to secure critical data.

Training Requirements

- Administrators and end-users must receive detailed user manuals and training materials.
- Interactive tutorials should be provided in the system to help new users navigate essential features.

2.6 Design Constraints

Environmental Constraint: The hosting environment must include geographically distributed data centers to reduce latency and improve access speed for rural users.

Hardware Constraint: Servers with a minimum configuration of 500GB SSD and 16GB RAM are required to run the system. Modern browsers like Chrome and Firefox are required for the system to function on client devices.

Security Constraint: RBAC must be enforced for administrators and end users, and all user data must be secured during transmission using the TLS protocol.

Compatibility Constraint: To guarantee that data may be imported and exported without issues, the system must smoothly connect with the current KADA accounting software.

Performance Constraint: Within three minutes, the monthly reporting module should produce summaries for up to 50,000 transactions.

3. Architectural Rationale

3.1 Architecture Style and Rationale

The chosen architecture style for the kada cooperative system is Model-View-Controller (MVC) pattern. MVC separates the system into three main interconnected components, which include Model, Controller and View to allow for modular design and efficient management of system functions. The model handles the core business logic and data storage, the view manage the interface for users, and the controller serves as the intermediary, process user input and upload both of the model and view accordingly.

This architecture style is suitable for kada cooperative system because it provide great scalability. With distinct components, the system can accommodate future changes in functionalities. For example, in our system, both user, admin, loan, and member has their own controller and model. Each component can be updated independently without significantly impact other components. Furthermore, each component can be reused across various module like the member profile management, loan application and reporting module. By applying MVC, our proposed system can efficiently supports its core functionalities, such as membership management, loan applications, saving account management, financial reporting and administration, while ensuring a maintainable structure.

4. Architectural Views

4.1 Use Case View

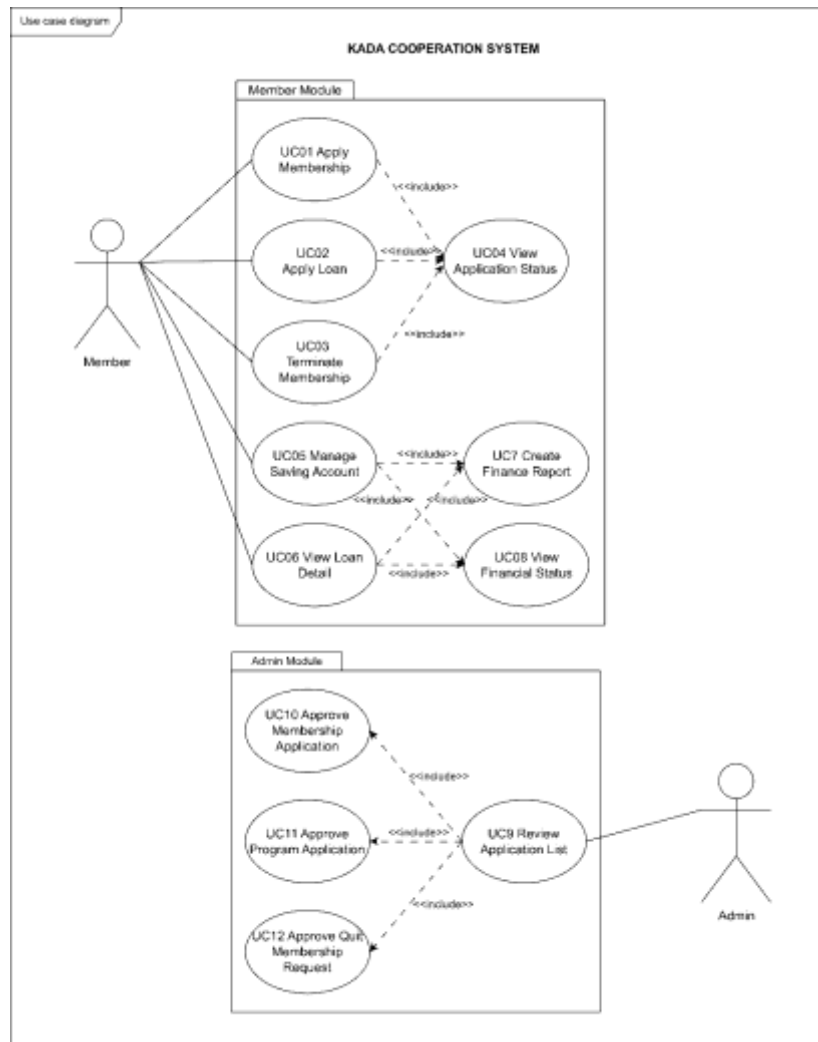


Figure 4.1.1: Use Case Diagram for <KADA Cooperation System>

4.2 Implementation View

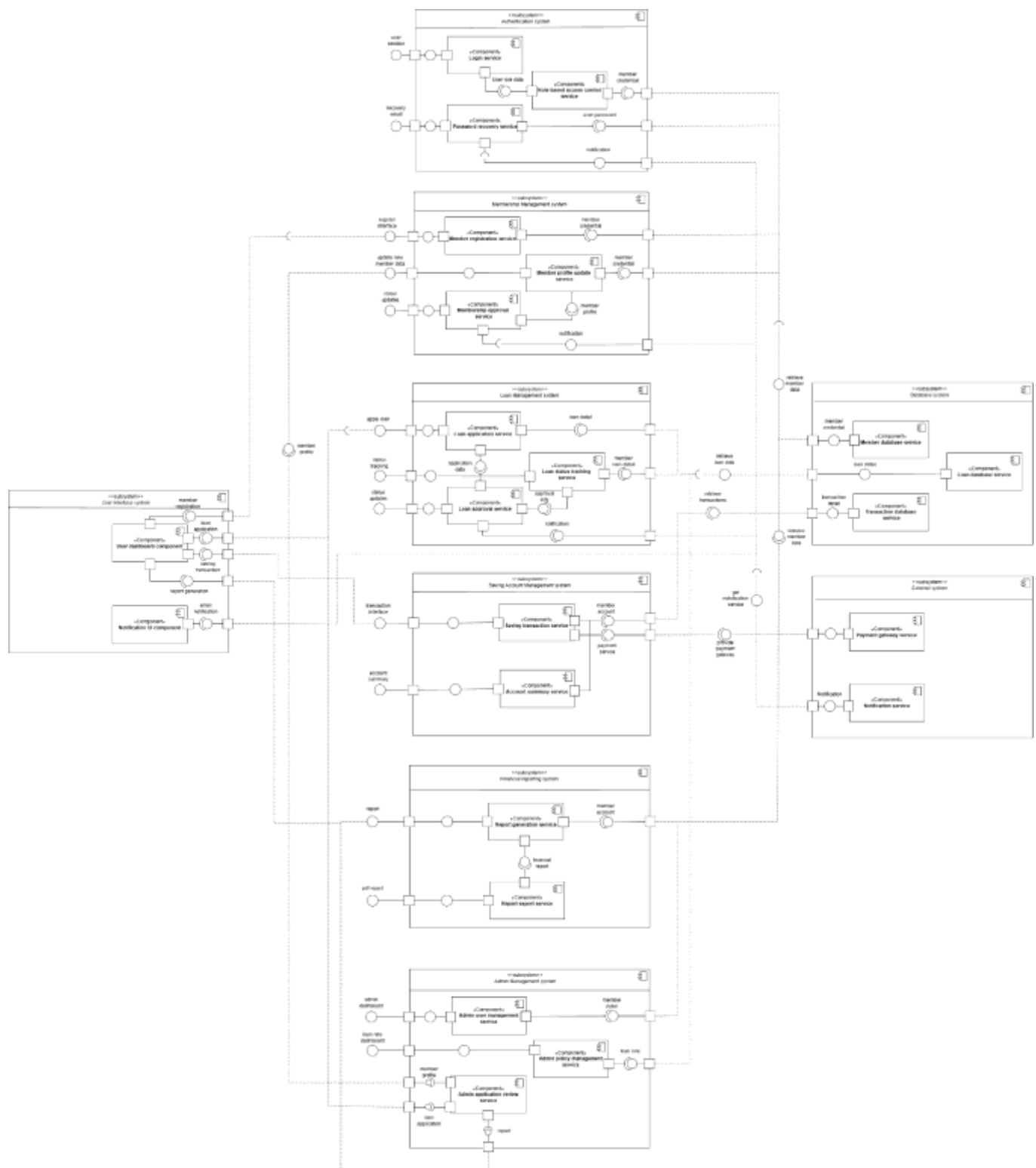


Figure 4.2.1: Component Diagram of <KADA Cooperation System>

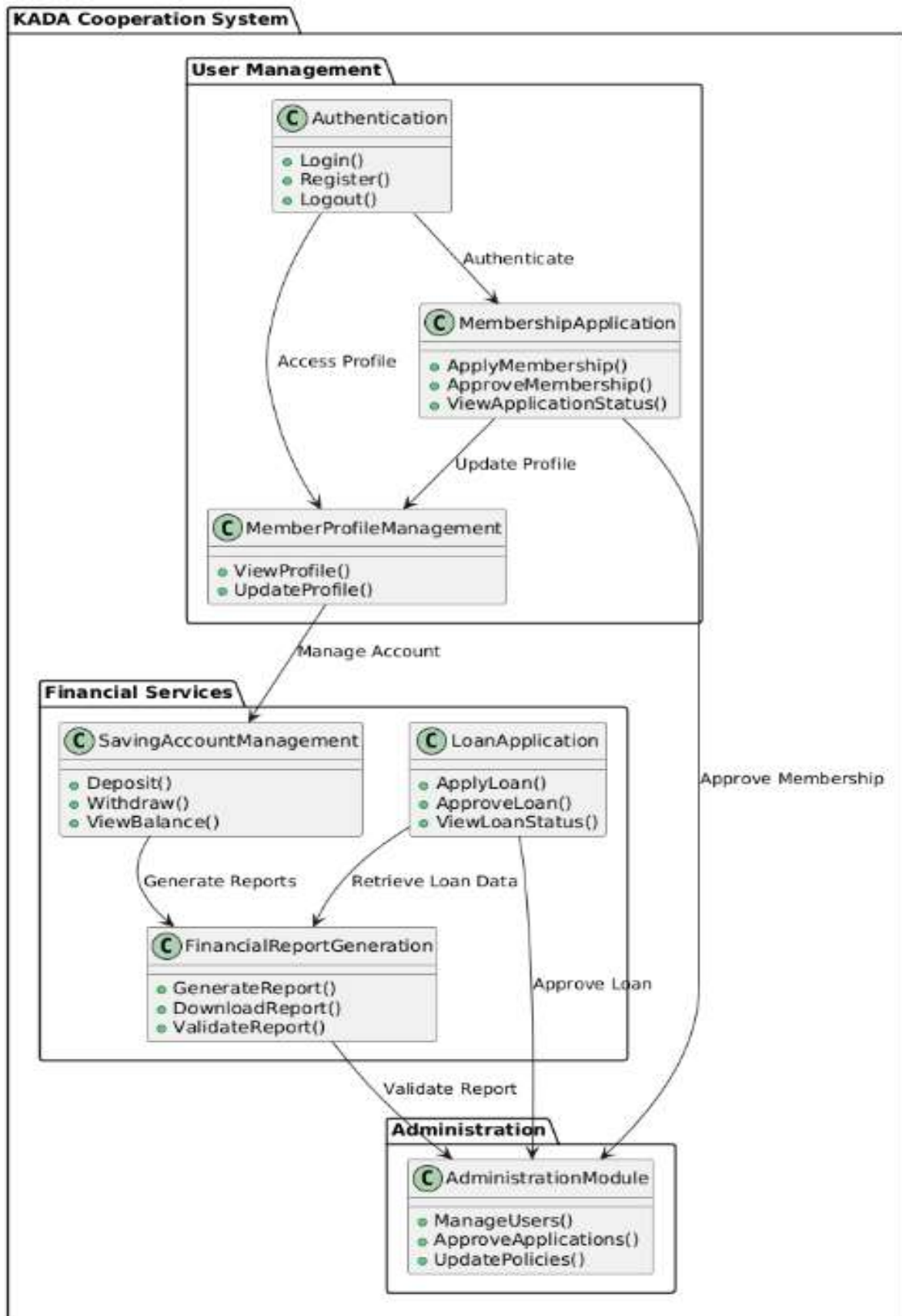


Figure 4.2.2.1: Package Diagram for <KADA Cooperation System>

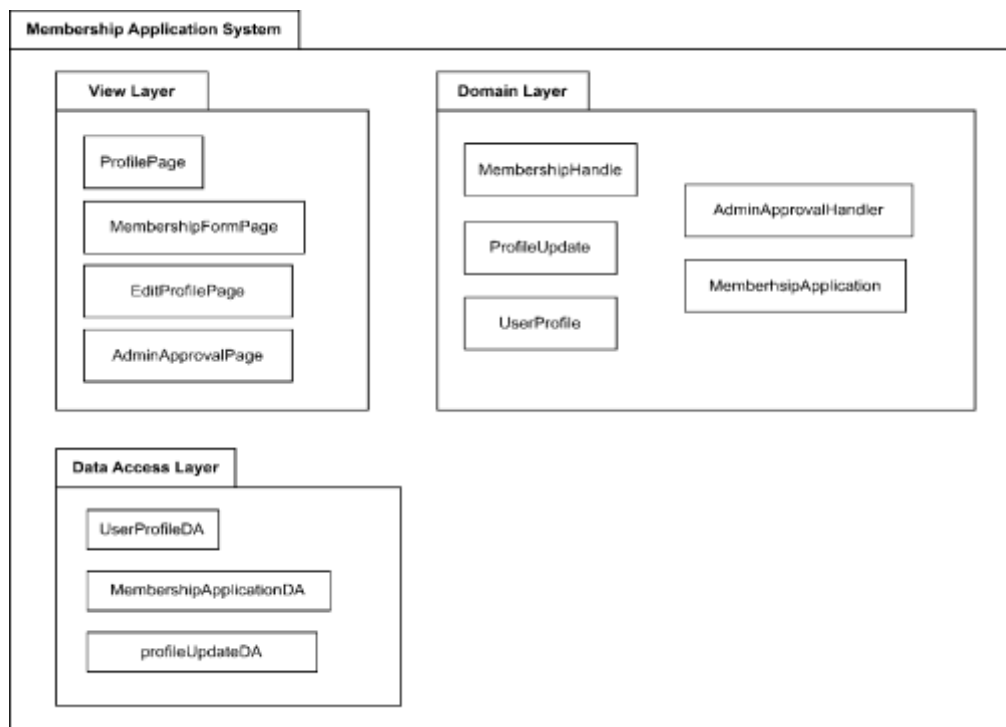


Figure 4.2.2.2: Package Diagram for < Membership Registration> Subsystem

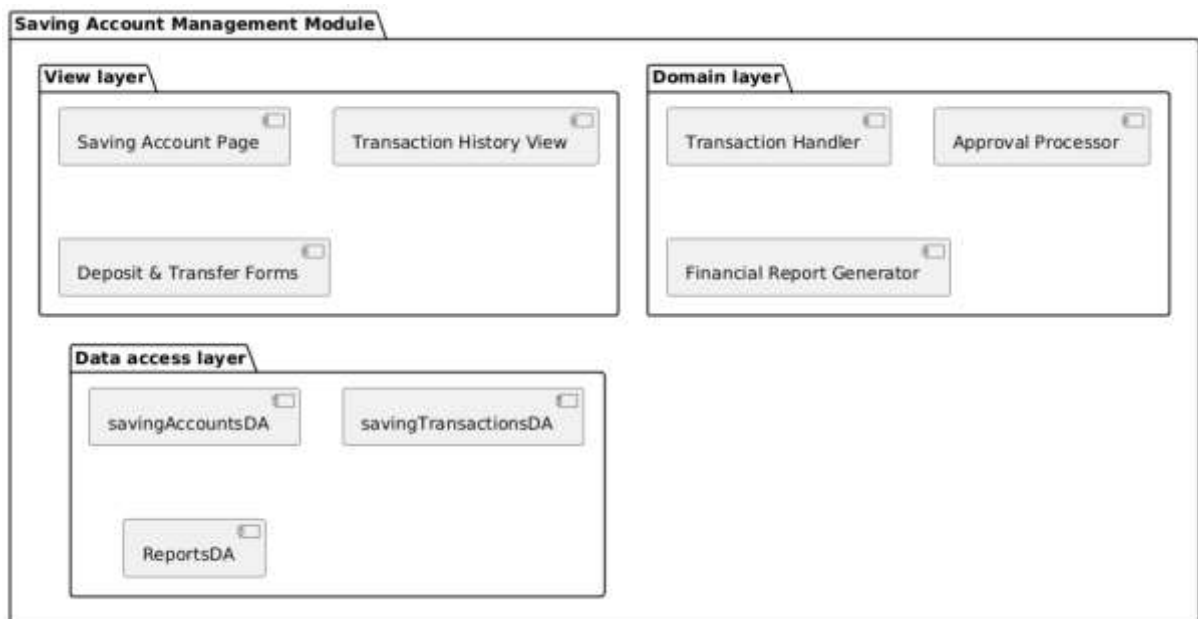


Figure 4.2.2.3 Package Diagram for <Saving Account Management> Subsystem

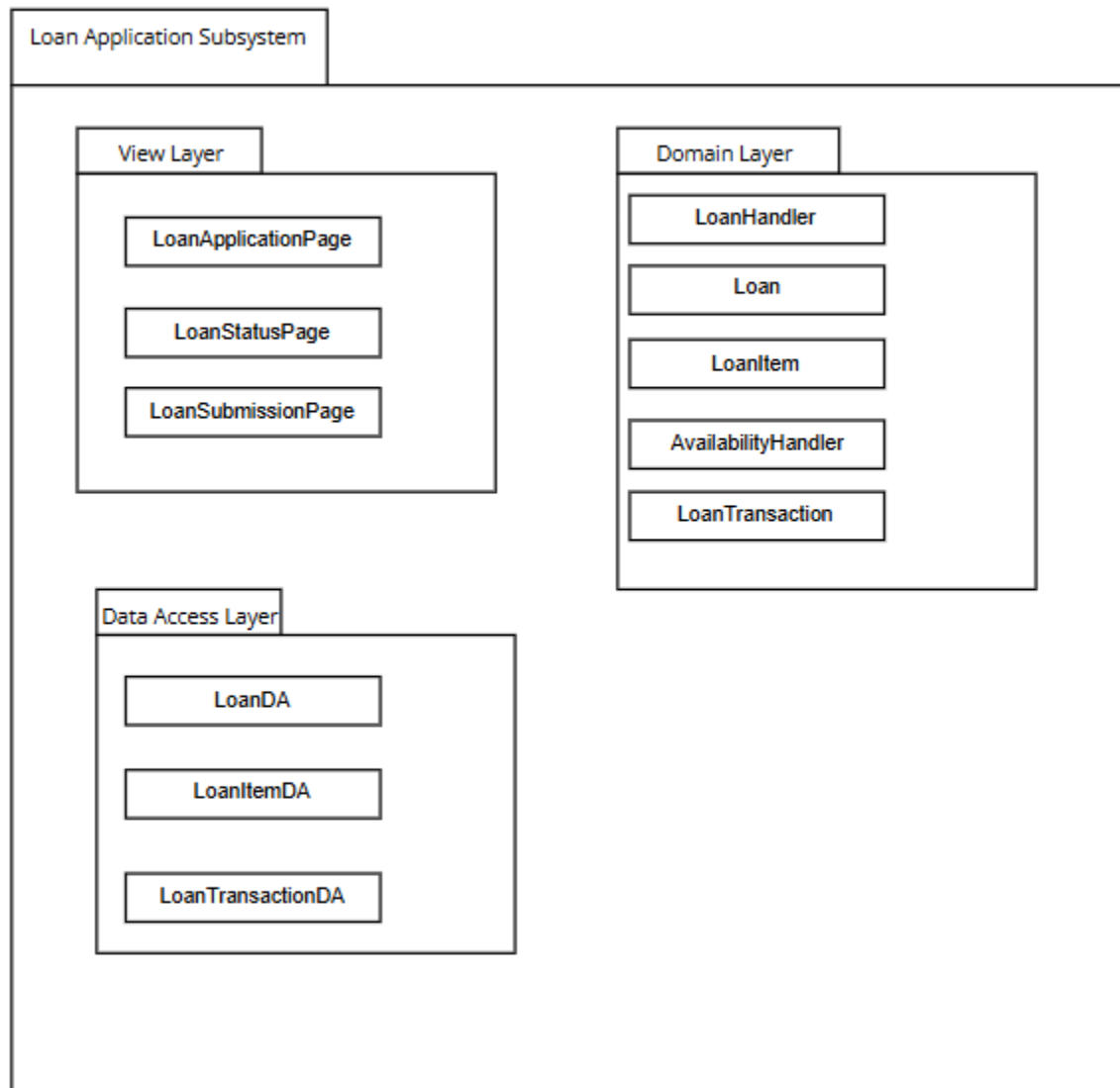


Figure 4.2.2.4: Package Diagram for <Loan Application > Subsystem

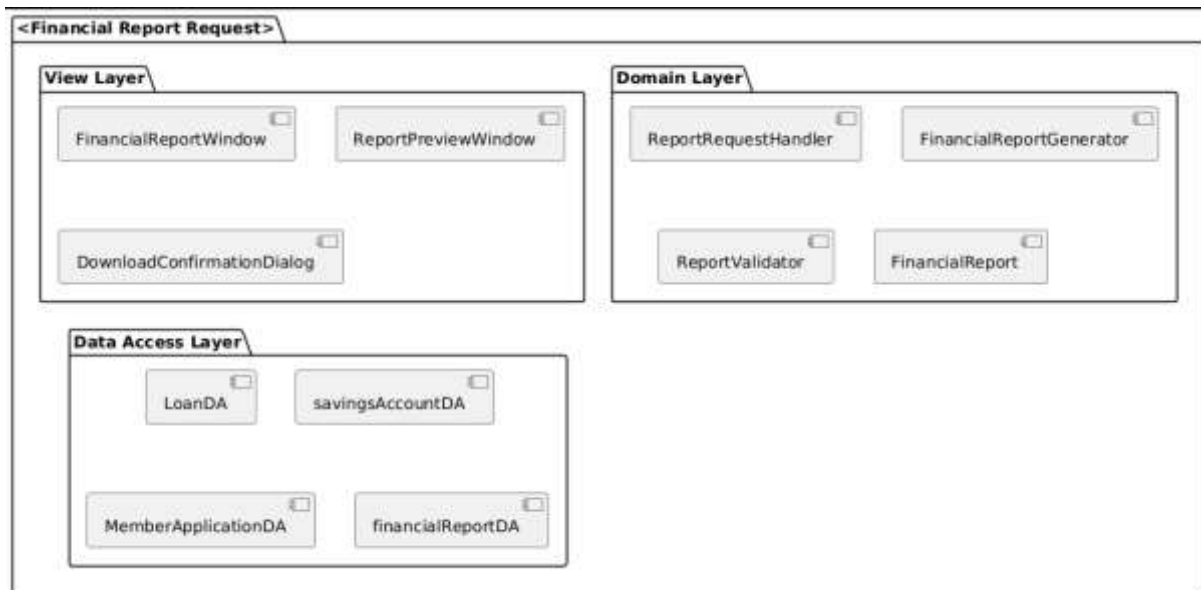


Figure 4.2.2.5: Package Diagram for < Financial Report Request > Subsystem

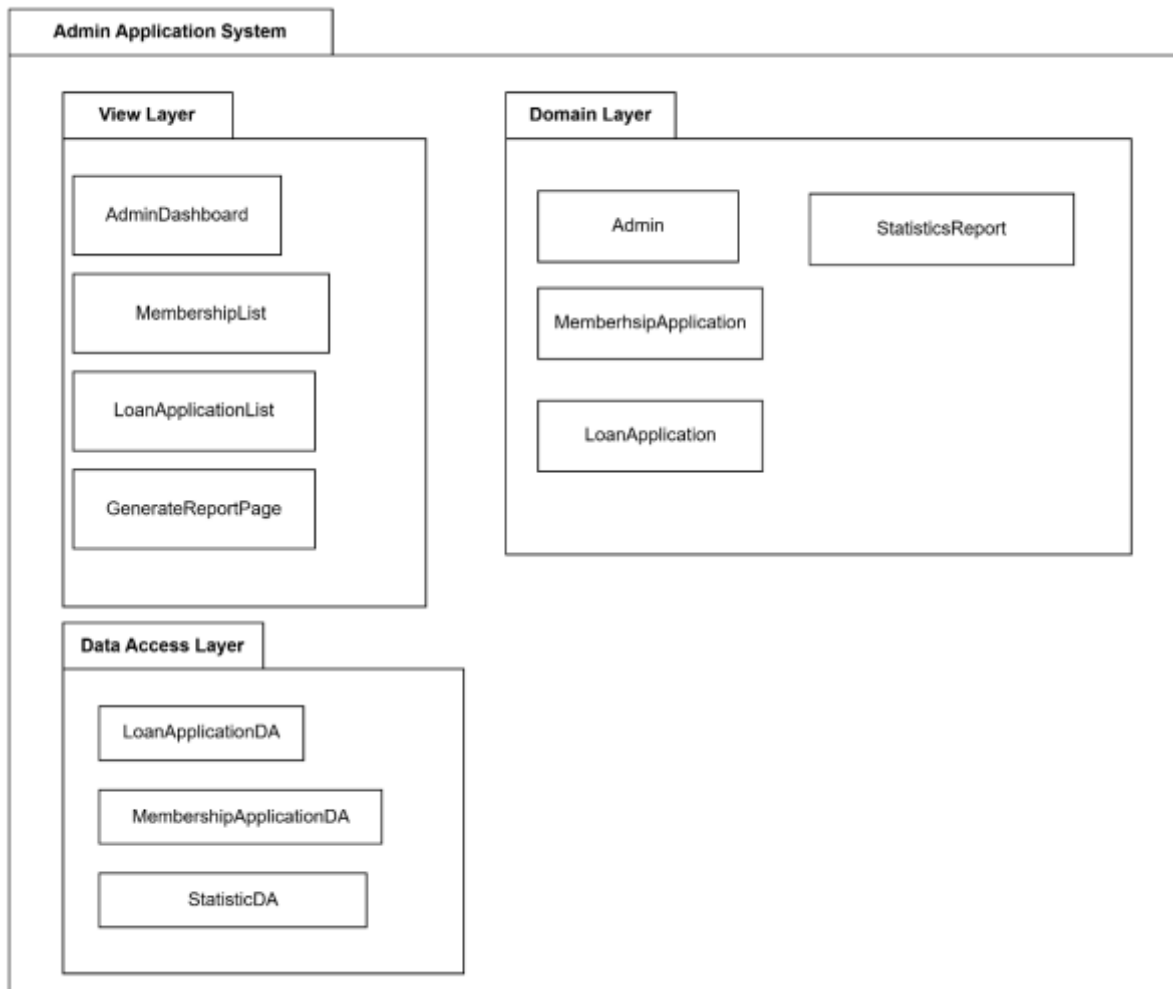


Figure 4.2.2.6: Package Diagram for <Administration Module > Subsystem

4.3 Logical View

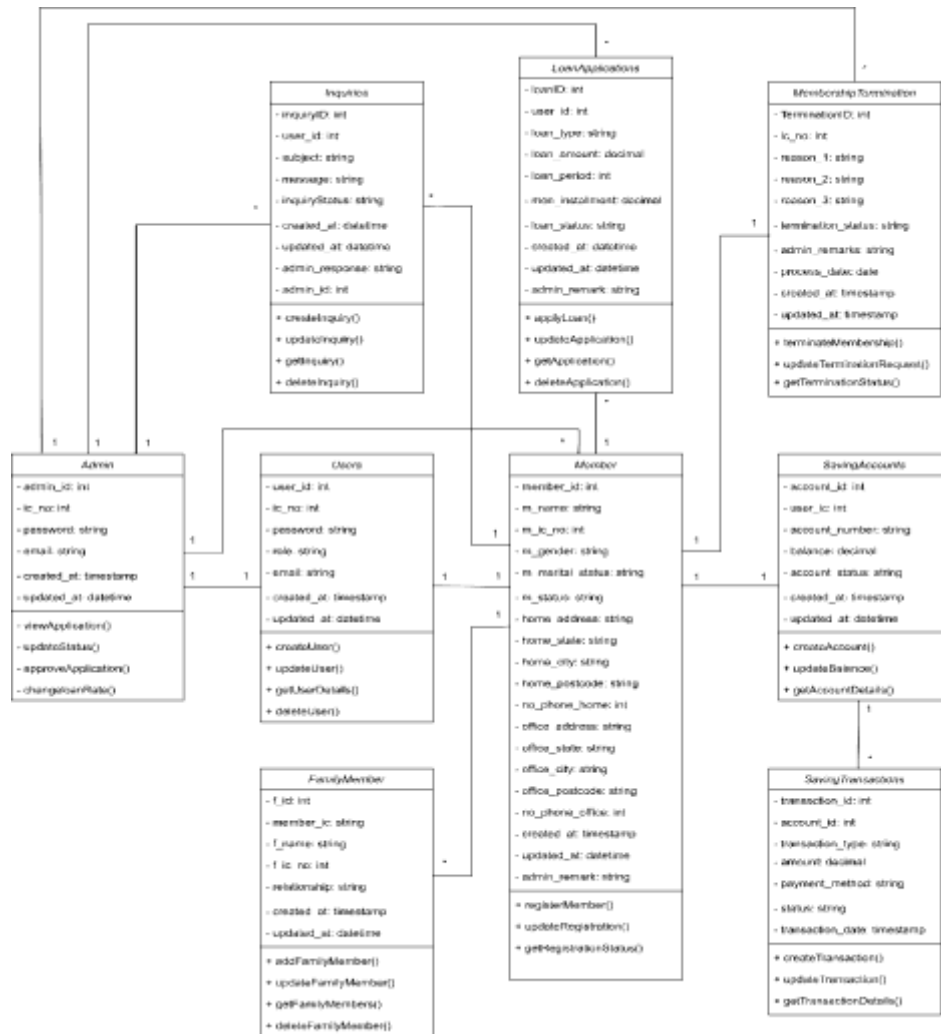


Figure 4.3.1: Class diagram for <KADA Cooperation> System

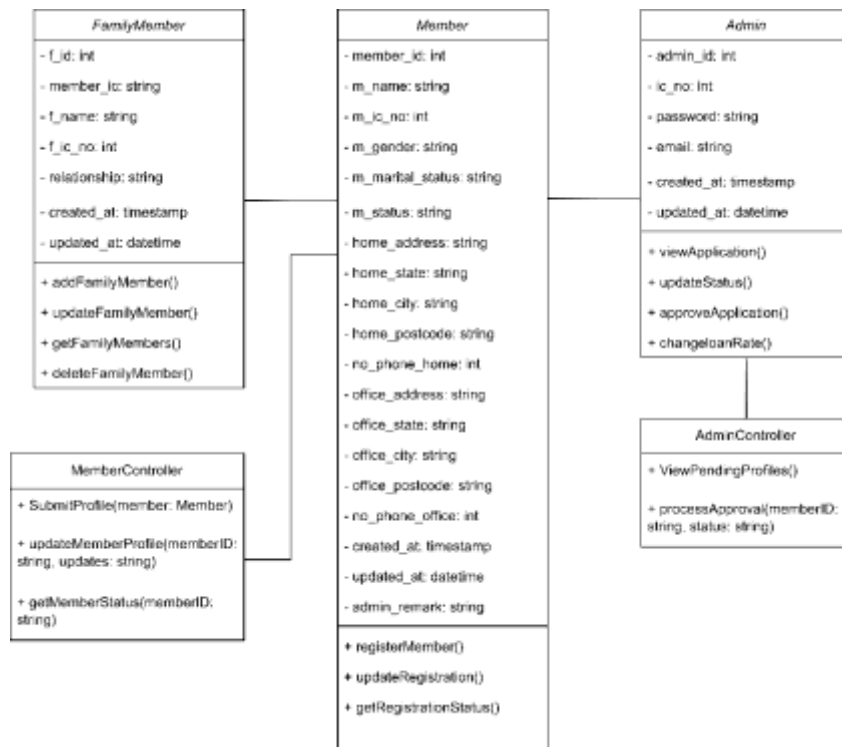


Figure 4.3.2: Class Diagram for <Membership Application> Subsystem

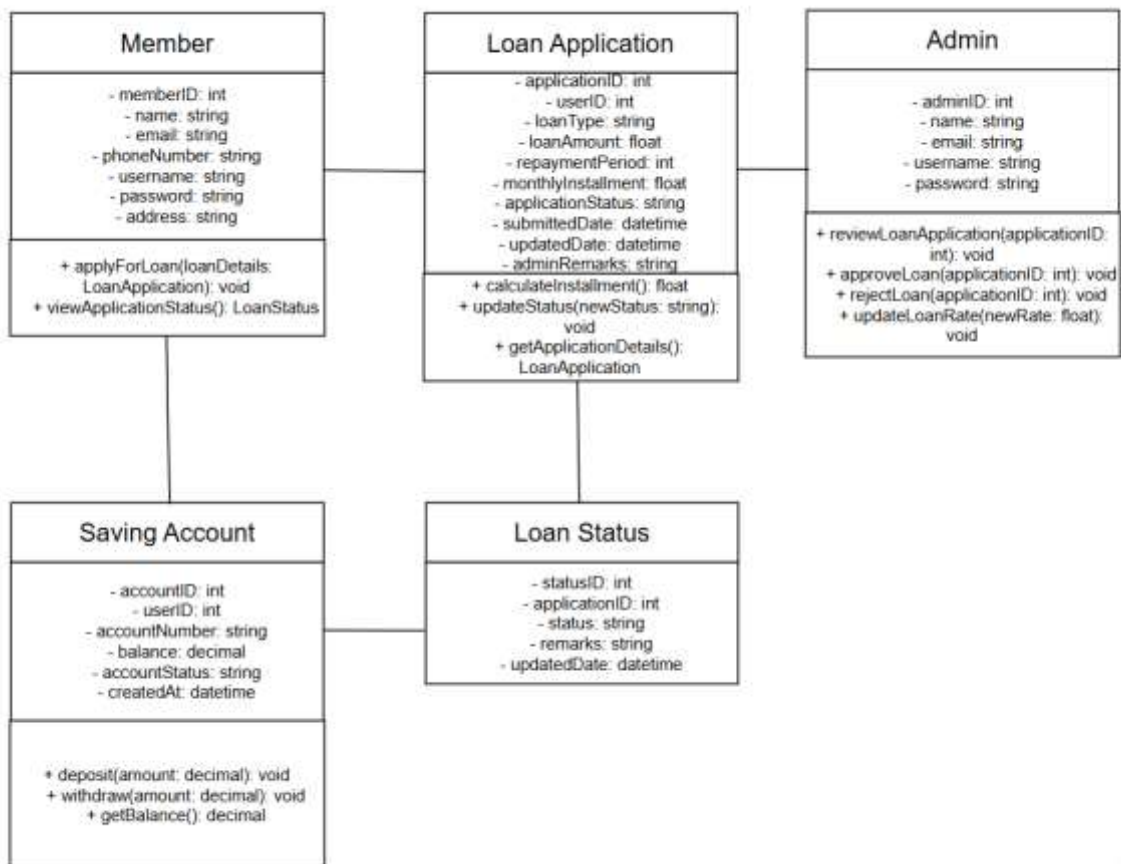


Figure 4.3.3: Class Diagram for <Loan Application> Subsystem

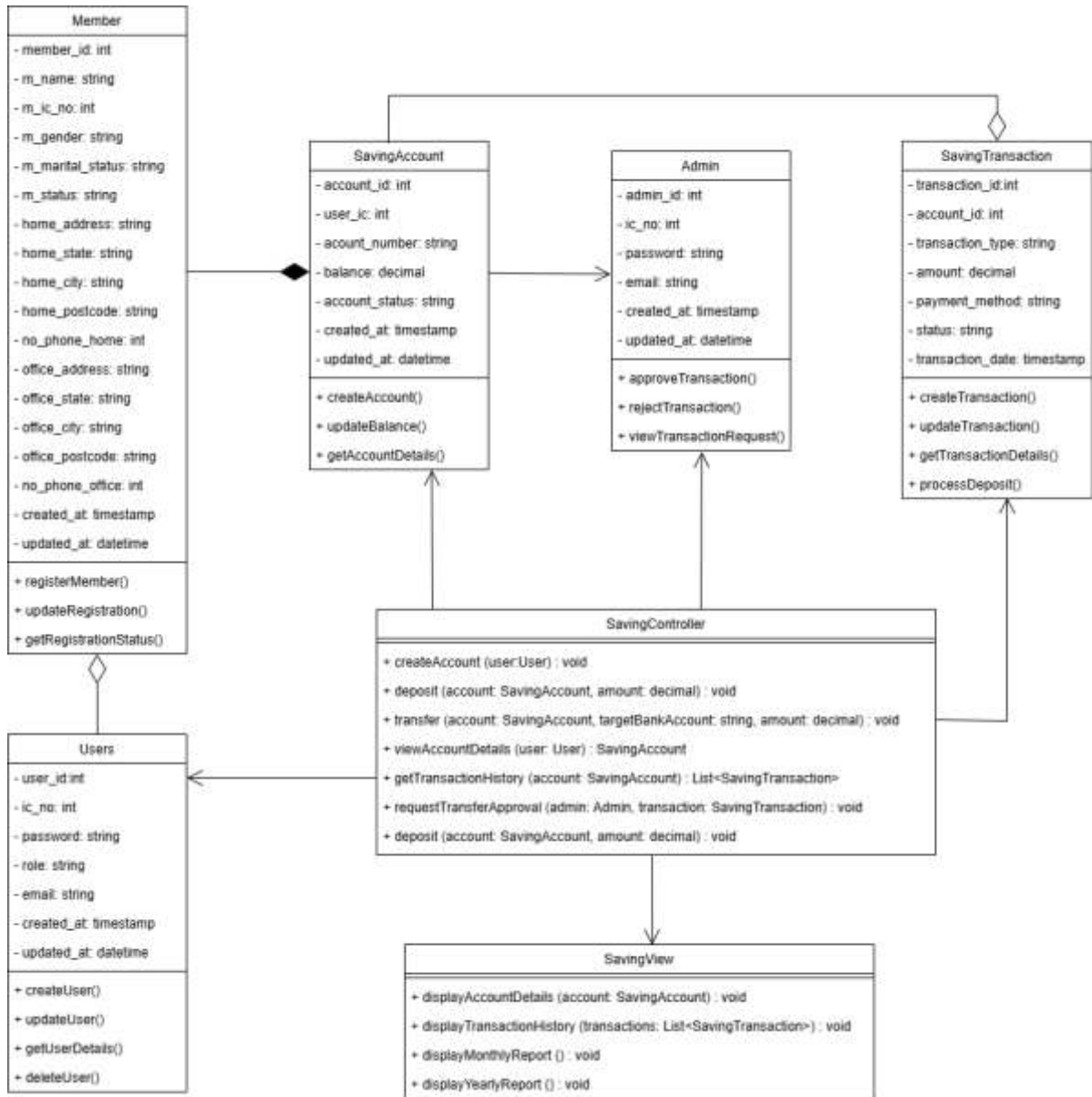


Figure 4.3.4: Class Diagram for < Saving Account Management Module > Subsystem

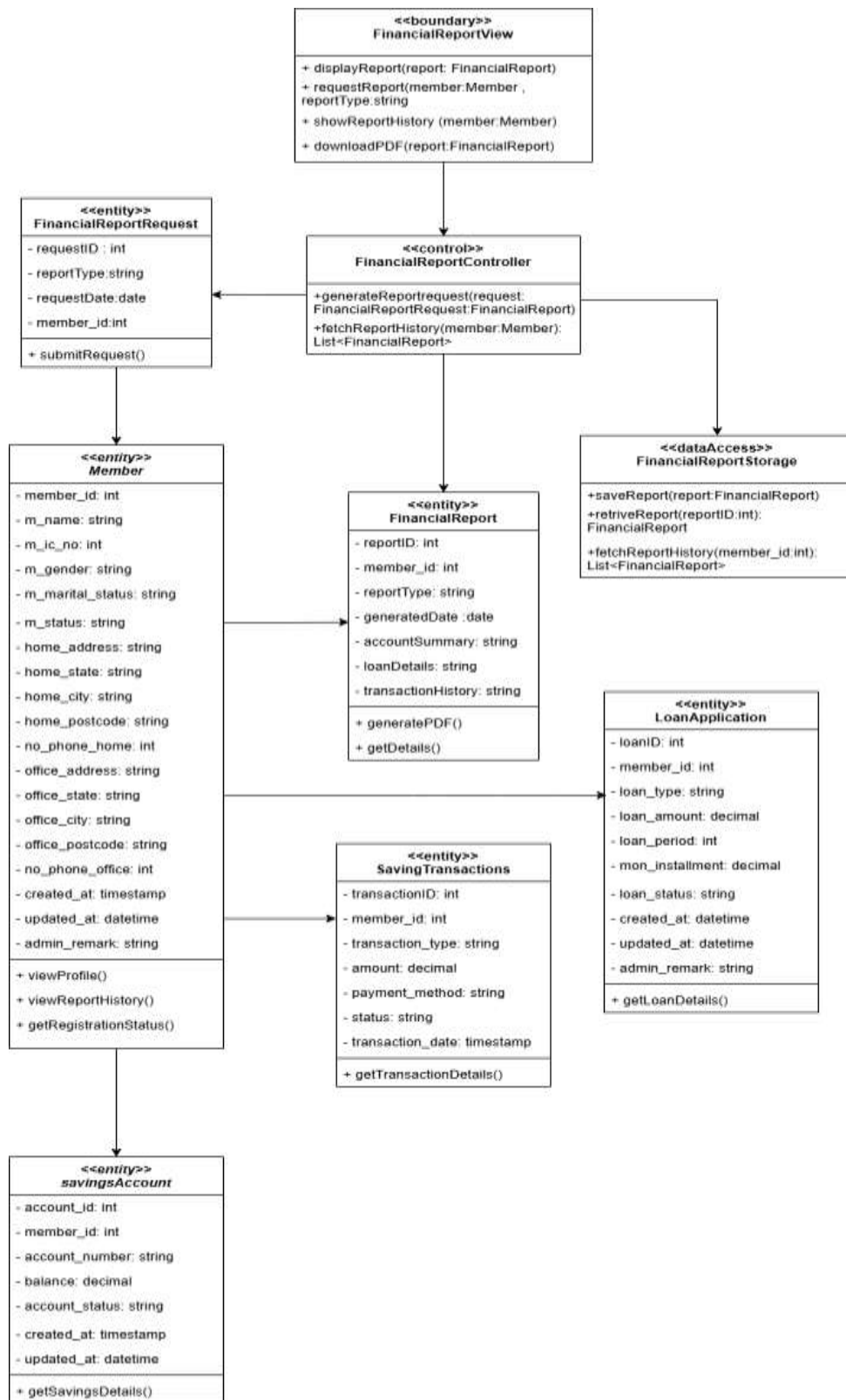


Figure 4.3.5: Class Diagram for < Financial Report Request Module > Subsystem

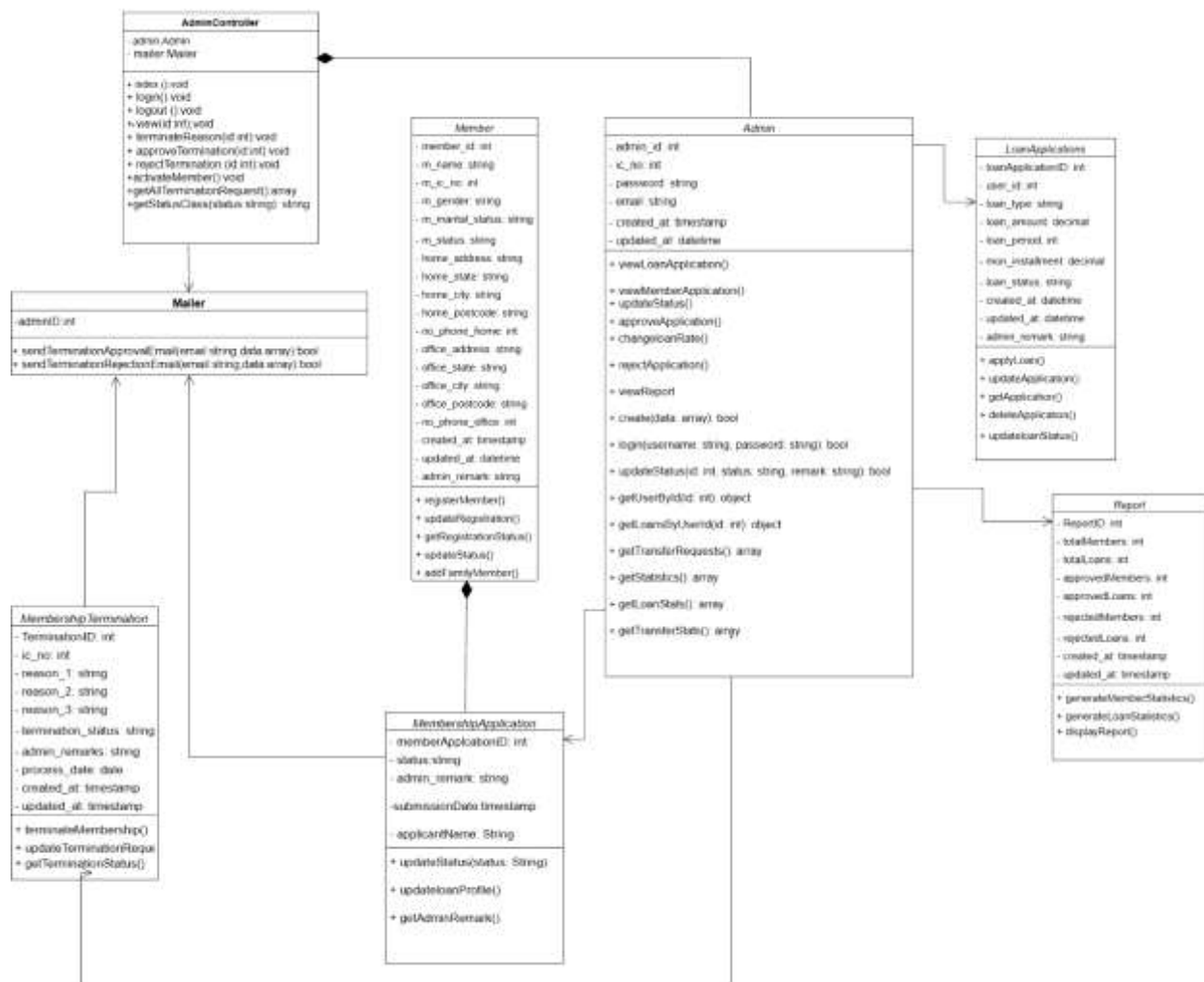


Figure 4.3.6: Class Diagram for < Administrative Module > Subsystem

4.4 Process View

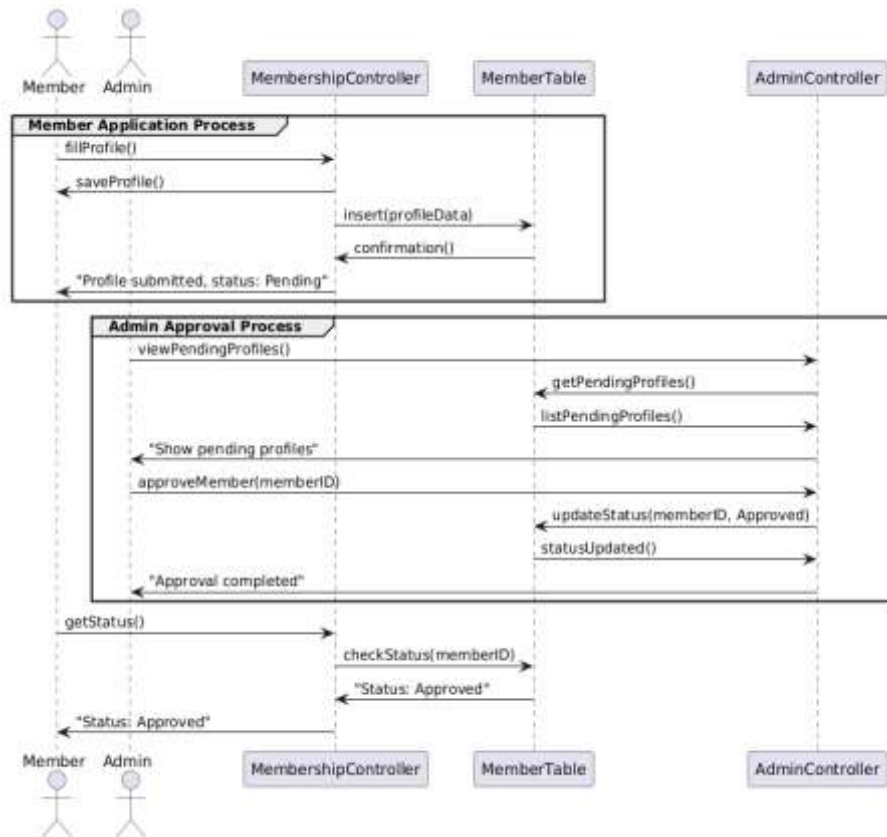


Figure 4.4.1: SD001 Sequence Diagram for <Apply for membership>

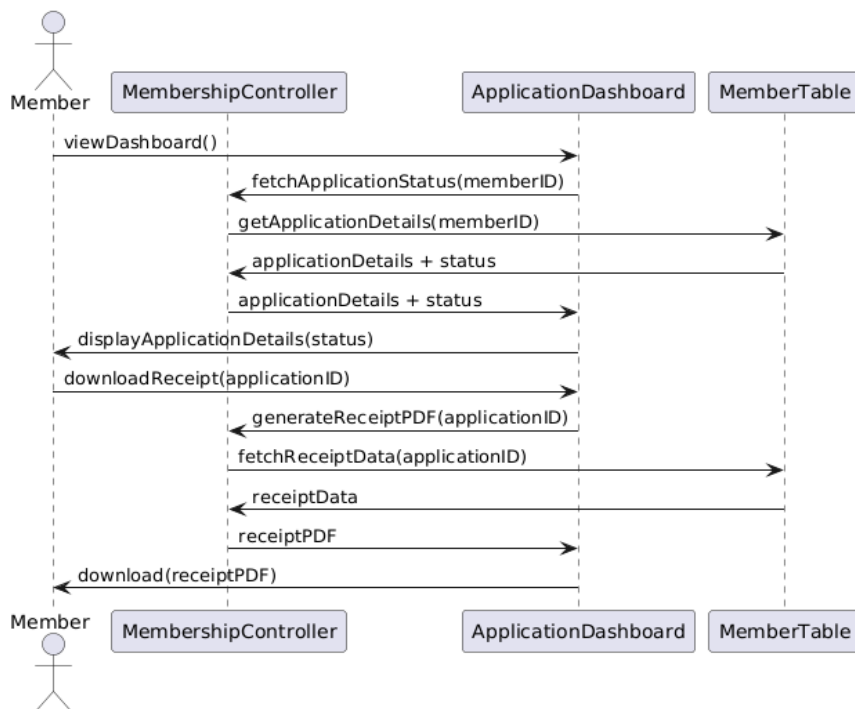


Figure 4.4.2: SD002 Sequence Diagram for <View Application Status>

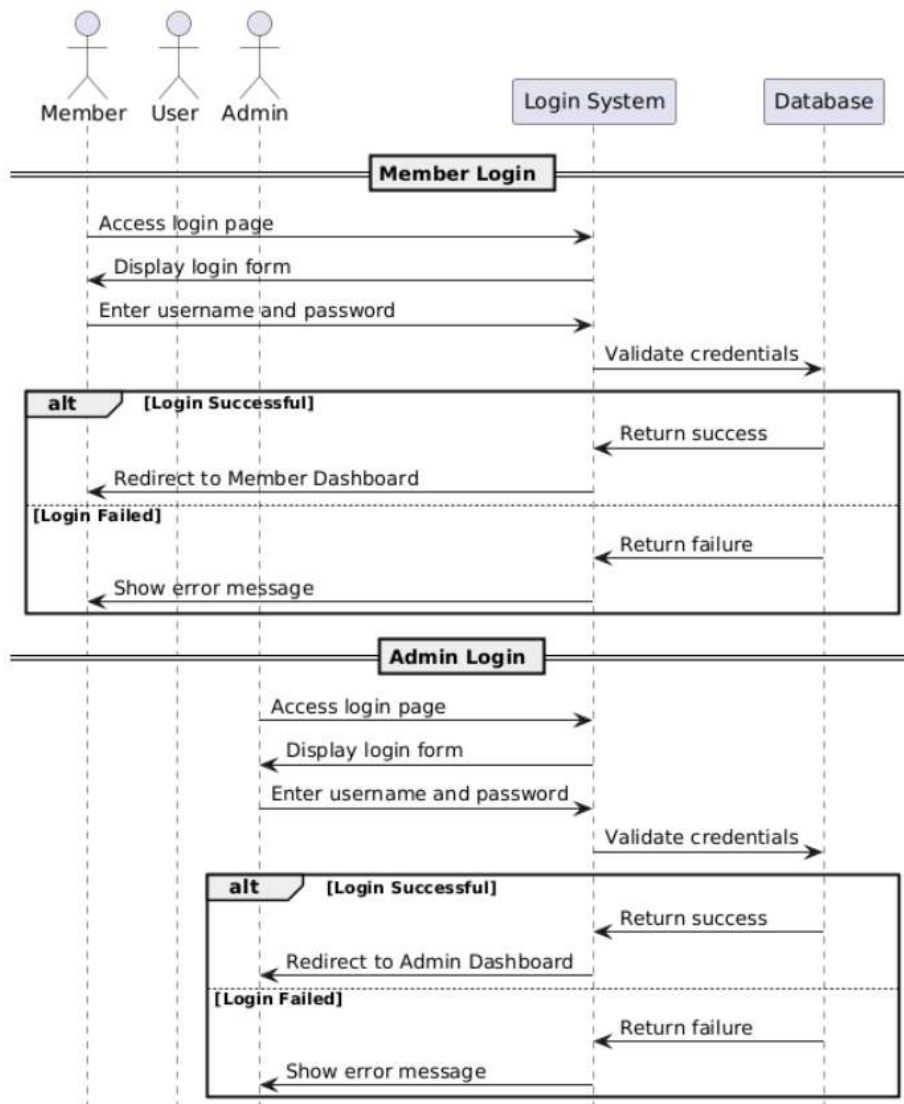


Figure 4.4.3: SD003 Sequence Diagram for <Log In >

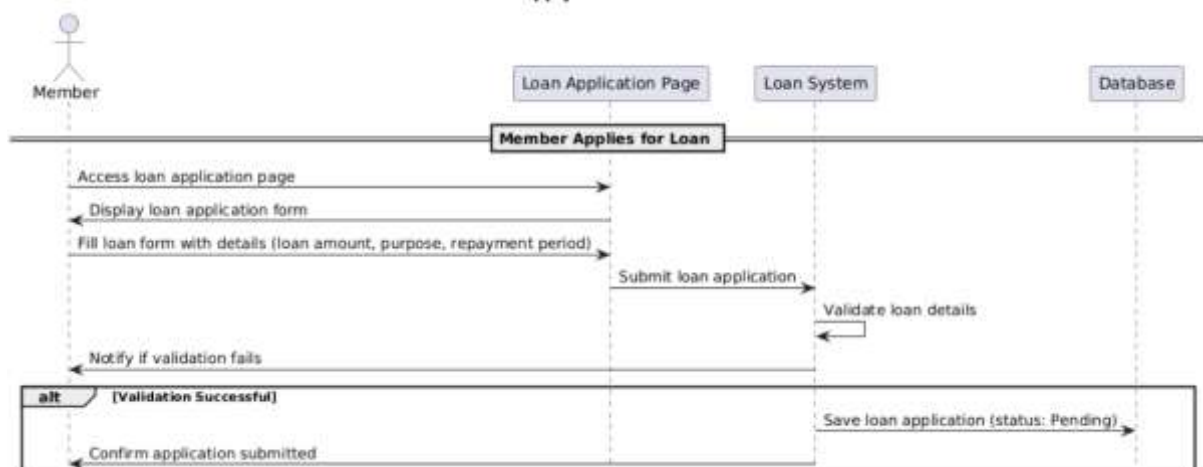


Figure 4.4.4: SD004 Sequence Diagram for < Loan Application >

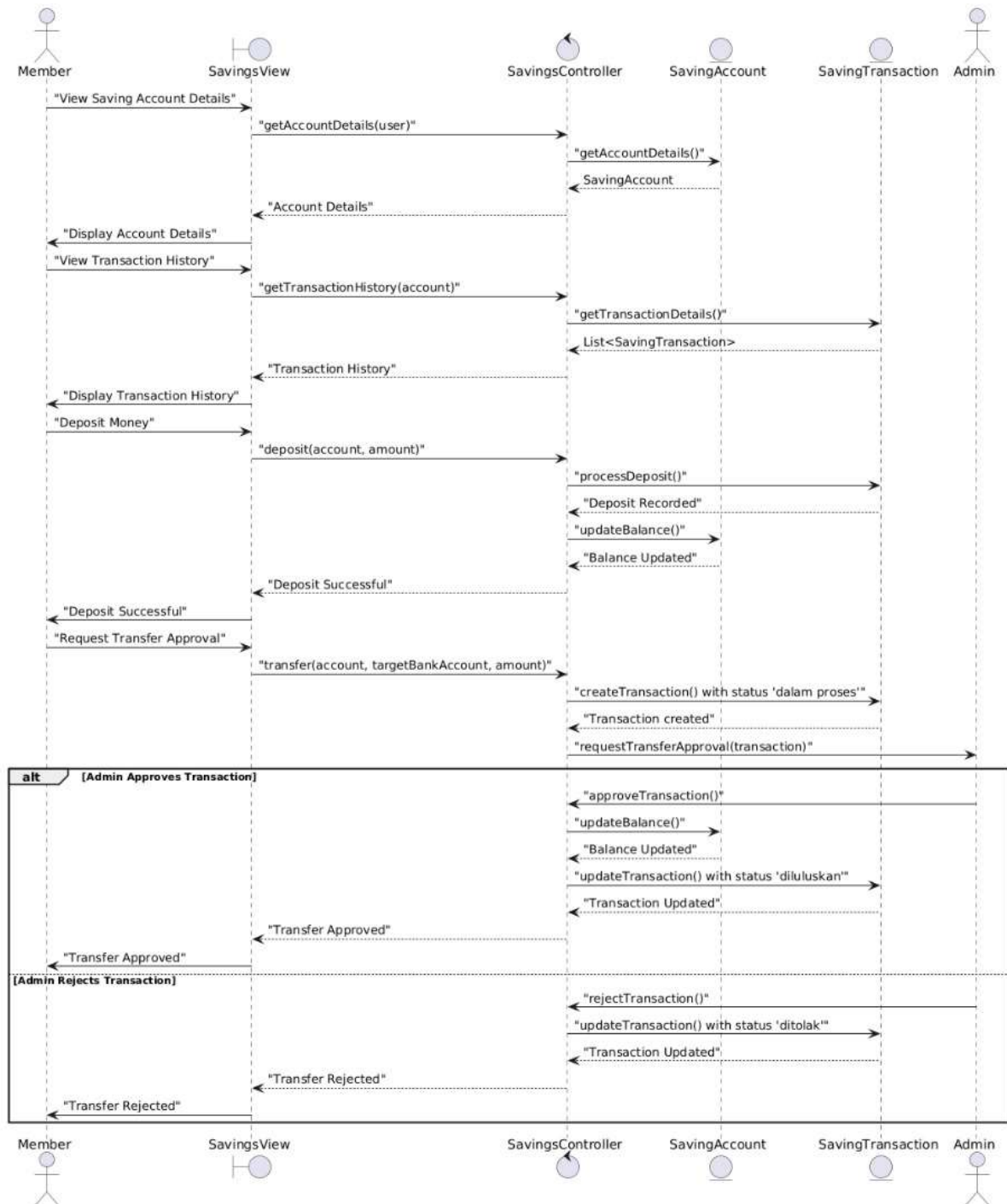


Figure 4.4.5: SD005 Sequence Diagram for < Manage Saving account >

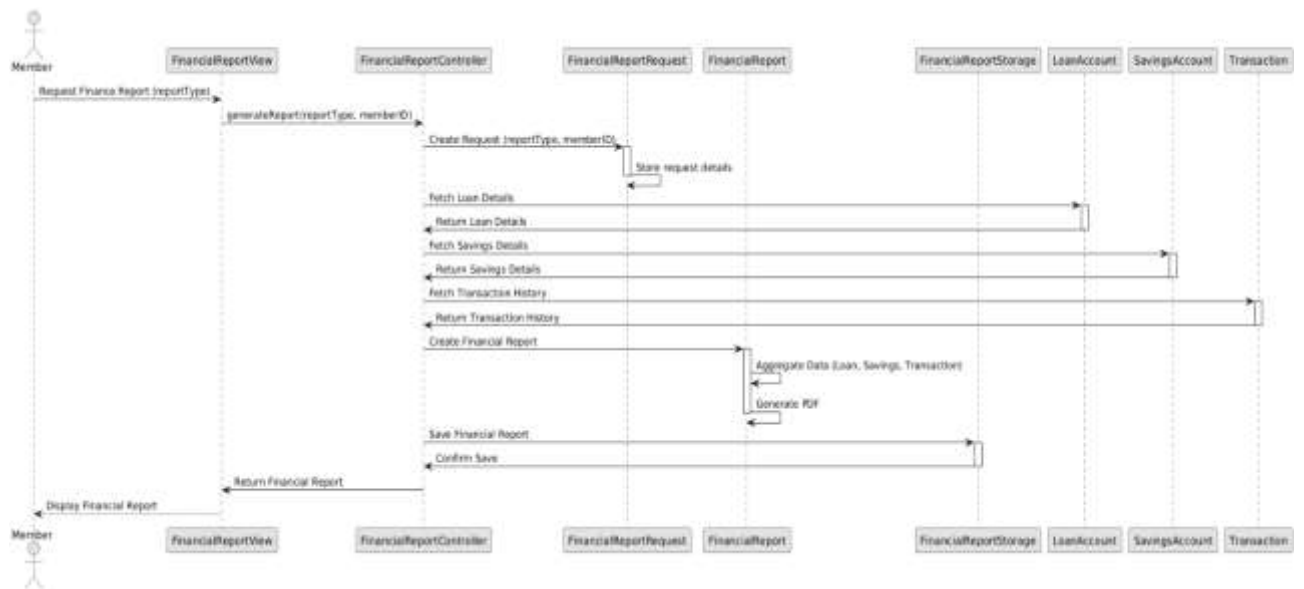


Figure 4.4.6: SD006 Sequence Diagram for < Request Finance Report >

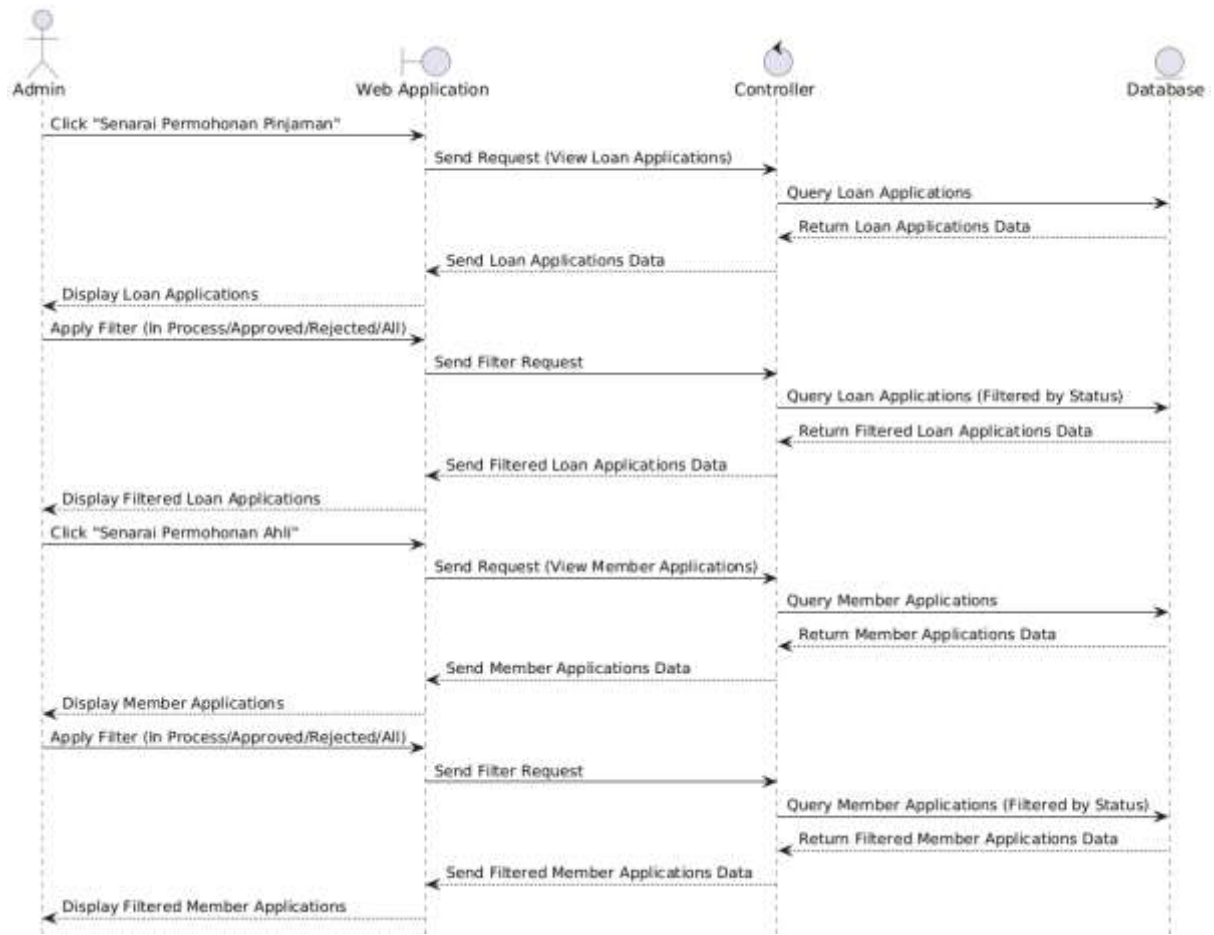


Figure 4.4.7: SD007 Sequence Diagram for < Admin view application list >

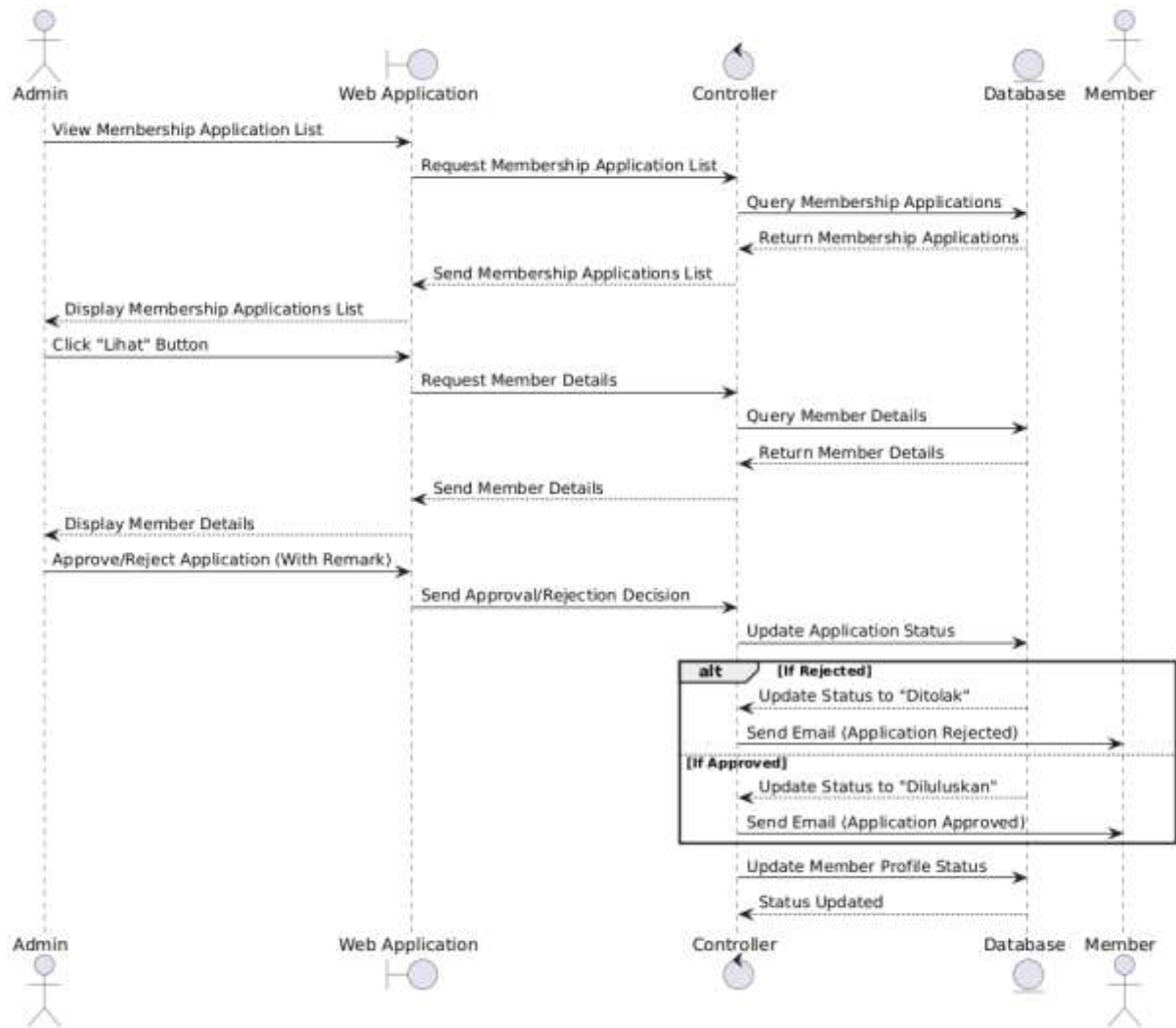


Figure 4.4.8: SD008 Sequence Diagram for < Admin approve membership application >

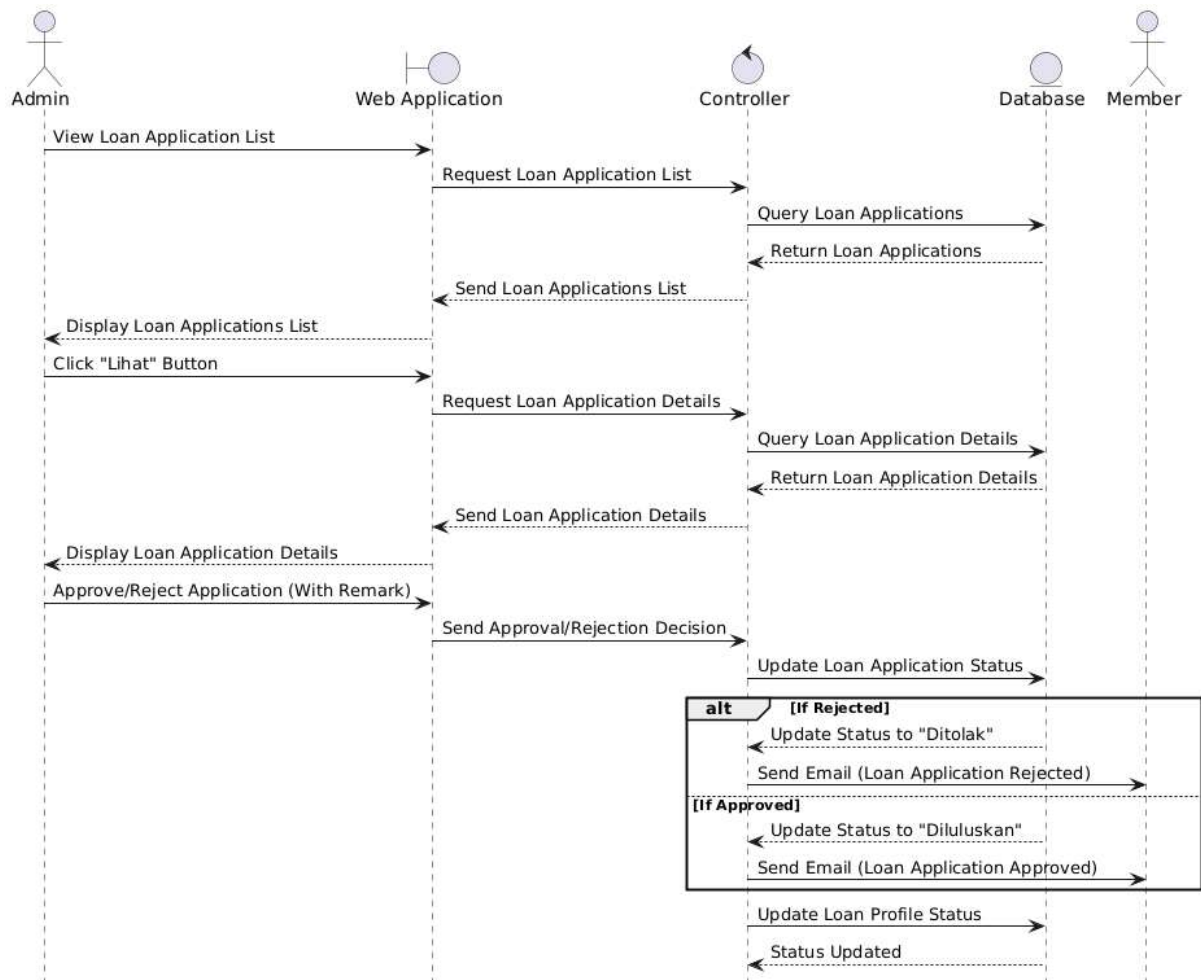


Figure 4.4.9: SD009 Sequence Diagram for < Admin approve loan application >

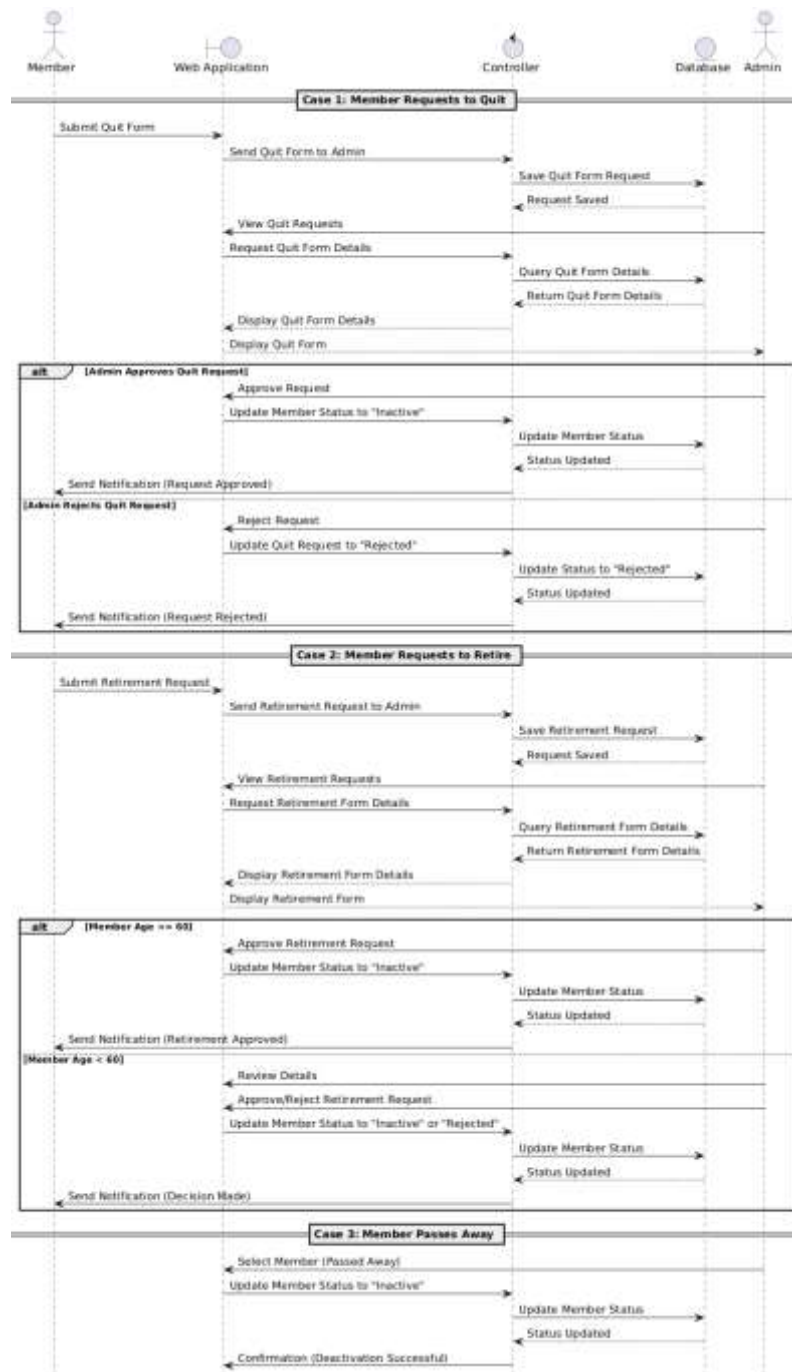


Figure 4.4.10: SD010 Sequence Diagram for < Admin deactivate member >

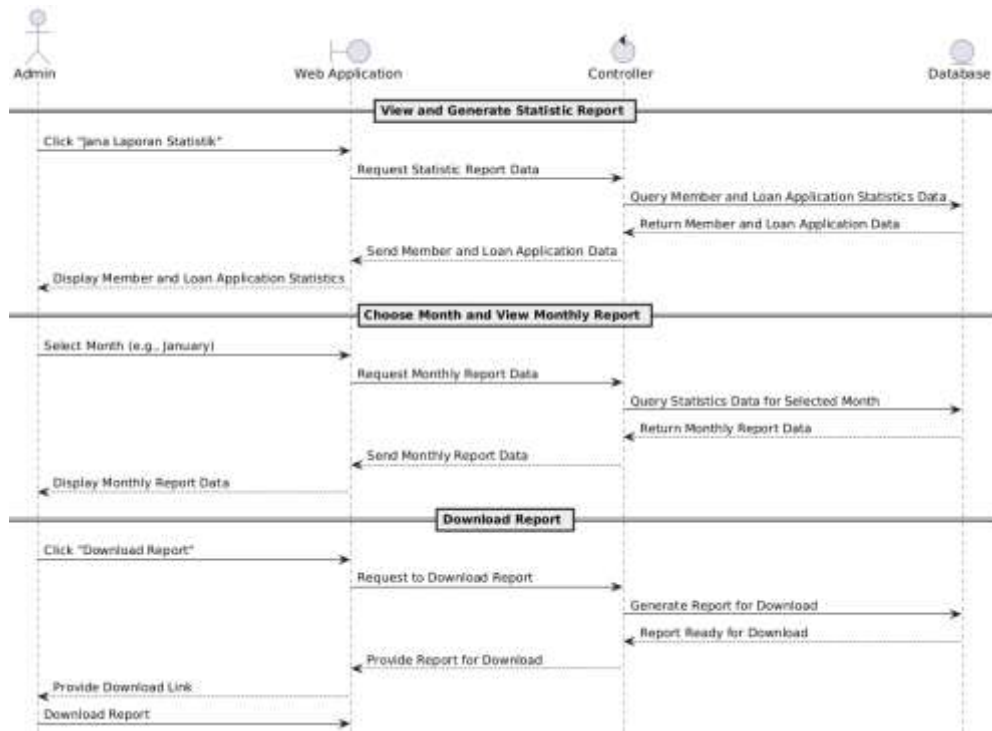


Figure 4.4.11: SD011 Sequence Diagram for < Admin view administrative report >

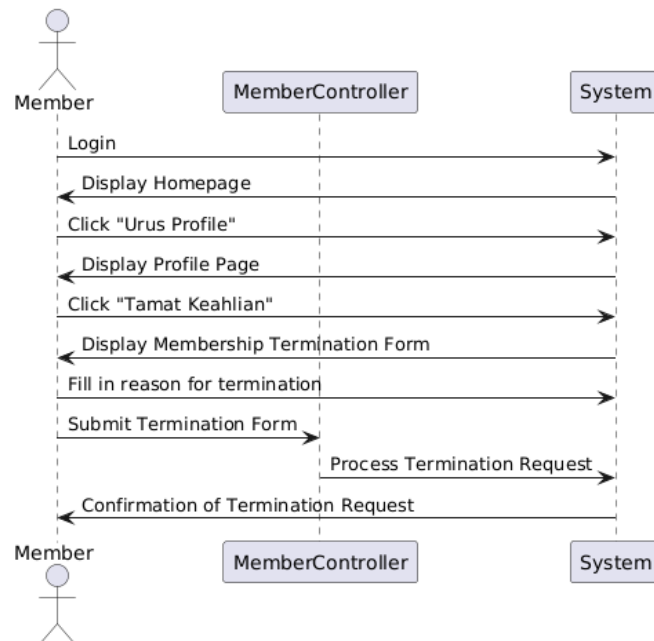


Figure 4.4.12: SD012 Sequence Diagram for < Member Terminate Membership >

4.5 Physical/ Deployment View



4.5.1 Deployment Diagram for KADA system

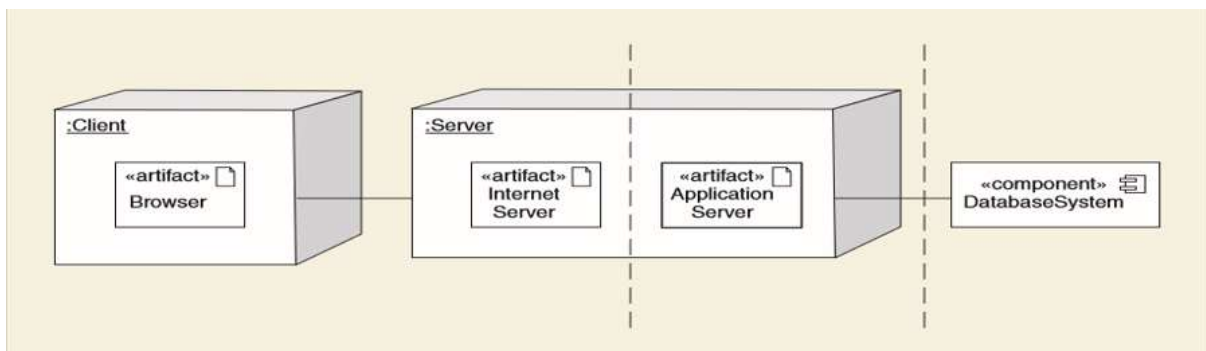


Figure 4.12: Deployment diagram for Internet-based System

5. Data Design

The major data or systems entities are stored into a relational database named as..., processed and organized into 10 entities as listed in Table 5.1.

Table 5.1: Description of Entities in the Database

Entity Name	Description	Aliases	Occurrence
User	Represents a user in the system, which could be an admin or a member.	Account Holder, User	Occurs when a user is created in the system.
Member	Represents an individual who has registered and been approved as a cooperative member.	Applicant, Member	Occurs when a user applies and is approved for membership.
Admin	Represents the administrator who manages applications, system info, and website content.	Admin, Manager	Occurs when an admin performs system operations.
Saving Account	Represents the savings account of a member, including balance, loan balance, share capital, and fixed savings.	Account, Savings	Occurs when a member opens or manages their savings account.
Transaction	Records transactions related to a member's saving account, such as deposits or payments for loans and programs.	Payment, Deposit	Occurs during savings account operations.
Financial Report	Represents monthly or annual financial reports generated based on saving account operations.	Report, Statement	Occurs when a member requests a financial report.
Application	Represents an application for membership, loans, or programs by a member.	Request, Form	Occurs when a member submits an application.
Membership Application	Represents an application for membership, including status and approval date.	Membership Request	Occurs when a user applies for membership.
Loan Application	Represents a loan application, including loan type, amount, period, and approval details.	Loan Request	Occurs when a member applies for a loan.
System Info	Represents system-level information, such as loan policies, membership requirements, and interest rates.	System Settings, System Info	Occurs when an admin updates or views system configurations.

5.1 Data Dictionary

5.1.1 Entity: < Member >

Attributes	Data Type & Length	Description	Null	Multi-valued	Example
memberID	int	Uniquely identifies the member	No	No	1
username	Varchar (50)	Username for member login	No	No	“Audrey”
password	Varchar (50)	Password for member login	No	No	“abcabc”
name	Varchar (100)	Full name of the member	No	No	“Audrey Sim”
IDNumber	Varchar (20)	Identification number	No	No	“A123456789”
gender	Varchar (10)	Gender of the member	No	No	“Female”
religion	Varchar (20)	Religion of the member	yes	No	“Christianity”
race	Varchar (20)	Race of the member	Yes	No	“Asian”
maritalStatus	Varchar (20)	Marital status of the member	Yes	No	“Single”
address	text	Address of the member	No	No	“123 Street Name”
phoneNumber	Varchar (15)	Phone number of the member	No	No	+60123456789
email	Varchar (50)	Email address of the member	No	No	“ abcabc@gmail.com ”
position	Varchar (50)	Job position of the member	No	No	“Manager”
positionGrade	Varchar (10)	Grad level of the position	Yes	No	“G7”

officeAddress	text	Office address of the member	No	No	“456 Office Street”
MonthlySalary	Decimal (10,2)	Monthly salary of the member	No	No	3500.50
applicationDate	date	Date of application	No	No	2025-01-03
approvalStatus	Varchar (20)	Approval status of the application	No	No	“Approved”

5.1.2 Entity: < Admin >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
adminID	int	Uniquely identifies the admin	No	No	1
username	Varchar (50)	Username for admin login	No	No	“admin01”
password	Varchar (50)	Password for admin login	No	No	“admin123”
name	Varchar (100)	Full name of the admin	No	No	“John Danny”
lastLogin	datetime	Last login time of the admin	Yes	No	2024-12-25 14:00:00

5.1.3 Entity: < Account >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
accountID	int	Uniquely identifies the account	No	No	101
memberID	int	Foreign key linking to	No	No	1

		Member entity			
accountStatus	Varchar (20)	Status of the account	No	No	“Active”
accountCreationDate	datetime	Date and time the account was created	No	No	2024-05-05 19:00:00

5.1.4 Entity: < SavingAccount >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
balance	Decimal (10,2)	Current balance of the savings account	No	No	5000.00
shareCapital	Decimal (10,2)	Share capital amount	Yes	No	200.00
fixedSavings	Decimal (10,2)	Fixed savings amount	Yes	No	1000.00
interestRate	Decimal (5,2)	Interest rate for savings	Yes	No	2.50
withdrawalLimit	Decimal (5,2)	Maximum amount of withdrawal allowed	No	No	5000.00
lastUpdated	datetime	Last update timestamp for savings account	No	No	2024-12-25 12:00:00

5.1.5 Entity: < LoanAccount >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
loanBalance	Decimal (10,2)	Outstanding loan balance	No	No	10000.00

overdueAmount	Decimal (10,2)	Amount overdue on the loan	Yes	No	500.00
penaltyRate	Decimal (5,2)	Penalty rate for overdue payments	Yes	No	1.50
nextPayment Rate	date	Date of next loan payment	No	No	2025-02-02
lastUpdated	datetime	Last update timestamp for loan account	No	No	2024-12-12 12:00:00

5.1.6 Entity: < Transaction >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
transactionID	int	Uniquely identifies the transaction	No	No	1
accountID	int	Foreign key linking to Account entity	No	No	101
transactionType	Varchar (20)	Type of transaction (e.g., Deposit, Withdraw)	No	No	“Deposit”
amount	Decimal (10,2)	Transaction amount	No	No	1000.00
transactionDate	datetime	Date and time of the transaction	No	No	2024-12-12 10:00:00
transactionStatus	Varchar (20)	Status of the transaction (e.g., Completed)	No	No	“Completed”

5.1.7 Entity: < FinancialReport >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
reportID	int	Uniquely identifies the financial report	No	No	1
memberID	int	Foreign key linking to Member entity	No	No	1
reportType	Varchar (50)	Type of the financial report (e.g., Savings, Loan)	No	No	“Savings Report”
generatedDate	datetime	Date and time the report was generated	No	No	2024-12-25 15:00:00
details	text	Detailed description of the report	Yes	No	“Report content”

5.1.8 Entity: < Application >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
applicationID	int	Uniquely identifies the application	No	No	1
memberID	int	Foreign key linking to Member entity	No	No	1
applicationDate	date	Date the application was submitted	No	No	2024-12-12

5.1.9 Entity: < LoanApplication >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
loanType	Varchar (50)	Type of loan applied	No	No	“Personal”
loanAmount	Decimal (10,2)	Amount of loan applied	No	No	40000.00
loanPeriod	int	Duration of the loan in months	No	No	60
monthlyInstallment	Decimal (10,2)	Monthly repayment amount	No	No	1000.00
bankAccNo	Varchar (30)	Bank account number for loan disbursement	No	No	“123456789”
bankName	Varchar (50)	Name of the bank	No	No	“Maybank”
approvalDate	date	Date the loan was approved	Yes	No	2024-12-31

5.1.10 Entity: < LoanStatus >

Attribute	Data Type & Length	Description	Null	Multi-valued	Example
statusID	int	Uniquely identifies the loan status	No	No	1
applicationID	int	Foreign key linking to Application entity	No	No	1
loanAccountID	int	Foreign key linking to LoanAccount entity	No	No	201

statusType	Varchar (50)	Type of the status	No	No	“Pending”
statusRemarks	text	Remarks or comments on the status	Yes	No	“Awaiting review”
statusTimestamp	datetime	Date and time of the status update	No	No	2025-01-01 16:00:00

6. User Interface Design

The KADA system assures the interface for the user and administrator's ease of use. It is highly navigable, where core functions such as membership applications, loan applications, savings account management, and financial reporting are accessed without any problem. The user interface ascertains that users interact with it easily regardless of how 'tech-savvy' they might be through a combination of straightforward layout, helpful prompts, and clear labels that enhance their experience.

To **join the membership**, new users are registered through an online form, which captures their essential information, such as ID numbers, email addresses, and telephone numbers. The interface guides the user through every step to ensure no field is left unfilled. If the user misses any field or enters invalid input, the system throws an error message to prompt them to make a correct entry. After successful submissions, a confirmation mail is sent with a verification link to confirm the account. An account should be activated by clicking the verification link. After activation, it will allow the users to log in and further track the process directly from the dashboard. In case the user forgets their password, they can request a reset of it, with a link for setting up a new one to be sent on their registered e-mail

The **loan application** feature allows members to view different types of loans through an organized interface. Members can select the desired loan type, view the details, and submit their application. It also contains a loan calculator to help in the estimation of interest rates and amounts repayable by the borrower. Such interfaces will immediately display any errors committed-for example, exceeding the limit or failure to provide some mandatory details. Once submitted, users are provided with confirmation messages regarding successful submission, while they can track the status of their applications anytime on the "Papan Pemuka" section of the dashboard.

The **Savings Account Management** interface provides various facilities to members for better management of their finances. It allows the member to view detailed information about their accounts, including the current balance, transaction history, and outstanding loan amounts. It supports online transactions securely, such as depositing money or paying off loans, and viewing past transactions. The system will always give feedback right after any transaction has been made, whether it was successful or there was an issue, such as insufficient funds. It keeps a history of transactions, making it easy for users to monitor their financial activities over time. There is also a receipt that can be download when member successfully making the payment.

The **financial reporting** feature enables members to generate correct and comprehensive reports. Members can generate financial summaries that include transaction history, loan details and savings contribution while administrators can generate reports on the performance of their cooperatives, including new membership and loan applications. These reports can be visualized directly in the interface and downloaded in a format easily understandable by users, such as in PDF format. Real-time accuracy is assured with this system as it fetches the latest data from the database to keep things transparent for better decision-making.

The **administrative interface** is built to optimize management tasks. For instance, membership, loan, and membership termination applications can be viewed and approved in both sortable and filterable lists for added efficiency. Batch processing options are also provided to help deal with several applications simultaneously. Besides this, the admin panel depicts critical metrics including the total number of applications and summaries of financial performance. There is also an admin report feature whereby administrators will have access to further detailed statistics on loan applications, membership registrations, membership terminations, and transaction amounts for a complete trend overview of the applications and financial activities. This streamlined interface ensures that administrators have their task load minimized while the integrity of the system is maintained

The interface incorporates several key features that go a long way in enhancing the user experience. There's a navigation bar through which one can easily access all the main modules such as Membership, Loans, Savings Accounts, and Reports. A customized dashboard displays user-specific information, pending actions, notifications, and status updates. For new users, interactive tutorials and on-screen guides will be embedded to help them get familiar with the system's features. It also includes responsive optimization for consistency and ease of access on desktops, tablets, and mobile devices.

Feedback mechanisms are very important in enhancing user interaction. Error messages appear when there are invalid inputs or system errors, such as wrong login credentials or a failed transaction. Confirmation messages assure the user of actions just performed, such as form submissions or payments. Real-time updates, such as the status of loan approval or the success of a transaction, are displayed on the user's dashboard to keep him updated on his activities within the system.

The KADA system's interface combines clear navigation, responsive design, and robust feedback mechanisms to ensure that users can efficiently complete all expected features. Whether registering as a member, managing financial tasks, or performing administrative duties, the system is designed to provide an efficient and enjoyable user experience.

6.1 Screen Images

6.1.1 General

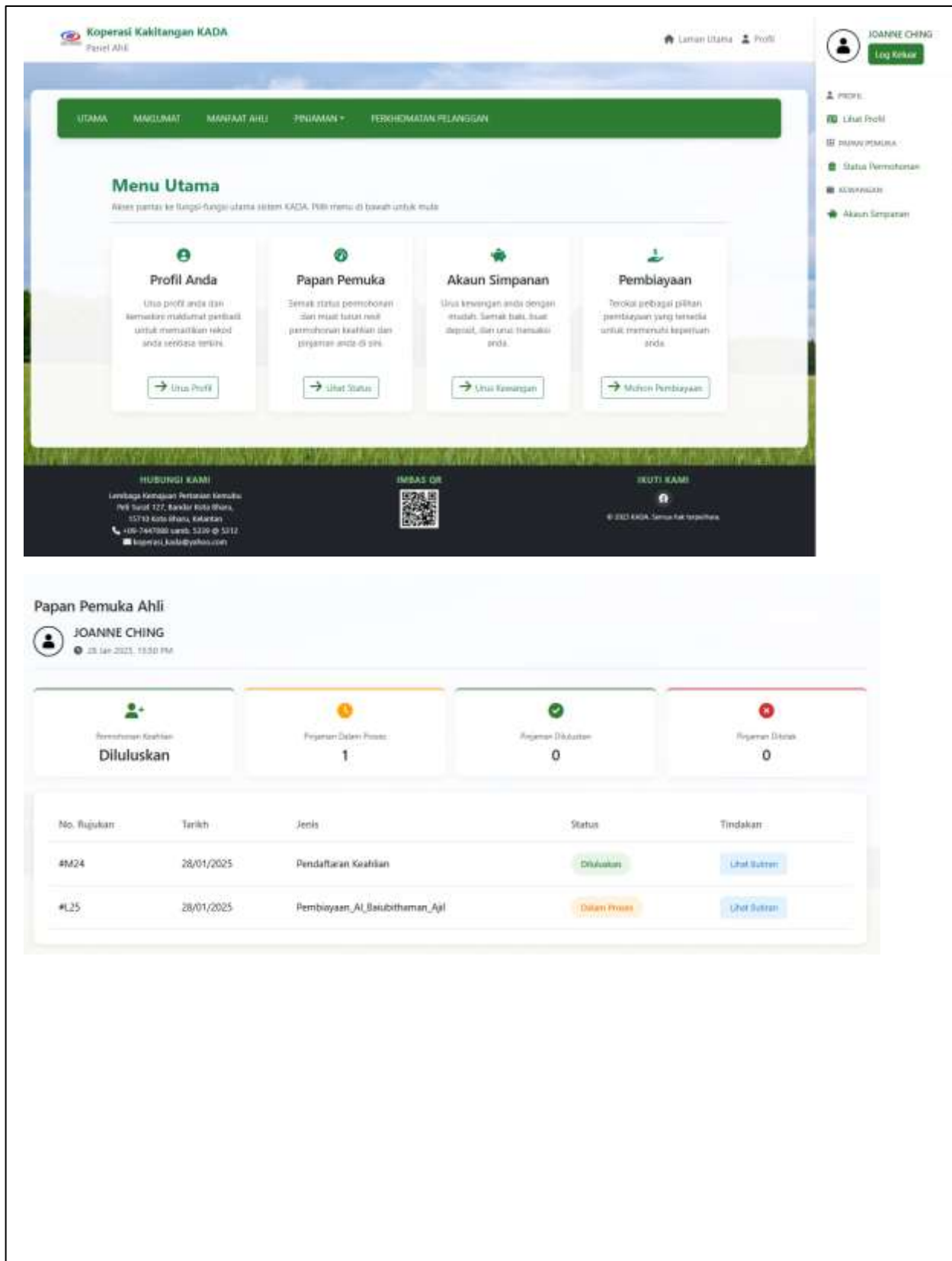
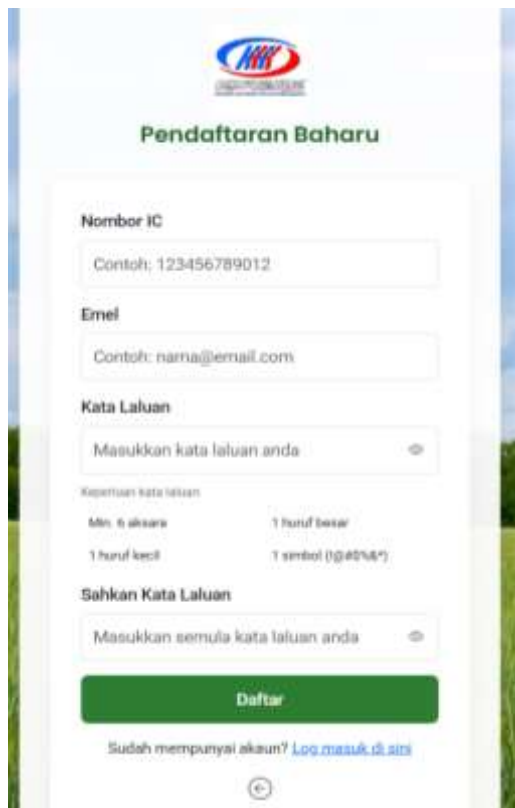


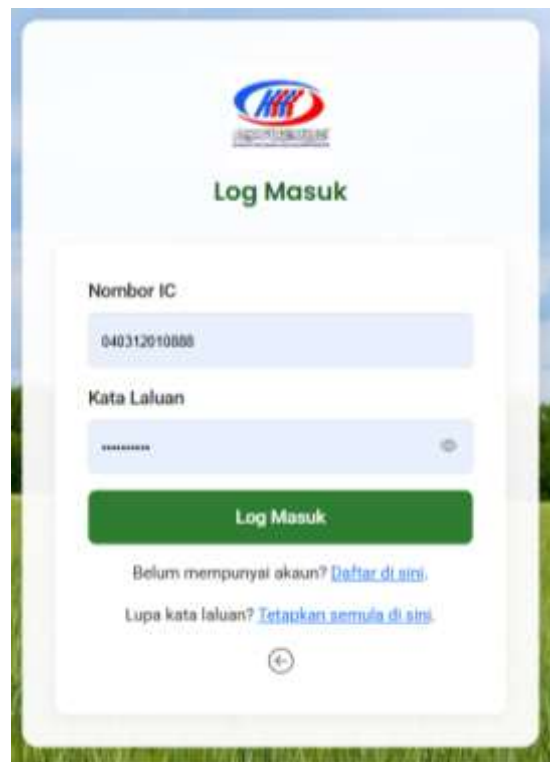
Figure 6.1: Interface for <Home Page>

6.1.2 User Authentication



The 'Pendaftaran Baharu' (New Registration) form includes fields for 'Nombor IC' (IC Number) with a sample '123456789012', 'Emel' (Email) with a sample 'nama@email.com', and 'Kata Laluan' (Password). The password field has a strength indicator and requirements: minimum 6 characters, at least one uppercase letter, one lowercase letter, and one symbol. A second field 'Sahkan Kata Laluan' (Confirm Password) is also present. A green 'Daftar' (Register) button is at the bottom, with a link to 'Log masuk di sini' (Log in here) for existing users.

New Registration



The 'Log Masuk' (Login) form includes fields for 'Nombor IC' (IC Number) with a sample '040312010888' and 'Kata Laluan' (Password). A green 'Log Masuk' button is at the bottom. Below the button are links for 'Daftar di sini' (Register here) for new users and 'Tetapkan semula di sini' (Reset here) for forgotten passwords. A back arrow icon is at the very bottom.

Login

Status Permohonan Keahlian KADA

KS KADA System <elijahshe04@gmail.com>
To: You

Status Permohonan Keahlian KADA

Assalamualaikum dan Salam Sejahtera,
Tahniah! Permohonan keahlian anda telah **DILULUSKAN**.

Butiran Keahlian:

Nama: JOANNE CHING YIN XUAN
No. Ahli: 1234545

Anda kini boleh:

- Mengakses semua kemudahan ahli KADA
- Memohon pinjaman
- Membuat simpanan
- Menikmati faedah-faedah keahlian

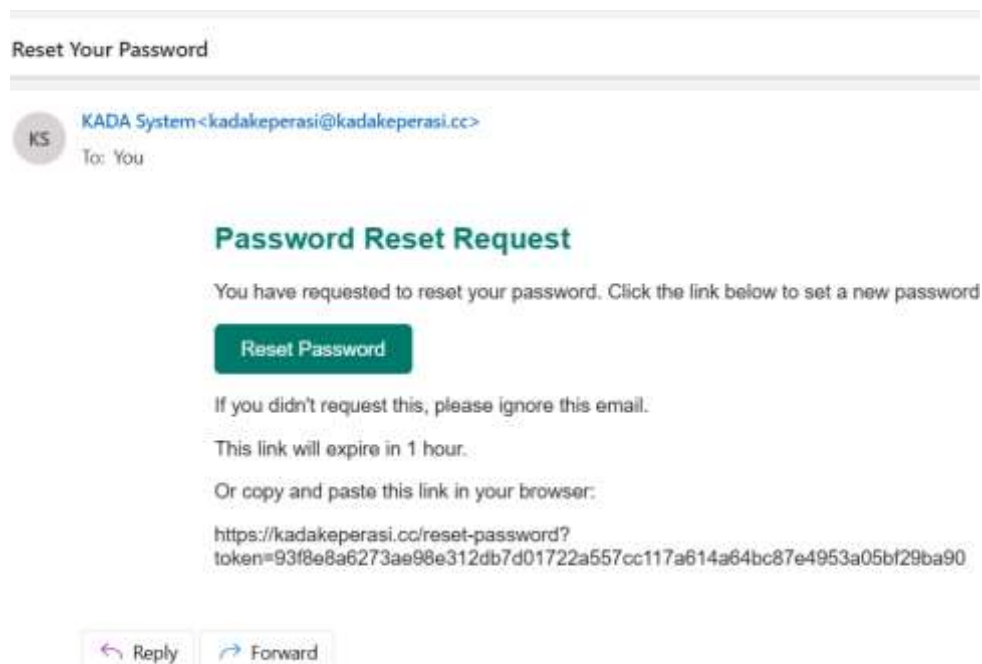
Selamat datang ke keluarga besar KADA!

Yang benar,
Pihak Pengurusan KADA

[Reply](#) [Forward](#)

Approved Email

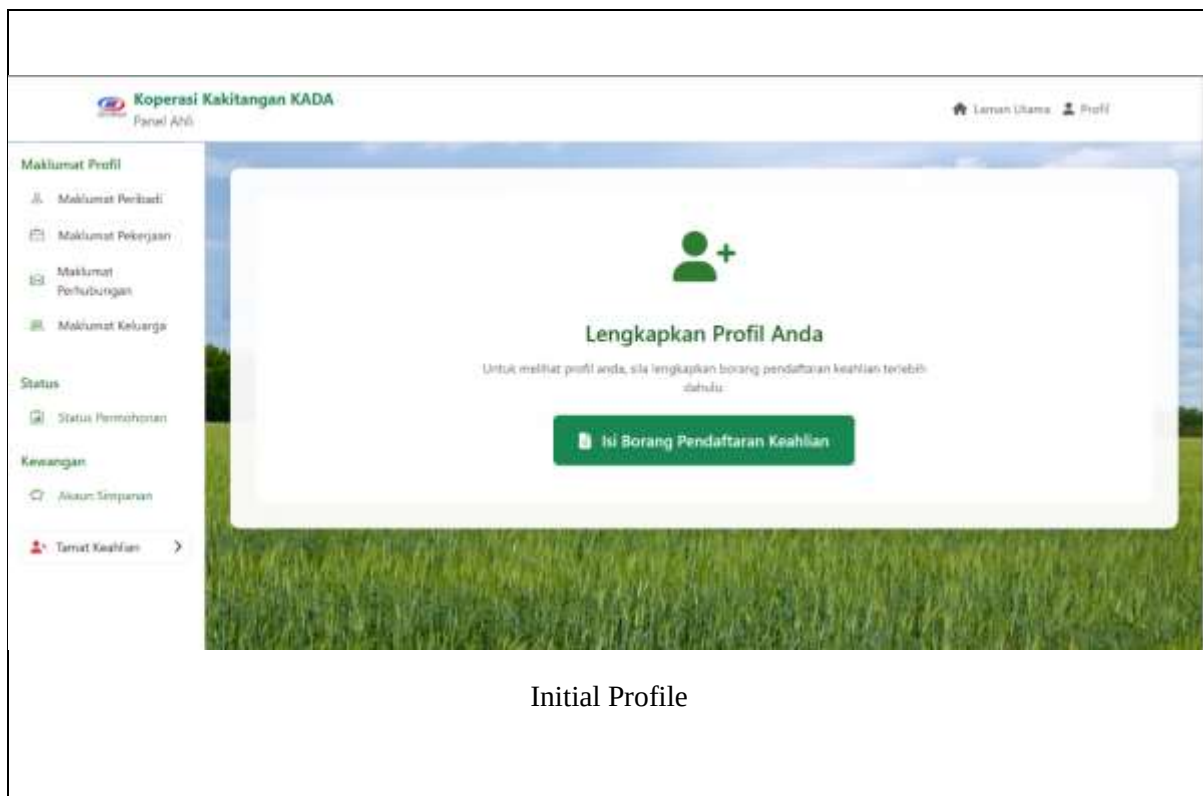
Reset password



Email to register password

Figure 6.2: Interface for < User authentication>

6.1.3 Membership Application and Profile Management



The screenshot shows the 'Pendaftaran Profil Ahli' form. At the top, there's a 'Kembali ke Halaman Utama' button and a progress indicator with five steps. The first step, 'Maklumat Peribadi', is active. Below the progress bar, there are five tabs: 'Maklumat Peribadi', 'Maklumat Pekerjaan', 'Maklumat Perhubungan', 'Maklumat Keluarga', and 'Maklumat Keuangan'. The 'Maklumat Peribadi' tab is selected, showing a form with the following fields: 'Nama Penuh' (text input), 'No. Kad Pengenalan' (text input with a placeholder '33333333333'), 'Jantina' (dropdown menu with 'Pilih Jantina'), 'Agama' (dropdown menu with 'Pilih Agama'), 'Bangsa' (dropdown menu with 'Pilih Bangsa'), and 'Status Perkahwinan' (dropdown menu with 'Pilih Status'). A green 'Setorannya' button is at the bottom right. The background is a green field under a blue sky.

Membership Application form

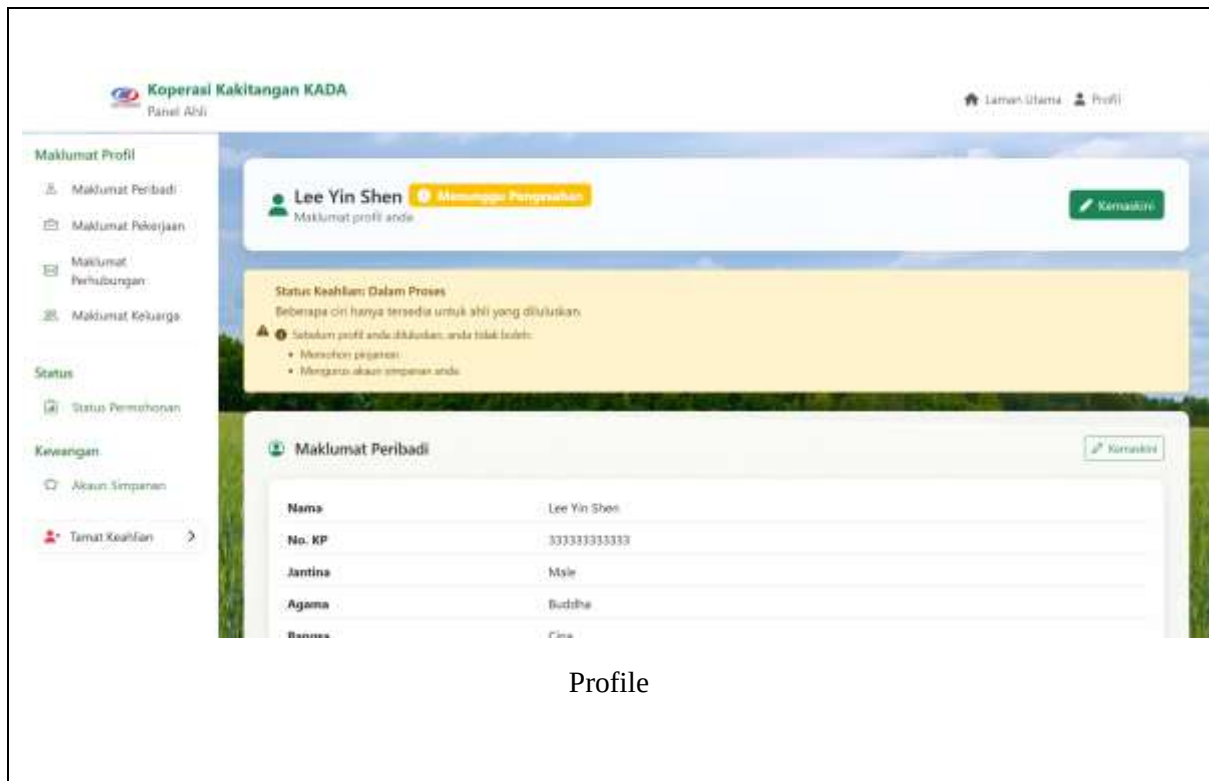


Figure 6.3: Interface for <Membership Application and Profile Management>

6.1.4 Membership Termination

[illegible]

Membership Termination Form

A screenshot of an email from KADA System. The email header shows the sender as 'KADA System <dkjshu04@gmail.com>' and the recipient as 'En: Yui'. The subject is 'Status Permohonan Menamatkan Keahlian KADA'. The body of the email states: 'Assalamualaikum dan Salam sejahtera, Kami ingin memaklumkan bahawa permohonan anda untuk menamatkan keahlian KADA telah **DITOLAK**.' Below this, there is a section titled 'Butiran Ahli:' containing the following details: 'Nama: JOVIANE CHING YIN XUAN', 'No. PF: 123456', and 'Catatan Admin: do'. At the bottom, there is a note: 'Anda juga perlu mengenalpastikan: Semua pinjaman telah diselesaikan, Tiada tunggakan yuran bulanan, Semua dokumen yang diperlukan telah diungkapkan.' and a contact number: 'Untuk sebarang pertanyaan, sila hubungi pihak kami di talian +60 97455388.' The footer of the email reads: 'Yang benar, Pihak Pengurusan KADA'.

Rejected Email

The screenshot shows an email interface. At the top, the subject is "Status Permohonan Penamatan Keahlian KADA". The sender is "KADA System <cfjahseh54@gmail.com>". The recipient is "To: You". The email body has a green header "Status Permohonan Penamatan Keahlian KADA". The main text says "Assalamualaikum dan Salam Sejahtera," followed by "Kami ingin memaklumkan bahawa permohonan penamatan keahlian anda telah **DILULUSKAN**." Below this is a grey box titled "Butiran Ahli:" containing the name "Name: JOANNE CHING YIN XUAN" and the number "No. Ahli: 650312010888". The text continues with "Keahlian anda telah dibatalkan berkuat kuasa serta-merta." and "Terima kasih atas sokongan anda sepanjang menjadi ahli KADA." At the bottom, it says "Yang benar, Pihak Pengurusan KADA". At the very bottom of the screenshot, there are two buttons: "Reply" and "Forward".

Approved Email

Figure 6.4: Interface for <Membership Termination>

6.1.5 Loan Application

Loan Type

Specific Loan Information

TAHAP	PRIBADI/PAK. DBK/PAK. DBK/PAK. DBK	KELOMPOK/PAK. DBK/PAK. DBK	PAK. DBK
1	RM 5,580.00	RM 231.00	RM 0.00

Loan Calculator

UTAMA MAKLUMAT MANFAAT AHLI PINJAMAN PERKHIDMATAN PELANGGAN

[← Kembali ke Skim Pembiayaan](#)

Borang Permohonan Pembiayaan

1 Maklumat Pembiayaan 2 Maklumat Peribadi 3 Maklumat Penjamin 4 Pengesahan

Maklumat Pembiayaan

Jenis Pembiayaan

Pilih Jenis Pembiayaan

Amaun Dipohon (RM) Tempoh Pembiayaan (Bulan)

Masukkan amaun

Ansuran Bulanan (RM)

Selanjutnya

Loan Application Form

UTAMA MAKLUMAT MANFAAT AHLI PINJAMAN PERKHIDMATAN PELANGGAN

[← Kembali ke Skim Pembiayaan](#)

Borang Permohonan Pembiayaan

1 Maklumat Pembiayaan 2 Maklumat Peribadi 3 Maklumat Penjamin 4 Pengesahan

Maklumat Pembiayaan

Jenis Pembiayaan

Pilih Jenis Pembiayaan

Sila pilih jenis pembiayaan

Amaun Dipohon (RM) Tempoh Pembiayaan (Bulan)

Masukkan amaun

Sila masukkan amaun yang dipohon

Sila masukkan tempoh pembiayaan

Ansuran Bulanan (RM)

Selanjutnya

Error messages appear when loan application mandatory field incomplete

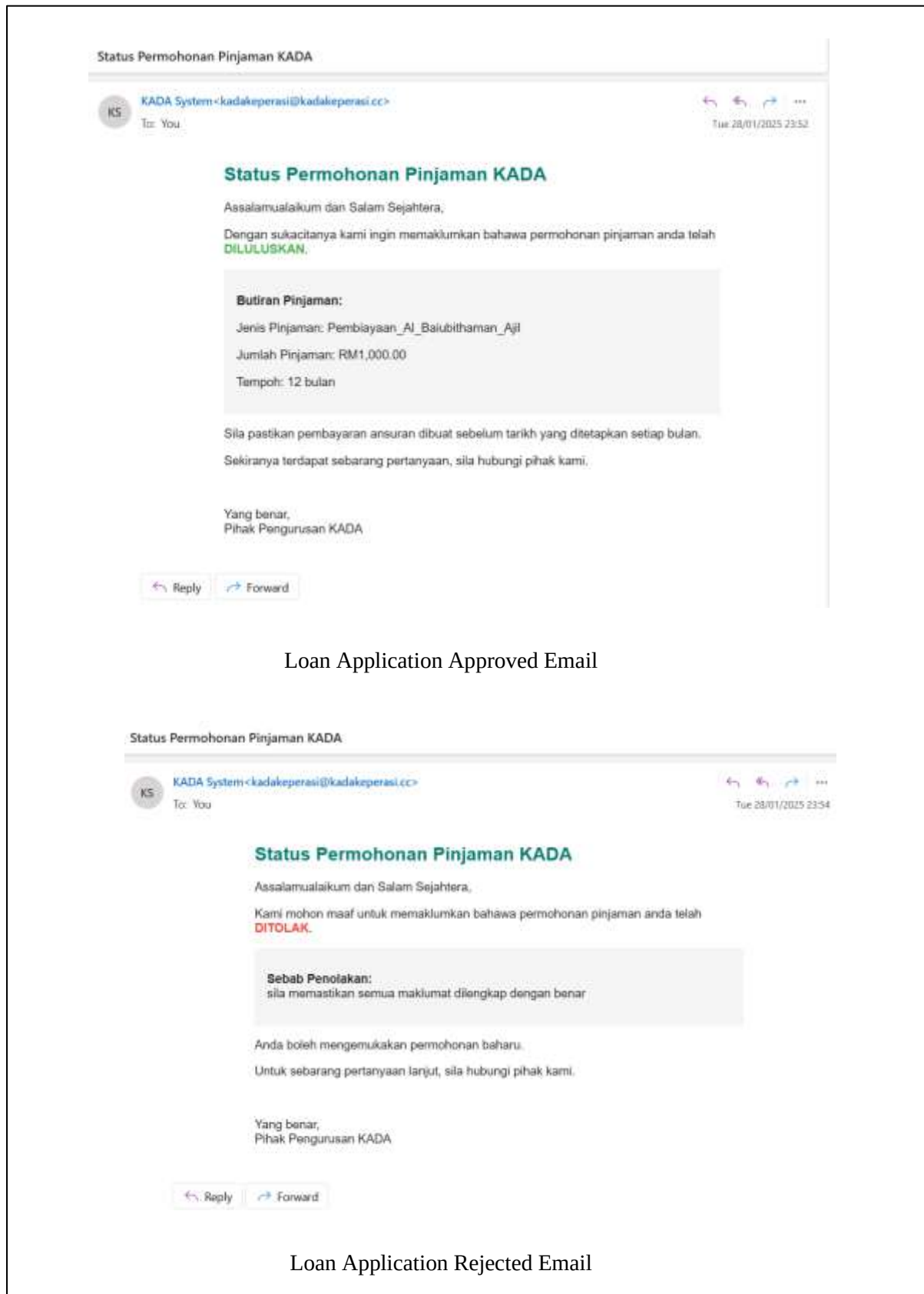
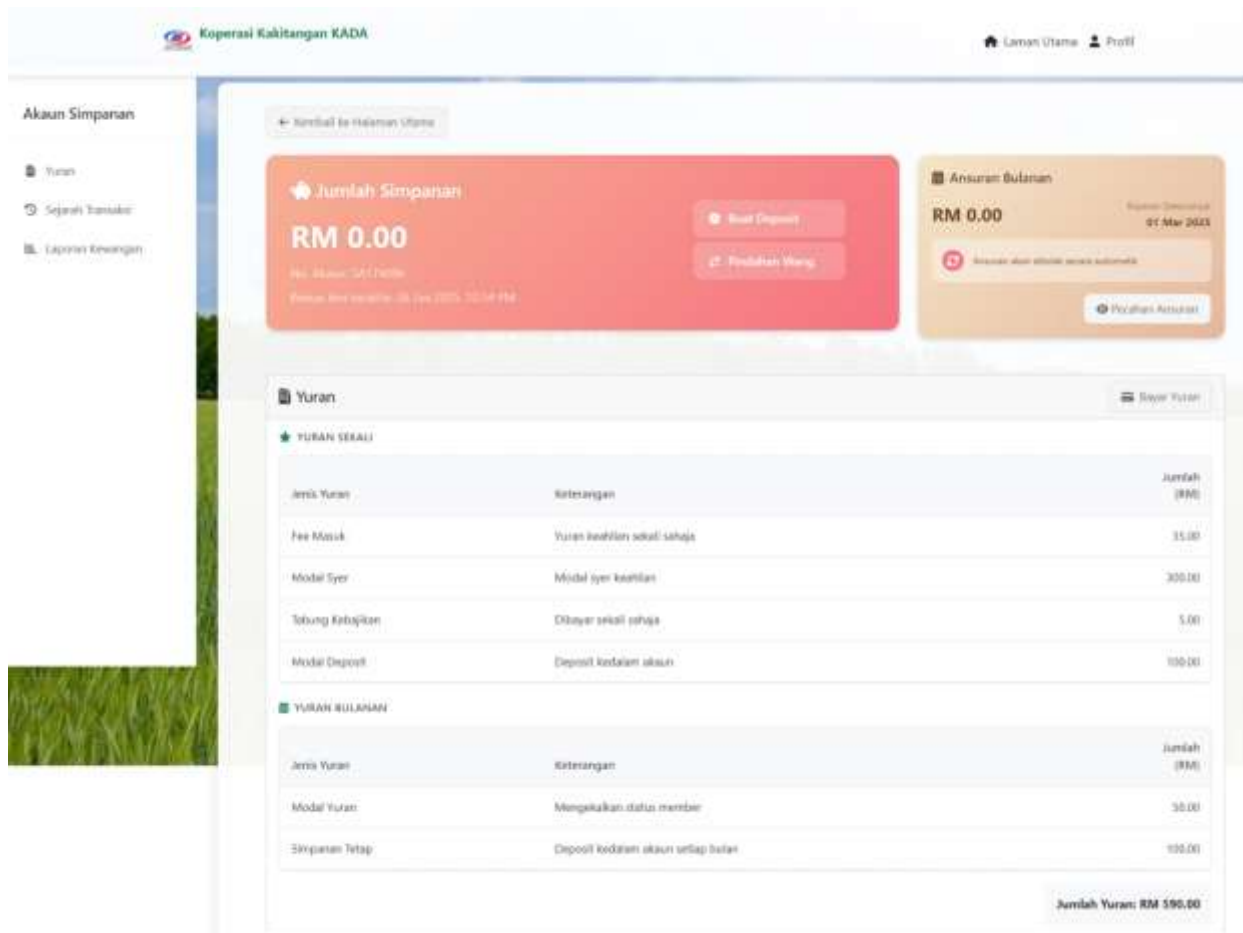


Figure 6.5: Interface for <Loan Application>

6.1.6 Saving Account



Saving Account Dashboard with navigation side bar



Loan Detail

Buat Deposit

Jumlah Deposit (RM)

RM 0.00

Kaedah Pembayaran

FPX

Kad

Pilih Bank

Masukkan nombor akaun bank

Catatan (Pilihan)

Tambah catatan untuk transaksi ini...

Teruskan Pembayaran

Deposit interface

Pindahan Wang

Jumlah Pindahan (RM)

RM 0.00

Nama Bank

Pilih Bank

Masukkan nombor akaun bank

Tujuan Pindahan

Pembayaran

Pemindahan

Pendidikan

Lain-lain

Catatan (Pilihan)

Tambah catatan untuk transaksi ini...

Teruskan Pembayaran

Transaction Interface

Pengesahan Deposit

No. Rujukan

TRX202501303116

Nama Ahli

JOANNE CHING YIN XUAN

No. Ahli

1234545

Jumlah Deposit

RM 10,000.00

Kaedah Pembayaran

fpx

Tarikh & Masa

30/01/2025 00:57

Baki akaun semasa: RM 0.00

Sahkan Pembayaran

Batal

Deposit Approval Interface

Pengesahan Pindahan

No. Rujukan	TRF202501304849
Nama Ahli	JOANNE CHING YIN XUAN
No. Ahli	1234545
Jumlah Pindahan	RM 10,000.00
Bank	Maybank
No. Akaun Bank	12345678
Tujuan Pindahan	payment
Tarikh & Masa	30/01/2025 01:04

Baki akaun semasa: RM 10,000.00

Sahkan Pindahan

Batal

Transaction Approval Interface



KOPERASI KADA

Lembaga Kemajuan Pertanian Kemubu
Peti Surat 127, Bandar Kota Bharu,
15710 Kota Bharu, Kelantan
Tel: +60 97455388

RESIT DEPOSIT

No. Resit: RES/2025/01/110
Tarikh: 30/01/2025 12:58 AM
Nama Ahli: JOANNE CHING YIN XUAN
No. Ahli: 1234545
Jumlah: RM 10,000.00
Kaedah Pembayaran: FPX Online Banking

*Terima kasih atas transaksi anda.
Resit ini dijana secara komputer dan tidak memerlukan tandatangan.
Dicetak pada: 30/01/2025 01:04:59 AM*

Receipt

Sejarah Transaksi					
Semua Transaksi					
Tarikh	Jenis	Jumlah (RM)	Status	Catatan	Tindakan
30/01/2025 12:58 AM	Deposit	RM 10,000.00	Dibulatkan		Muat Turun Resit
30/01/2025 12:56 AM	Penarikan	RM 1,000.00	Dalam Proses		-

Transaction History interface

Figure 6.6: Interface for <Saving Account>

6.1.7 Financial Report

Lembaga Kemajuan Pertanian Kemubu
KAPKEM

Kembali ke halaman Utama

NAMA: CHONG CHONG YIN SUAN

NO. K/P: 0001201980

NO. PH: 123456

NO. AHC: 123456

Penyata Kewangan

Tuan/Ruan:

PERKATAAN PERUSAHAAN BERSEKUTUAN KAPKEM KANDUNGAN SADA SELANGOR BERHAD SAGU SELATAN, Januari 2025

Untuk perincian terperinci, kami dengan ini menyatakan bagi akaun tahunan adalah sebagaimana berikut:

SUMBER DANA DULU

Modal Tiap:	Modal Tiap:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM 100.00	RM 50.00	RM 100.00	RM 10,000.00	RM 10,000.00

SUMBER DANA BUKU

Pembayaran Al Bar:	Pembayaran Al Iman:	Pembayaran Sim Sihat:	Pembayaran Road Tax & Insuran:	Pembayaran Al Qardhul Hasan:
RM 0.00	RM 0.00	RM 0.00	RM 0.00	RM 0.00

SEJARAH TRANSAKSI Januari 2025

Tarikh	Jenis	Jumlah (RM)	Status	Keterangan
01/01/2025	Deposit	10,000.00	Dikuburkan	

Laporan Pinjaman Januari 2025

Kategori Pinjaman

Jenis Pinjaman	Jumlah Pinjaman (RM)	Tempoh (Bulan)	Ansuran Bulanan (RM)	Status
Tidak terdapat pinjaman untuk 01/01/2025 - 31/01/2025				

[Masuk Tindakan Segera](#)

KONTAK KAMI

Lembaga Kemajuan Pertanian Kemubu
Lot 1001 101, Jalan Kudu Bharu,
10100 Kudu Bharu, Kulim
+60 174711388
lankap@kankem.gov.my

QRIS QR

HEUTY KAMI

© 2025 KAPKEM. Semua hak terpeliharakan.

Figure 6.7: Interface for <Financial Report>

6.1.8 Administration Site for Loan Approval

Tarikh	Nama	Jenis Pinjaman	Jumlah (RM)	Tindakan
2025-01-28 17:44:22	ELIJAH SHE YU SHENG	Pembayaran_Ai_Baiubithaman_Aji	RM 15,000.00	Lihat Lulus
2025-01-28 17:28:17	Lee Yin Shen	Pembayaran_Ai_Baiubithaman_Aji	RM 15,000.00	Lihat Lulus

Loan Approval List with drop down for filtration

Tolak Permohonan Pinjaman

Catatan Admin

Sila masukkan sebab penolakan...

Batal Hantar

Maklumat Permohonan Pinjaman

Maklumat Pribadi

Nama: Lee Yin Shen No. ID: 000001 No. HP: 0123456789 No. NIK: 000001 No. NKP: 000001 No. NRE: 000001

Jantina: Laki-laki Agama: Buddha Wangsa: Cina

Tarikh Lahir: 2004-12-30

Alamat

Alamat Rumah

No. 123, Jalan 123, Taman 123, 45678, Kuala Lumpur, Selangor, 12345

Alamat Pejabat

No. 123, Jalan 123, Taman 123, 45678, Kuala Lumpur, Selangor, 12345

Maklumat Pinjaman

Jenis Pinjaman: Pembayaran_Ai_Baiubithaman_Aji

Jumlah Pinjaman: RM 15,000.00

Tempoh (Bulan): 36

Ansuran Bulanan: RM 444.44

Maklumat Penjamin

Penjamin 1

Nama: A.A. No. ID: 000 No. HP: 000 No. NIK: 000 No. NKP: 000 No. NRE: 000

Penjamin 2

Nama: B.B. No. ID: 000 No. HP: 000 No. NIK: 000 No. NKP: 000 No. NRE: 000

Lulus Tolak Batal

Admin rejection remark

Applicant details

Figure 6.8: Interface for <Loan Approval>

6.1.9 Administration Site for Membership Approval

Senarai Ahli

Semua Ahli

Nama	No. KP	Jantina	Jawatan	Gaji Bulanan	Tindakan	Status
ELIJAH SHE YU SHENG	77777777777	Male	1	RM 80,000.00	Lihat Lulus Tolak	Dibulkan
JOANNE CHING	040312010888	Female	Ahli	RM 10,000.00	Lihat Lulus Tolak	Dibulkan
Choh Jing yi	040809050165	Male	Student	RM 3,000.00	Lihat Lulus Tolak	Dibulkan
Lee Yin Shen	11111111111	Male	A123b	RM 88,888.00	Lihat Lulus Tolak	Dibulkan

Membership Approval List with drop down for filtration

Tolak Permohonan Keahlian

Catatan Admin

Sila semak semula profil anda

Batal Hantar

Senarai Ahli

Nama	No. KP	Jantina	Jawatan	Gaji Bulanan	Tindakan
ELIJAH SHE YU SHENG	77777777777	Male	1	RM 80,000.00	Lihat Lulus Tolak
JOANNE CHING	040312010888	Female	Ahli	RM 10,000.00	Lihat Lulus Tolak
Choh Jing yi	040809050165	Male	Student	RM 3,000.00	Lihat Lulus Tolak

Admin rejection remark

Figure 6.9: Interface for <Membership Approval>

6.1.10 Administration Report



Administration Report that can be downloaded

Figure 6.10: Interface for <Administration Report>

6.1.11 Administration Site for Membership Termination Approval

Senarai Penamatan Keahlian

Semua Ahli

Nama	No. KP	Jantina	Tarikh Permohonan	Alasan Penamatan	Tindakan	Status
ELIJAH SHE YU SHENG	777777777777	Male	28/01/2025	Pencen Awal	Lihat	Ditolak

Kembali

Jumlah Ahli Aktif
4

Jumlah Ahli Tidak Aktif
0

Cadangan Pencen (60+)
0

Senarai Ahli Aktif

Semua Umur

☒ Ahli Aktif
☐ Ahli Tidak Aktif

No.	Nama	No. Ahli	No. IC	Umur	Status	Tindakan
1	ELIJAH SHE YU SHENG	1233	777777777777	48	Aktif	Lihat Nyahaktif
2	JOANNE CHING	12345	040312010888	21	Aktif	Lihat Nyahaktif
3	Choh Jing yi	A2	040809050185	21	Aktif	Lihat Nyahaktif
4	Lee Yin Shen	P865443	1111111111	14	Aktif	Lihat Nyahaktif

Figure 6.11: Interface for <Membership Termination Approval>

7.0 Traceability

Sprint #	Subsystem / Package	User Story ID
Sprint 1	Membership Registration	US001
		US002
		US003
	Terminate Membership	US012
Sprint 2	Loan Application	US004
Sprint 3	Saving Account Management	US005
Sprint 4	Financial Report Request	US006
		US007
Sprint 5	Administration (Application Handling)	US008
		US009
		US010
Sprint 6	Administrative (Reports)	US011

Table 7.1: Traceability

